Honeywell

Asset Performance Management

Installation Guide

Release 540.1

August 2023

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CHAPTER

1 ABOUT THIS GUIDE

1.1 Purpose

This guide describes the procedures for installing Asset Performance Management (previously Uniformance Asset Sentinel) in the necessary server and client systems.

1.2 Audience

This document is intended for the project engineers and system administrators who are responsible for installing Asset Performance Management.

1.3 Revision History

Revision	Date	Description
1	August 2023	Asset Performance Management R540.1 General Availability (GA) release

1.4 References

The following list identifies all documents that may be sources of reference for material discussed in this publication.

> **NOTE**
> Asset Performance Management documentation is available on the installation media in the following location:
>
> **[installation Media]:\Installer\Components\US\Documents**

Document Name	Document Number
Asset Performance Management Upgrade Guide	APMDOC-X389-en-R540.1
Asset Performance Management Configuration Guide	APMDOC-X391-en-R540.1
Asset Performance Management User's Guide	APMDOC-X392-en-R540.1

For the most up-to-date version of the documentation, see https:// www.honeywellprocess.com (click the Support tab). User registration and login are required to access the Honeywell Process site.

CHAPTER

2 PLANNING FOR INSTALLATION

This chapter describes the planning requirements for Asset Performance Management installation, which include, the installation scenarios, prerequisite knowledge required for installation, hardware and software requirements, deployment topologies, and the licenses required for Asset Performance Management.

> NOTE
> If an earlier version of Asset Performance Management is already installed, refer to **Upgrading a Single Server** in Asset Performance Management upgrade guide.

2.1 Installation Scenarios

Following are the installation scenarios of Asset Performance Management:

1. Fresh Installation
2. Migration from earlier releases (Refer to Asset Performance Management Upgrade Guide for more information)

2.1.1 Fresh Installation

If you are new to Asset Performance Management and you are installing the application for the first time, it is recommended that you read the information in the following sections in the order specified.

Section	Provides information on
Software and Hardware Requirements	Hardware and software requirements that should be available to install Asset Performance Management.
Single Server Installation	How to install Asset Performance Management in a single server.
Split Server Installation	How to install Asset Performance Management in a split server.
Primary and Shadow Server Installation	How to install Asset Performance Management in a primary and shadow server.

2.2 Knowledge Required

Asset Performance Management installation assumes that:

- You have read the Software Change Notice that accompanied this release.
- You are familiar with the concepts and components of the Intuition Core. If not, refer to the **Honeywell Intuition Core Installation Guide** and **Intuition Core Software Change Notice**.
- You are familiar with the concepts and components of Asset Performance Management. If not, refer to **Asset Performance Management User Guide**.
- You are familiar with Microsoft Windows system administration, including security configuration. If not, contact your

Windows network or system administrator.

- You are familiar with the SQL Server Management Console and have the required SQL administrative privileges.
- You are familiar with the administration and configuration of Experion, which is necessary to implement asset synchronization. If not, contact your control system administrator.

2.3 Software and Hardware Requirements

2.3.1 Hardware Requirements

Connect with the Honeywell project team to determine the hardware sizing based on project requirements. The below table lists the hardware requirements for the servers and the client computers.

Hardware	Minimum Requirement	Recommended Requirement
Asset Performance Management Primary or Shadow Server		
Processor	16 Logical Processors 3GHz, 64-bit (x64)	16 Logical Processors 3GHz, 64-bit (x64)
RAM	16 GB Memory	32 GB Memory
Hard disk drive	500 GB, 15K RPM	500 GB, 15K RPM
Asset Performance Management Database, Application, or Web Server		
Processor	8 Logical Processors 3GHz	16 Logical Processors 3GHz
RAM	16 GB Memory	64 GB Memory
Hard disk drive	150 GB, 15K RPM	500 GB, 15K RPM
Asset Performance Management Calculation Server		
Processor	16 Logical Processors 3GHz	16 Logical Processors 3GHz
RAM	16 GB Memory	32 GB Memory
Hard disk drive	250 GB, 15K RPM	500 GB, 15K RPM
Asset Performance Management Client (plain client without any Asset Performance Management tools)		
Processor	Dual Core Processor, 1GHz	-
RAM	8 GB Memory	-
Experion Flex Station with Asset Synchronizer		
Processor	Dual Core Processor, 1GHz	-
RAM	8 GB Memory	-
Hard disk drive	40 GB frees pace	-

2.3.2 Software Requirements

The software specified in this section must be installed on the computer before you start installing Asset Performance Management.

NOTE
- Microsoft components can be freely downloaded from the Microsoft website.
- All servers must support TLS 1.2.

Operating System and Software Requirements

The below table lists the operating system and software requirements for the Asset Performance Management server and client computers.

Software Requirements	Details
Server Requirements	• Windows Server 2017/2019 • Microsoft SQL Server 2019 Standard Edition/Enterprise Edition (64bit) or Microsoft SQL Server 2017 Standard Edition/Enterprise Edition (64bit) • Microsoft Office 2013 or Microsoft Office 2016 (64bit)(optional) • Microsoft.NET Framework 3.5 Service pack 1 • Microsoft.NET Framework 4.8
Client Requirement (plain client without any Asset Performance Management tools)	• Windows 8.1 (32bit and 64bit) or Windows 10 (32bit and 64bit) • Microsoft.NET Framework 3.5 Service pack 1 • Microsoft.NET Framework 4.8 • Google Chrome 74.0.3729.108 and above • Microsoft Edge 91.0.864.54 and above
APM Configuration Excel Plugin	• Windows 8.1 (32bit) or Windows 10 (64bit) • Microsoft Office 2013 or Microsoft Office 2016 (64bit) or Microsoft Office 365 • Microsoft Office should be installed using the **All Users** option in the installation wizard.

Other Software Requirements

The below table lists the available tools and the settings that should be available on the computer before you start installing Asset Performance Management.

Requirements	Points to note
Microsoft Office 2013, Microsoft Office 2016 (64-bit), or Microsoft Office 365	Installing Microsoft Excel is mandatory
Microsoft.NET Framework 3.5 Service pack 1	-
Microsoft.NET Framework 4.8	If Microsoft SQL Server is installed on the same server, .NET features are enabled automatically.
Microsoft SQL Server 2019 Standard Edition/Enterprise Edition (64bit) or Microsoft SQL Server 2017 Standard Edition/Enterprise Edition (64bit)	During installation, enable the following Instance and Shared features. **Database Engine Services** • Client Tools Connectivity • Client Tools Backwards Compatibility

Requirements	Points to note
	• Management Tools - Basic • Management Tools - Complete • Reporting Service • Replication Service for L3.5 replication
	During the SQL Server installation, set the **Startup Type** to **Automatic** for the following SQL services: • SQL Server Browser • SQL Server (Instancename) This ensures that the SQL service starts automatically each time the server computer is restarted.
Intuition Core	R302.1
InfluxDB Shell	Version 1.8.3-1
ULM	Version 6.6
EPHD	Version 410
The required roles, role services, and features for Asset Performance Management are installed automatically during installation.	

Domain Controller

- Ensure the user installing **Uniformance Asset Manager** (installation user) on all servers has **read** permissions on the domain controller.
- Ensure the domain controllers and all Asset Performance Management servers have the same network link speed.

2.3.3 EPHD Native Historian

Asset Performance Management comes with an EPHD installation for use with systems that do not have an Enterprise PHD available on-site, or where Asset Performance Management cannot write to the current plant historian.

> **NOTE**
> We recommend using Enterprise PHD when possible.

2.3.4 Additional Software Compatibility

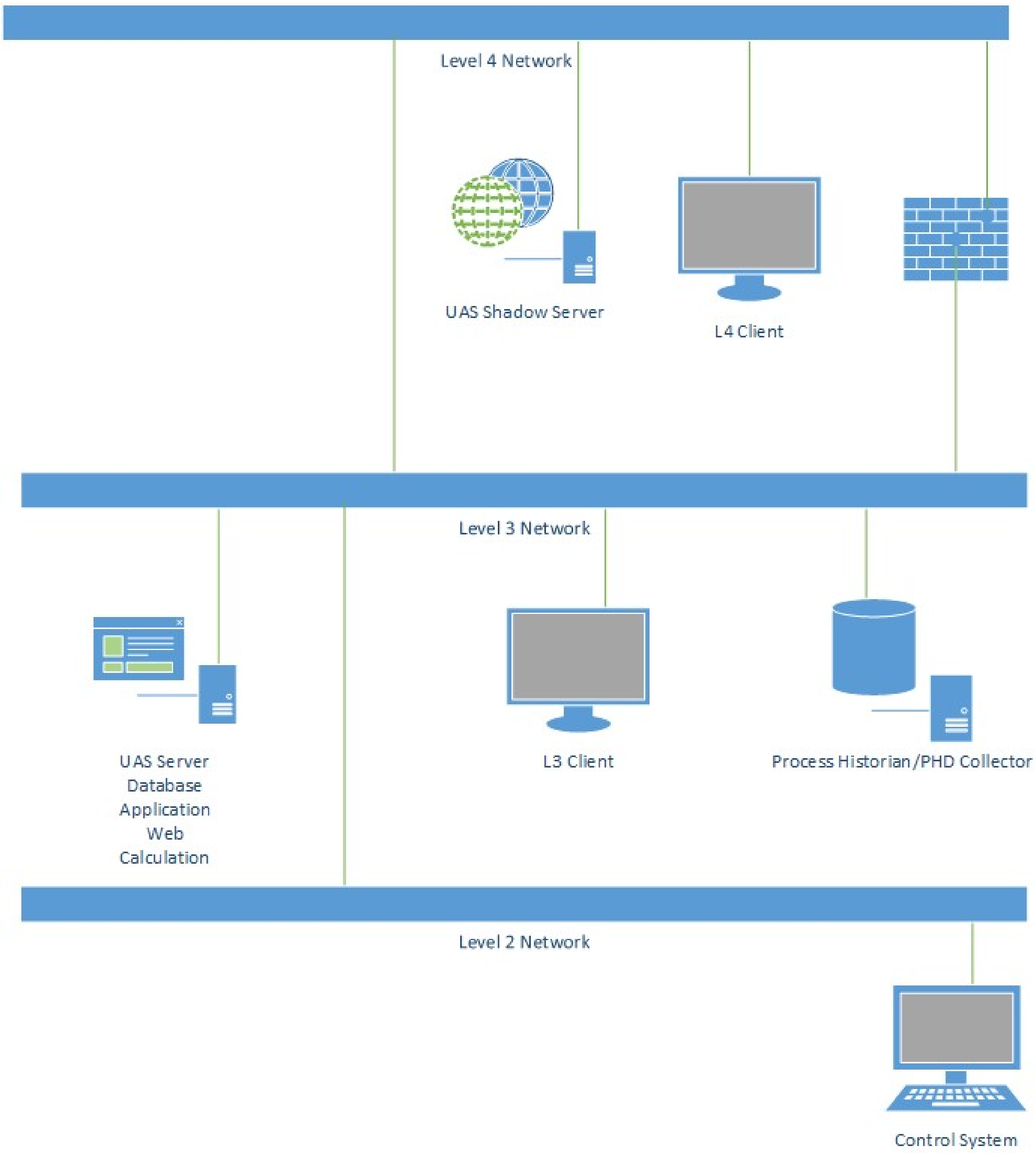
Asset Performance Management is compatible with the listed versions of the following software:

Product	Version
Uniformance Insight	R240
Field Device Manager (FDM)	R520.1 and R521.1
Sure Sense	6.1.11.40 and 6.2

2.4 Asset Performance Management Topology

2.4.1 Single Server Topology

In this topology, all the components of Asset Performance Management (Application Server, Database Server, Web Server, and Calculation Server) are installed on an L3 network. The Asset Performance Management client is also installed on an L3 network to access the Asset Performance Management application.

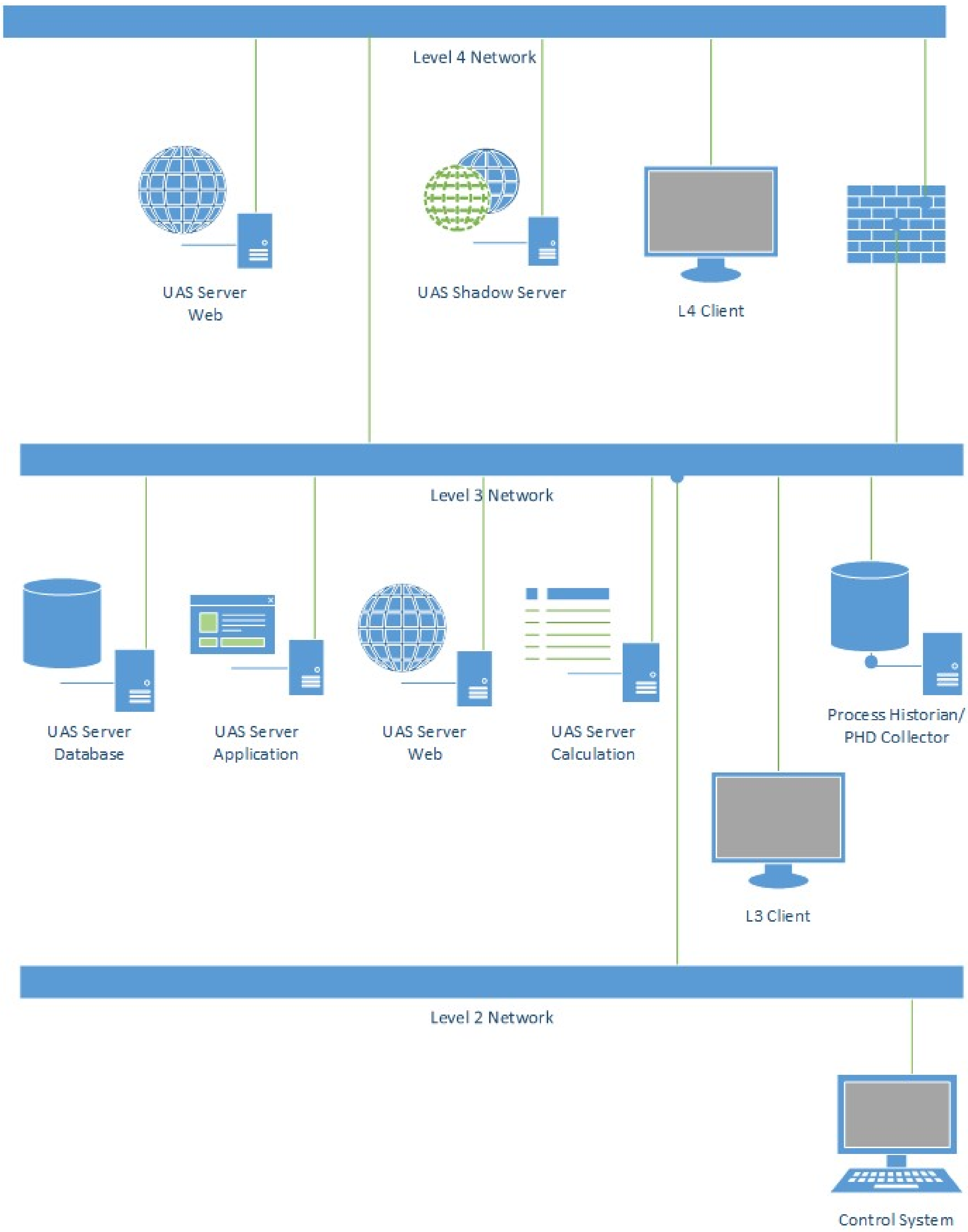


2.4.2 Split Server Topology

This topology separated the Asset Performance Management components out, most commonly into 3 Servers: Database Server, Application/Web Server, and Calculation Server.

Split Server Topology can also be installed in a 2-server arrangement (with the Application, Web, and Calculation components installed on the same computer) or a 4-server arrangement (with Database, Application, Web, and Calculation components all installed on separate servers). The number of servers selected depends on the resource requirements of your system. For example, we recommend the following arrangements:

- 2 Servers: This arrangement is useful when there are a high number of users browsing the site, but a low number of calculations or assets.
- 3 Servers: This arrangement is typical and supports a large number of users, but also a significant number of calculations and assets.
- 4 Server: This arrangement is recommended for systems with a large number of calculations.



2.4.3 Across Firewall Topology

Level 4 network users with assigned APM roles can access the Asset Performance Management server using clients located in the same network. By placing a shadow server in the Level 4 network as shown in the topology diagram below, the following Asset Performance Management functions can be accessed from L3.5/L4:

- Add/Update Recommendation
- Change Fault or Symptom status
- Add/Update Asset Logs
- Add/Update Equipment status
- Enter manual observation data
- Insert Manual Fault
- Manual initiation of CMMS notification
- Equipment attribute real-time data read/write
- Read Trend data
- Reports

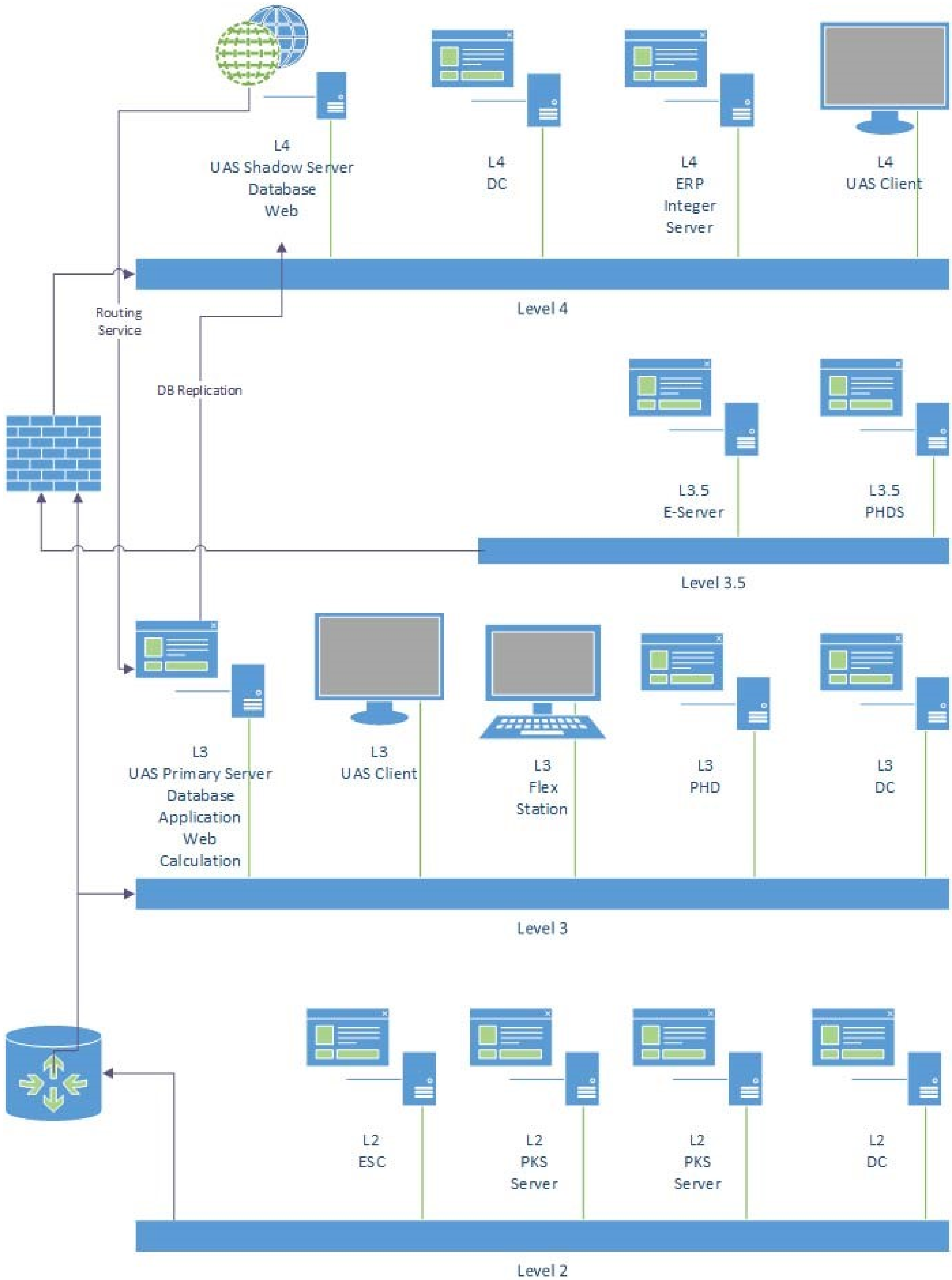
All write operations from Level 4 are redirected to the Level 3 server using the Service Routing capability in L4. For read operations, the SQL replication service is used to transfer data from L3 Asset Performance Management Server to L4 Asset Performance Management using replication service via a TCP port.

For service routing, if there is trust between the L3 and L3.5/L4 domains, a Windows user account can be used for authentication between the Primary and Shadow Servers. If no trust exists between the domains, security certificates must be used for authentication between the domains.

For more information about certificate configuration, refer to Configuring Security Certificates.

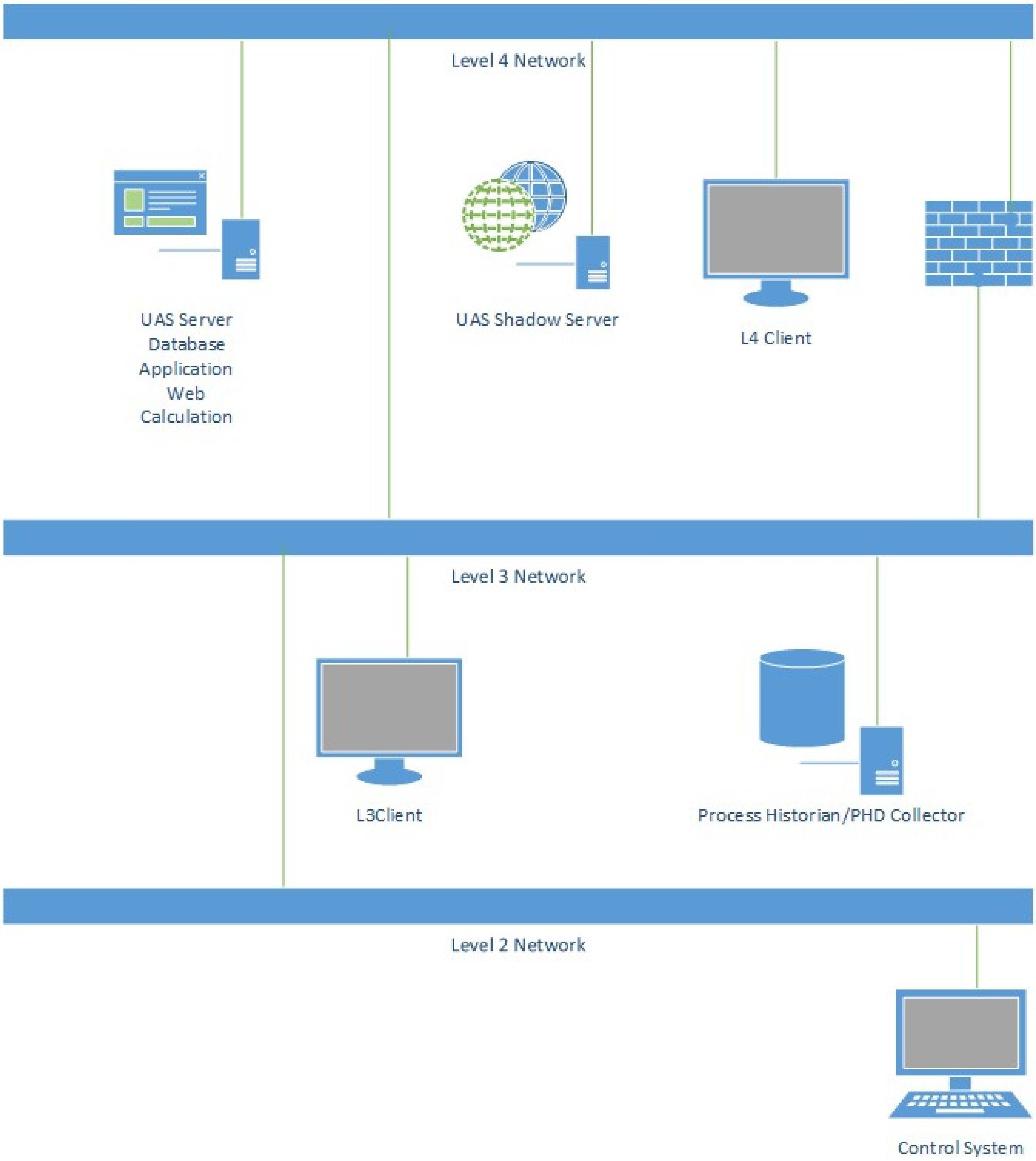
For more information about configuring the replication service, refer to Configuring SQL Replication.

For more information on the ports that are opened between L3 and L3.5 networks, refer to Open Ports in the Firewall.



2.4.4 Asset Performance Management Installed on L4 Network

In this topology, the components of Asset Performance Management such as the Application Server, Database Server, and Web Server are installed on an L4 network. The application server accesses real-time data through a PHD Shadow Server, which is installed on L4.



2.4.4.1 Limitations of Asset Performance Management Installed on L4 Network

The following are the limitations of Asset Performance Management installed on the L4 network:

- It is not possible for Asset Performance Management to directly connect to the Experion datasource.
- The Shadow PHD server is required to provide real-time data values to the Asset Performance Management Server.
- It is not possible to view Asset Performance Management alerts on the Experion station.
- Any changes to faults made on the L4 network will appear in the Fault History as the changes are made by the L4 Administrator, not the actual user.
- Scope-based security on the L3 network cannot be enforced on the L4 network.

2.5 Review Licensing Requirements

You should review the available Asset Performance Management licenses, to determine the licensing requirements of your planned system. For a list of the license types, refer to Asset Performance Management Licenses.

2.6 Network Planning

2.6.1 Ports

The Asset Performance Management application uses the following ports for various internal functions:

> **NOTE**
> The below ports can be configured as per your requirement.

Service	Listening Port	Source	Destination	Direction
AM Application Server	9020	Application Server	Calculation Server	Uni direction
AM Application Server	9021	Application Server	Calculation Server	Bi Direction
AM Application Server	9022	Application Server	Calculation Server	Uni direction
AM Application Server	9024	Application Server	Calculation Server	Bi Direction
AM Application Server	9025	Application Server	Calculation Server	Bi Direction
AM Application Server	9026	Application Server	Calculation Server	Uni direction
AM Application Server	9028	Application Server	Calculation Server	Uni direction
AM Application Server	9029	Application Server	Calculation Server	Bi Direction
AM Application Server	9030	Application Server	Calculation Server	Bi Direction
AM Application Server	9031	Application Server	Calculation Server	Uni direction
AM Application Server	9032	Application Server	Calculation Server	Bi Direction
AM Application Server	9033	Application Server	Calculation Server	Uni direction
AM Application Server	9034	Application Server	Calculation Server	Uni direction
AM Application Server	9038	Application Server	Calculation Server	Uni direction
AM Application Server	9041	Application	Calculation Server	Uni

Service	Listening Port	Source	Destination	Direction
		Server		direction
AM Application Server	9042	Application Server	Calculation Server	Uni direction
AssetManagerIntegration Service	9898	web clients	Web Server	Uni direction
HDA Service	9040(HDA Service)	Application Server	Application Server	Uni direction
Thermo Service	9043(Thermo Service)	Application Server	Application Server	Uni direction
Fault Service	9044(Fault Service)	Application Server	Calculation Server	Uni direction
	HTTPS PORT (default443)	Web server	Application Server	Uni direction
	HTTPS PORT (default443)	Web clients	Web Server	Uni direction
	SQL Port(default1443*	Application Server Calculation Server	Database Server	Uni direction
	HTTPS PORT(default-443)*	Web Client	SSRS Report (Database Server)	Uni direction
	SQL Port(default- 1443	SSRS Report	Database Server	Uni direction
	8086 influx DB	Application Server Calculation Server	InfluxDB (Database Server)	Uni direction
	RabbitMQ 5672 and 5671	Additional Calculation Server Application Server	Primary Calculation Server	Bi Direction

2.6.2 EPHD Servers

Service	Port	Source	Destination	Direction
APIServer	TCP 3100	Application Server Calculation Server	EPHD Server	Uni direction
APIServer	TCP 3101	Application Server Calculation Server	EPHD Server	Uni direction

Service	Port	Source	Destination	Direction
UDBServer	53100	Application Server Calculation Server	EPHD Server	Uni direction

2.6.3 Open Ports in the Firewall

The following TCP ports should be opened in the firewall for **L3** and **L3.5** communication:

Service	Port	Source	Destination	Function
AMHostApplication AM Historian Service	808*	Shadow APM server	Primary Calculation Server	Uni direction
SQL Server, SQL replication service	55901*	Shadow APM server	Primary Database Server	Bi Direction

> **NOTE**
> The above ports can be configured as per your requirement.

2.7 Asset Performance Management User Groups and Users

The Asset Performance Management application can create a default local windows user **AMUser** during installation. The user **AMUser** is then used to run all the Asset Performance Management applications.

If you do not want to use the default user, during installation select a different services user.

The table below describes the various groups created in Asset Performance Management and their assigned permissions.

The following services run under **AMUser** (can also be run as a domain user, or a different local user).

Groups	Permissions
AMUsersGroup	Full control is given to access Asset Performance Management applications
Product Administrators	Full Control is given to access EPHD applications
AMViewOnlyUsers	Refer to Predefined Roles and Tasks in Asset Performance Management.
Maintenance Engineers	Refer to Predefined Roles and Tasks in Asset Performance Management.
Maintenance Planners	Refer to Predefined Roles and Tasks in Asset Performance Management.
Maintenance Technicians	Refer to Predefined Roles and Tasks in Asset Performance Management.
AMSystemAdmin	Full control of Asset Performance Management applications.
Reliability Engineers	Refer to Predefined Roles and Tasks in Asset Performance Management.

Services	Functionality
AMApplicationServer	Hosting application for Asset Performance Management
AssetManagerIntegration Service	Hosting application for integration services

Services	Functionality
Native Historian Services	Native Historian applications
Uniformance PHD Server	
Uniformance Database Server	
Uniformance RDI Server	
Uniformance DB Security Server	
Uniformance Legacy API server	
Uniformance remote API server	
Uniformance PHD Startup	

2.7.1 Predefined Roles and Tasks in Asset Performance Management

The following pre-defined roles and tasks performed by the Asset Performance Management roles are created during installation. These roles can be used to grant access to Asset Performance Management to additional users and groups on your system. If users are now assigned these roles or not a member of a group assigned to these roles, they will not be able to access the related Asset Performance Management features.

> **NOTE**
> Tasks under the Application Server node are required for services and should not be changed for any of the predefined roles.

AM System Admin

Category	Task Name	Task Description
Configuration	ConfigureExternalDatasource	Configuring external data sources.
Navigate	Base	Access to the APM link.
Navigate	DataSource Configuration	Access to the Datasources link on the Contents tree.

Maintenance or Reliability Engineer

Category	Task Name	Task Description
Configuration	ConfigureClassTypeLibrary	Add, modify, and delete asset classes, types, and asset type attributes.
Configuration	ConfigureAssets	Add, modify, and delete asset hierarchy and assets.
Configuration	SynchronizeAssets	Import asset data.
Configuration	BulkConfiguration	Use the APM Configuration Excel Plugin to update asset information.
Configuration	TrendConfiguration	Configure asset attribute trends.
Configuration	BaselineConfiguration	Set baseline data for the EHM algorithm.
Configuration	APMConfiguration	Add, modify, and delete pre-processors and custom functions, and import Thermo configuration.
View	AssetsView	View the state of all equipment.
View	DashboardView	Access equipment health dashboard.

Category	Task Name	Task Description
View	TrendsView	View asset attribute trends.
View	AssetInformationView	View asset information.
View	FaultModelConfigurationView	View fault model configuration.
View	AssetConfigurationView	View asset and asset attribute configuration.
View	FaultViewer	View the Event Viewer.
View and Analyze	ViewAssets	View the state of all equipment.
View and Analyze	AccessDashboard	Asset overview and view equipment health index.
View and Analyze	ViewFaultSummary	Change fault status, eliminate faults, and unlock faults.
View and Analyze	AnalyzeVibrationData	Analyze vibration data.
View and Analyze	ViewTrends	Use the asset trends.
View and Analyze	ViewAssetInformation	Use asset details.
View and Analyze	ViewFaultHistory	Use the event history.
View and Analyze	ViewFaultModelConfiguration	Use model configuration.
View and Analyze	ViewAssetConfiguration	Use asset configuration.
View and Analyze	SetBaseline	Use and create baselines.
View and Analyze	AnalyzeVibrationData	Analyze vibration data.
View and Analyze	ViewReports	View standard reports.
Navigate	Base	Access to the APM link.
Navigate	Build Workspaces	Access to the Workspaces link on the Contents tree.
Navigate	Dashboard	Access to the dashboard link on the Contents tree.
Navigate	Calculation Configuration	Access to the Calculations link on the Contents tree.
Navigate	Asset Report	Access to the Asset Report link on the Contents tree.
Navigate	Event History	Access to the Event History link on the Contents tree.
Navigate	Fault Tree Status	Access to the Fault Tree link on the Contents tree.
Navigate	Performance Monitoring	Access to the Performance Overview link on the Contents tree.
Navigate	Datasource Status	Access to the Datasource Status link on the Contents tree.

Category	Task Name	Task Description
Navigate	DataSource Configuration	Access to the Datasources link on the Contents tree.
Navigate	Manual Data Entry	Access to the Manual Data Entry link on the Contents tree.
Navigate	Asset Monitoring	Access to the Asset Status link on the Contents tree.
Navigate	Template Configuration	Access to the Templates links on the Contents tree.
Navigate	Asset Attributes	Access to the Asset Attributes links on the Contents tree.
Navigate	Calculation Status	Access to the Calculation Status link on the Contents tree.
Navigate	KPI Report	Access to the KPI Report link on the Contents tree.
Navigate	Fault Report	Access to the Fault Report link on the Contents tree.
Navigate	Investigate Events	Access to the Event Investigation link on the Contents tree.
Navigate	Asset Detail	Access to the Asset Details link on the Contents tree.
Navigate	Asset Trend	Access to the Asset Trends link on the Contents tree.
Navigate	Asset Configuration	Access to the dashboard link on the Contents tree.
Navigate	Event Monitor	Access to the Reports links on the Contents tree.

Maintenance Planner

Category	Task Name	Task Description
View	AssetsView	View the state of all equipment.
View	DashboardView	Access equipment health dashboard.
View	Analyze > ReadOnlyView	View in ReadOnly mode: attribute values or time stamps cannot be edited and the asset status cannot be changed.
View	TrendsView	View asset attribute trends.
View	AssetInformationView	View asset information.
View	FaultViewer	View the Event Viewer.
View and Analyze	ViewAssets	View the state of all equipment.
View and Analyze	AccessDashboard	Asset overview and view equipment health index.
View and Analyze	ViewTrends	Use the asset trends.
View and Analyze	ViewAssetInformation	Use asset details.
View and Analyze	ViewFaultHistory	Use the event history.
View and Analyze	AnalyzeVibrationData	Analyze vibration data.
View and Analyze	ViewReports	View standard reports.

Category	Task Name	Task Description
Navigate	Base	Access to the APM link.
Navigate	Dashboard	Access to the dashboard link on the Contents tree.
Navigate	Asset Report	Access to the Reports links on the Contents tree.
Navigate	Event History	Access to the Event History link on the Contents tree.
Navigate	Fault Tree Status	Access to the Fault Tree link on the Contents tree.
Navigate	Performance Monitoring	Access to the Performance Overview link on the Contents tree.
Navigate	Datasource Status	Access to the Datasource Status link on the Contents tree.
Navigate	Asset Monitoring	Access to the Asset Status link on the Contents tree.
Navigate	Asset Attributes	Access to the Asset Attributes links on the Contents tree.
Navigate	Calculation Status	Access to the Calculation Status link on the Contents tree.
Navigate	KPI Report	Access to the KPI Report link on the Contents tree.
Navigate	Fault Report	Access to the Fault Report link on the Contents tree.
Navigate	Investigate Events	Access to the Event Investigation link on the Contents tree.
Navigate	Asset Detail	Access to the Asset Details link on the Contents tree.
Navigate	Asset Trend	Access to the Asset Trend link on the Contents tree.
Navigate	Event Monitor	Access to the Event Monitor link on the Contents tree.

Maintenance Technician

Category	Task Name	Task Description
Configuration	SynchronizeAssets	Import asset data.
View	AssetsView	View the state of all equipment.
View	DashboardView	Access equipment health dashboard.
View	TrendsView	View asset attribute trends.
View	AssetInformationView	View asset information.
View	FaultViewer	View the Event Viewer.
View and Analyze	ViewAssets	View the state of all equipment.
View and Analyze	AccessDashboard	Asset overview and view equipment health index.
View and Analyze	ViewTrends	Use the asset trends.
View and Analyze	ViewAssetInformation	Use asset details.
View and Analyze	ViewFaultHistory	Use the event history.
View and Analyze	AnalyzeVibrationData	Analyze vibration data.
View and Analyze	ViewReports	View standard reports.
Navigate	Base	Access to the APM link.

Category	Task Name	Task Description
Navigate	Dashboard	Access to the dashboard link on the Contents tree.
Navigate	Asset Report	Access to the Reports links on the Contents tree.
Navigate	Event History	Access to the Event History link on the Contents tree.
Navigate	Fault Tree Status	Access to the Fault Tree link on the Contents tree.
Navigate	Performance Monitoring	Access to the Performance Overview link on the Contents tree.
Navigate	Datasource Status	Access to the Datasource Status link on the Contents tree.
Navigate	Asset Monitoring	Access to the Asset Status link on the Contents tree.
Navigate	Asset Attributes	Access to the Asset Attributes links on the Contents tree.
Navigate	Calculation Status	Access to the Calculation Status link on the Contents tree.
Navigate	KPI Report	Access to the KPI Report link on the Contents tree.
Navigate	Fault Report	Access to the Fault Report link on the Contents tree.
Navigate	Investigate Events	Access to the Event Investigation link on the Contents tree.
Navigate	Asset Detail	Access to the Asset Details link on the Contents tree.
Navigate	Asset Trend	Access to the Asset Trend link on the Contents tree.
Navigate	Event Monitor	Access to the Event Monitor link on the Contents tree.

AM View Only

Category	Task Name	Task Description
Configuration	TrendReadOnly	Allows you to view asset attribute trends.
View	AssetsView	View the state of all equipment.
View	DashboardView	Access equipment health dashboard.
View	TrendsView	View asset attribute trends.
View	AssetInformationView	View asset information.
View	Analyze > ReadOnlyView	View in ReadOnly mode: attribute values or time stamps cannot be edited and the asset status cannot be changed.
View	FaultViewer	View the event viewer.
Navigate	Base	Access to the APM link.
Navigate	Asset Report	Access to the Reports links on the Contents tree.
Navigate	Event History	Access to the Event History link on the Contents tree.
Navigate	Fault Tree Status	Access to the Fault Tree link on the Contents tree.
Navigate	Performance Monitoring	Access to the Performance Overview link on the Contents tree.
Navigate	Asset Monitoring	Access to the Asset Status link on the Contents tree.
Navigate	Asset Attributes	Access to the Asset Attributes links on the Contents tree.

Category	Task Name	Task Description
Navigate	Calculation Status	Access to the Calculation Status link on the Contents tree.
Navigate	KPI Report	Access to the KPI Report link on the Contents tree.
Navigate	Fault Report	Access to the Fault Report link on the Contents tree.
Navigate	Investigate Events	Access to the Event Investigation link on the Contents tree.
Navigate	Asset Detail	Access to the Asset Details link on the Contents tree.
Navigate	Asset Trend	Access to the Asset Trend link on the Contents tree.
Navigate	Event Monitor	Access to the Event Monitor link on the Contents tree.

CHAPTER

3 SPLIT SERVER INSTALLATION

The Split Server Topology installs the required Asset Performance Management components of multiple servers, instead of one.

ATTENTION

- If there is an installation failure, you need to perform password encryption before resuming the installation, for more information refer to Password Encryption .
- Ensure you clear the browser cache after installation for the UX changes to reflect.

ATTENTION

If the system prompts for a reboot during installation, click **No** and proceed with Installation.

If you want to proceed with the reboot, click **Yes**, and perform the following steps

1. Mount the media once the reboot is completed.
2. Perform password encryption, for more information refer to Password Encryption .
3. Restart the installation.

   The installation will resume from the point before reboot.

We recommend the following split server deployments:

- 2 Server: the database is installed on one server, and the Asset Performance Management application, web, and calculations components are on a second server.
- 3 Server: the database is installed on one server, application, and calculation components on the next, and web components on the third server.
- 4 Server: the database, application, web, and calculation components are installed on individual servers.

NOTE

If you are installing Asset Performance Management on systems with existing Honeywell software installed in a split server arrangement with the database, application, and web components installed on separate servers, we recommend installing Asset Performance Management with the 4-server deployment.

If the recommended deployments cannot be used, please contact Honeywell for options to meet the requirements of your site.

See Split Server Topology for more information on the split server arrangements.

Install the Asset Performance Management servers in the following order:

1. Database Server (On the Database Server)
2. Application Server (On the Application Server)
3. Web Server (On the Web Server)
4. Calculation Servers (On the Calculation Server)

> NOTE
> When entering a server name, use the computer name, not the Fully Qualified Domain Name (FQDN), or IP address for all servers except the DAS Server, unless otherwise mentioned. (The DAS server requires the use of the FQDN.)

> NOTE
> This chapter describes a 4-server split installation. If you are installing in a 3-server or 2-server deployment, some steps will differ from those described.

3.1 Split Server Pre-install

Ensure the following pre-requisite conditions are met during the installation of Asset Performance Management:

1. Ensure all Windows updates are completed and the system is restarted before starting the installation.
2. Disable automatic Windows updates in the system during the installation of Asset Performance Management.
3. Configuring SPNs on Multi-server is completed.
4. Ensure Powershell Scripts Execution policy is enabled.
5. Powershell Scripts are unblocked.

> ATTENTION
> 1. If the system restarts during Asset Performance Management installation, ensure the ISO is mounted on the same drive before proceeding further with the installation.
> 2. Do not click Next twice in the installation dialog manager window as this will lead to skipping of successive inputs screens.
> 3. Installation of RabbitMQ in a custom location is not possible. By default, RabbitMQ will be installed in the C drive.

3.1.1 Install all Microsoft Updates

On all servers:

1. Run Microsoft Updates to ensure all current updates are installed.
2. Restart the server.

3.1.2 Configuring SPNs on Multi-server

To ensure the multi-server functions as expected, configure the SPNs on the **Application**, **Web**, and **Calculation** Servers.

> **ATTENTION**
> To complete this procedure, the user running the command must be a member of **Domain Admins** or **Enterprise Admins** in an active directory, or have the authority to modify SPNs through the **Validated write to service principal name** permission.

1. Type the following in the Command Prompt window, after replacing **ServerName** with the actual name of your application server and **domain\ServicesUser** with the user running the Asset Performance Management Services:

   ```
   setspn -u -s http/ServerName.fully-qualified-domain-name domain\ServicesUser
   ```

2. If using a non-default port, also execute the following command, after replacing **ServerName** with the actual name of your application server, **PortNo** with the port number, and **domain\ServicesUser** with the user running the Asset Performance Management Services:

   ```
   setspn -u -s http/ServerName.fully-qualified-domain-name:PortNo
   domain\ServicesUser
   ```

   > **NOTE**
   > After executing the above commands, if the duplicate SPNs are displayed, delete the duplicate SPNs and run the command again. (Run the command **setspn -l domain\ServicesUser** to check for duplicates).

3. Type the following in the Command Prompt window, after replacing **ServerName** with the actual name of your application or web server and **domain\ServicesUser** with the user running the Asset Performance Management Services:

   ```
   setspn -u -s http/ServerName domain\ServicesUser
   ```

4. If using a non-default port, also execute the following command, after replacing **ServerName** with the actual name of your application server, **PortNo** with the port number, and **domain\ServicesUser** with the user running the Asset Performance Management Services:

   ```
   setspn -u -s http/ServerName:PortNo domain\ServicesUser
   ```

3.2 On the Database Server

This section contains the procedure for installing the Asset Performance Management Database Server.

> **NOTE**
> If you are using an existing installation of SQL Server, and are not installing on a server with other Honeywell applications installed (for example, Uniformance Insight), you do not need to run the HAS install on the Database Server and can skip ahead to installing the Application Server (On the Application Server).

To install SQL Server or to use a server with other Honeywell products (and therefore the Database Server role) already installed as the HAS Database server, complete all of the following.

3.2.1 Install All Microsoft Updates

On all servers:

1. Run Microsoft Updates to ensure all current updates are installed.
2. Restart the server.

3.2.2 Check Services User Permissions

Follow the below steps in the **Database Server** before you begin the installation:

- Ensure you have a **Services User** that does not require login permissions to the Asset Performance Management computer.

> **ATTENTION**
> Do not use the installation (current) user as the Services User. If the selected Services User has logon permissions, they will be removed as a part of the Asset Performance Management installation.
>
> Once the installation is complete, the Services User will no longer be able to log on to the computer locally or remotely.

> **NOTE**
> When installing SQL Server, if **NT SERVICE\ALL SERVICES** no longer has **LogOnAsAService** permissions, the installation cannot complete as expected. To ensure the SQL Server installation succeeds, **NT SERVICE\ALL SERVICES** should have **LogOnAsAService** permissions in either the local or domain group policy.

3.2.3 Stop Services for Honeywell Products

If you are installing or upgrading a system with other Honeywell products installed, before beginning the Asset Performance Management installation or upgrade, stop all services related to the installed Honeywell products.

For example, if Honeywell Uniformance KPI is installed on the system, stop the following services:

- Honeywell Intuition Calculation Service
- Honeywell Intuition Event Processing Service
- Honeywell Uniformance KPI Processing Service.

Once the services are stopped, perform an **IISRESET**.

3.2.4 Install the Asset Performance Management Database Server

> **NOTE**
> This chapter describes a 4-server split installation. If you are installing in a 3-server or 2-server deployment, some steps will differ from those described.

> **NOTE**
> When entering a server name, use the computer name, not the **Fully Qualified Domain Name** (FQDN), or IP address for all servers except the DAS Server, unless otherwise mentioned. (The DAS server requires the use of the FQDN.)
>
> We recommend you use **Web Server Fully Qualified Domain Name**.

1. Perform an **IISRESET**.
2. On the installation media of the new database server, right-click **setup.exe**, and then select **Run as Administrator**.
3. In the RabbitMQ prompt, click **No**.
4. On the Welcome page, click **Next**.

5. Read through the license agreement and click **I accept the terms in the license agreement** if you agree to the license terms and then click **Next**.
6. Do all of the following:
    a. Ensure the following is selected:
        - **Database components**

    b. Clear the following:
        - **Web components**
        - **Application components**
        - **Calculation Components**

    c. (Optional) Select the following:
        - **Reports**: Configure the SQL Server Reporting Service for HAS Reports.

    d. Select the **Asset Analytics** check box to install Asset Analytics and select **Create InfluxDB Instance**.

    > ATTENTION
    > Asset Analytics cannot be enabled in Asset Performance Management unless the checkbox is selected during the initial installation. Separate installation of asset analytics features is not possible.

    > ATTENTION
    > By selecting the Asset Analytics feature along with Database components, the installer will create InfluxDB user with Runtime account credentials provided during Asset Performance Management installation. InfluxDB user name will not be prefixed with the domain name.

    e. (Optional) To install to a non-default location, in the **Installation Path** area, click **Browse** and then select the new location.
    f. (Optional) To change the MES Data Files location, in the **Data Files Path** area, click **Browse** and then select the location in which to save **MES Data Files**.
    g. Click **Next**.

7. Do the following:
    a. (Optional) If you do not want to install the EPHD, clear the following:
        - **Embedded PHD**: Used when a full PHD server is not available.

        > NOTE
        > Use of EPHD with Asset Performance Management requires familiarity with PHD.EPHD does not support SQL Clustering and non-default SQL collation.

    b. Click **Next**.

8. Do one of the following:

- To install SQL Server along with Asset Performance Management, select **Install SQL**.
- To use an existing SQL server, in the SQL Server Name box, type or select the name of the SQL server instance (including instance name if using a named instance) used by Asset Performance Management.

9. Do all of the following:
   a. In the **Login Information for Application Services** area, in the **Runtime Domain\User Name** box, type the user name for the Asset Performance Management services user, including the domain name.

      In the Runtime Account Password box, type the password for the Asset Performance Management services user.

      > **NOTE**
      > Ensure that the below unsupported characters are not entered in the password:
      > - **$** Dollar sign
      > - space
      > - **"** Quote
      > - **'** Single quote
      > - ` Back quote.

   b. (Optional) If you are using an existing EPHD installation, in the **EPHD Server Details** area, in the **EPHD Server Name** box, type the database server name.
   c. Enable the SQL cluster check box only if u have an SQL cluster. If SQL cluster is enabled, enter the **Domain User Group Name**.

      The domain user group name should be in the **domain\UserGroupName** format.

      For example, l4-dc\ UASServiceGroup.
   d. In the **Service Account Permissions** area, select I accept the service account must be denied the ability to log on to the desktop to continue.

      > **NOTE**
      > The installation cannot continue if the service account has login permissions to the desktop. This check box should be selected for the installation to succeed.
      >
      > Once this change is made, the service account can no longer log on to the computer (locally or remotely) and cannot be used to run programs that interact with the desktop, such as Internet Explorer.

   e. Click **Next**.
10. Click **Install** to start the installation.
11. If you have selected **Install SQL**, when the installer reaches the Microsoft SQL Server step, a prompt to insert the SQL Server installation media appears. Do all of the following:

a. Put the provided SQL Server installation media into the listed location/drive.

> **NOTE**
> When you are mounting the installation media, ensure that the media is mounted to the same drive after the server restart.

b. Click **Yes**.

   The Microsoft SQL Server installation begins.

c. When the SQL Server installation is complete, a prompt appears to reboot the computer.

d. If it was removed, remount the Asset Performance Management installation media to the installation location/drive.

e. Click **Yes** to reboot.

f. Click **Yes**.

12. A message appears prompting you to restart the computer. Click **OK** to restart.

> **NOTE**
> Ensure all other applications or folders are closed before confirming the system restart, as the restart will happen even if other applications have unsaved changes.

13. Once the restart is complete, ensure the Asset Performance Management services user is given the following permissions to the **\[InstallDrive\]:\ProgramData\Microsoft\Crypto\ RSA\MachineKeys** folder on the Primary Server:
    - Read & Execute
    - List Folder contents
    - Read

3.2.5 Update the Database Index

1. Open SQL Server Management Studio: **Start > All Programs > Microsoft SQL Server [Version Number] > SQL Server Management Studio**

   The **Microsoft SQL Server Management Studio** window appears.
2. Enter the details for the database instance to be used by Asset Performance Management, and then click **Connect**.
3. Run the following SQL script:

```
<Install
Location>:\Installer\Components\UAS\Installers\AssetCX\Scripts\BulkConfig_All_
ViewIndexs.sql
```

3.2.6 Complete Reports Installation

1. Navigate to the following location:

   <Install Location>:\Program Files (x86)\Honeywell\MES\AssetManager\Tools\AMReports
2. Copy the **AMLocalizedAssembly.dll** file.
3. Paste it to the following location:

   \Program Files\Microsoft SQL Server Reporting Services\SSRS\ReportServer\bin
4. Navigate to the below path and open **RS Scripter Load All Items.cmd** in notepad.

5. Update **SET REPORTSERVER** is report server fully qualified domain name and **SET RC** value with the file path to **RS.exe** in the install location.

   Example:

```
::Script Variables
SET LOGFILE="RS Scripter Load Log.txt"
SET SCRIPTLOCATION=
SET BACKUPLOCATION=
SET REPORTSERVER=https://REPORTINGSERVERFQDN:443/ReportServer
SET RS="C"\Program Files\Microsoft SQL Server Reporting Services\Shared Tools\RS.exe"
SET TIMEOUT=60
```

6. Run the following file:

   `<Install Location>:\Program Files (x86)\Honeywell\MES\AssetManager\Tools\AMReports\PublishReports\RS Scripter Load All Items.cmd`

3.2.6.1 Application Key Configuration for Reports

By default, HAS application is configured as **http**report configuration.

For HTTPS installation, and report configuration to work from the HAS application, you should modify the **Honeywell.UAS.reportsURL** key in the **System**> **System Configuration**> **Application**as shown below.

- Double-click the **Value**fieldto update the value for https configuration:

> **NOTE**
> You can change http to https, and provide the DBServer name along with the port number.
>
> example: https:// DBServer FQDN name:Port number/Reports/

3.2.7 Configure the Reports Data Sources

1. Open the **SQL Server Reporting Services** page:

   http://[ServerName]/reports/

   where **[ServerName]** is the name or IP address of the server on which Reporting Services is installed.
2. On the reporting services **Home** page, in the toolbar, click **Tiles**, and then select **Show hidden items**.
3. In the folders list, select **Data Sources**.
4. Select **Assets**.
5. On the **Properties** page, do all of the following:
   a. Scroll to the **Credentials** section.
   b. Ensure **Using the following credentials** are selected.
   c. In the **User name** box, type the name installation user.
   d. In the **Password** box, type the password of the installation user.
   e. Click **Apply**.

3.2.8 Add Group or User Access to SQL Reports

1. Open the **SQL Server Reporting Services** page:

   http://[ServerName]/reports/

   where [ServerName] is the name or IP address of the server on which **Reporting Services** is installed.
2. On the reporting services **Home** page, click **Manage Folder**.
3. Click **Add group or user**.
4. Do all of the following:
   a. In the **Group or user name** box, type the name of the group or user who requires access to view the reports.

      For example, we recommend adding the default Reliability Engineer, Maintenance Engineer, and Maintenance Planner groups.

   b. In the **Role** list, select **Browser**.
   c. Click **OK**.

      The user or group appears on the **Security** page for the home folder.

3.2.9 Refresh EPHD

After installing Asset Performance Management and updating the EPHD settings, the EPHD system is left in an Initializing state. To bring the state back to active, run the following commands:

```
phdctl stop /y

phdctl stop /y dbsecurity

phdctl stop /y udbserver

phdctl start cold
```

3.3 On the Application Server

This section contains the procedure for installing the Asset Performance Management Application Server.

> **NOTE**
> This chapter describes a 4-server split installation. If you are installing in a 3-server or 2-server deployment, some steps will differ from those described.

> **NOTE**
> - Ensure you do not install the Asset Performance Management server on the FDM server node.
> - Asset Performance Management installs the Data Access Service (DAS) and creates the default instance and connections.

> **NOTE**
> During the installation, TLS 1.2 is enabled on the server.

3.3.1 Install All Microsoft Updates

On all servers:

1. Run Microsoft Updates to ensure all current updates are installed.
2. Restart the server.

3.3.2 Check Services User Permissions

Follow the below steps in the **Database Server** before you begin the installation:

- Ensure you have a **Services User** that does not require login permissions to the Asset Performance Management computer.

> ATTENTION
> Do not use the installation (current) user as the Services User. If the selected Services User has logon permissions, they will be removed as a part of the Asset Performance Management installation.
>
> Once the installation is complete, the Services User will no longer be able to log on to the computer locally or remotely.

3.3.3 Stop Services for Honeywell Products

If you are installing or upgrading a system with other Honeywell products installed, before beginning the Asset Performance Management installation or upgrade, stop all services related to the installed Honeywell products.

For example, if Honeywell Uniformance KPI is installed on the system, stop the following services:

- Honeywell Intuition Calculation Service
- Honeywell Intuition Event Processing Service
- Honeywell Uniformance KPI Processing Service.

Once the services are stopped, perform an **IISRESET**.

3.3.4 Install Microsoft SQL Server 2012 Native Client

1. On the installation media, Navigate to the following folder:
   installer/prerequisites/sqlncli.msi
2. Right-click **sqlncli.msi** and click **install**.
3. Follow the on-screen instructions to complete the installation.

3.3.5 Install the Asset Performance Management Application Server

> NOTE
> This chapter describes a 4-server split installation. If you are installing in a 3-server or 2-server deployment, some steps will differ from those described.

> NOTE
> When entering a server name, use the computer name, not the **Fully Qualified Domain Name (FQDN)**, or IP address for all servers except the DAS Server, unless otherwise mentioned. (The DAS server requires the use of the FQDN.)

> **NOTE**
> Rabbit MQ should be installed only in **Calculation Server**. If you are installing Appserver and Calcserver in the same box, Rabbit MQ should also be installed in the same box.

1. Perform an **IISRESET**.
2. On the installation media of the new database server, right-click **setup.exe**, and then select **Run as Administrator**.
3. In the RabbitMQ prompt, click **No**.
4. On the Welcome page, click **Next**.
5. Read through the license agreement and click **I accept the terms in the license agreement** if you agree to the license terms and then click **Next**.
6. Do all of the following:
   a. Ensure the following is selected:
      - **Application components**

   b. Select the following as applicable:
      - If you are installing the Client Tools: **Clientcomponents**: the APM Configuration Excel Plugin, Model Tools, and/or PHD Utility.

      > **NOTE**
      > If you need to import models after installation, you should select **Client Components**.

   c. Click **Next**.
7. Select **Asset Analytic** if you want the asset analytics installed.
8. click **Next**.
9. Do all of the following:
   a. Clear the following if you do not require it on the server:
      - **Embedded PHD**: Used when a full PHD server is not available.

   b. (Optional) To install to a non-default location, in the **Installation Path** area, click **Browse** and then select the new location.
   c. (Optional) To change the MES Data Files location, in the **Data Files Path** area, click **Browse** and then select the location in which to save MES Data Files.
   d. Click **Next**.
10. In the **Intuition Environment Setup,** correct any issues, and then click **Exit**. (If the Exit button does not appear with the success message, close the browser window to continue.)

    > **ATTENTION**
    > You must select the **Application Components** checkbox and click **Configure** button in the **Certificate Configuration** tab even if the system passed all the validation and is ready for installation.

    For example, if Certificate Configuration is flagged, do all of the following:

a. Beside **Certificate Configuration**, click **Repair**.
b. Click **Configure**.
c. In the **Success** dialog box, click **OK**.
d. Click the **Home** tab and click **Repair all**.

11. Do all of the following:
    a. In the **SQL Server Name** box, type or select the name of the SQL server instance (including instance name if using a named instance) used by Asset Performance Management.
    b. In the **Login Information for Application Services** area, in the **Runtime Domain\User Name** box, type the user name for the Asset Performance Management services user, including the domain name.
    c. In the **Runtime Account Password** box, type the password for the Asset Performance Management services user.

> **NOTE**
> Ensure that the below unsupported characters are not entered in the password.
> - $ Dollar sign
> - space
> - " Quote
> - ' Single quote
> - ` Back quote.

    d. (Optional) If you are using an existing EPHD installation, in the **EPHD Server Details** area, in the **EPHD Server Name** box, type the database server name.
    e. Enable the SQL cluster check box only if u have an SQL cluster.
    If SQL cluster is enabled, enter the **Domain User Group Name**.
    The domain user group name should be in the **domain\UserGroupName** format. For example, l4-dc\UASServiceGroup

> **NOTE**
> UserGroupName is not a Windows Group.

    f. In the Service Account Permissions area, select **I accept the service account** should deny the ability to log on to the desktop to continue

> **NOTE**
> The installation cannot continue if the service account has login permissions to the desktop. This check box must be selected for the installation to succeed.
> Once this change is made, the service account can no longer log on to the computer (locally or remotely) and cannot be used to run programs that interact with the desktop, such as Internet Explorer.

    g. Click **Next**.

12. Do all of the following:
    a. In the **Website selection** box, select the website you are using for Asset Performance Management.
    b. In the **net.tcp Binding Information** box type the net.tcp port number.
    c. If the server uses HTTP, clear the **Enable HTTPs based secure installation**. If it uses HTTPS, leave this box selected.
    d. In the **Calc Server Name** box, type the name of the main Calculation Server.
    e. Click **Next**.

13. Do all of the following:
    a. In the **Web Server Host Name** box, type the name of the Web Server.

    > **ATTENTION**
    > If you are using https, enter the fully qualified domain name of the Web Server.

    b. In the **Web Server HTTP Port No** box, type the port number used on the Web Server.
    c. In the **Web Server Server Protocol Name** box, select the protocol (**http** or **https**) used on the Web Server.
    d. Click **Next**.

14. Do all of the following:
    a. In the **Sender's Email ID**, type the email address that will appear in the **From** field of emails sent by the system.
    b. In the **Default Email ID**, type the email address that will receive a notification if an email delivery fails.
    c. In the **SMTP Server Name** box, type the name of the SMTP server.
    d. You can also enable SSL, by clicking the **Enable SSL** check box and providing the following information
        - Server Name of SMTP server.
        - Port Number of the SMTP server.
        - Secure Socket Option - Enter socket option.

        For more details refer to Configuring Secure Socket Option

15. Click **Install** to start the installation.
16. If you selected to install the Asset Performance Management Configuration Excel Plugin, when the APM Excel Add-in Setup page appears, click **Accept**.
17. Click **Finish**.
18. A message appears prompting you to restart the computer. Click **OK** to restart.

> **ATTENTION**
> Ensure all other applications or folders are closed before confirming the system restart, as the restart will happen even if other applications have unsaved changes.

19. Once the restart is complete, ensure the Asset Performance Management services user is given the following permissions to the **<InstallDrive>:\ProgramData\Microsoft\Crypto\RSA\MachineKeys** folder on the Primary Server:

- Read & Execute
- List Folder contents
- Read

3.3.6 Add Users to the Reliability Engineers Group

Once the Application Server is installed, you need to ensure the services and install users are added to the Windows Reliability Engineers Group.

1. **Start** > **Server Manager**.
2. On the server tree, choose **Configuration** > **Local Users and Groups** > **Groups**.
3. Right-click the **Reliability Engineers** group, and click **Add to Group**.
4. In the **Reliability Engineers Properties** dialog box, click **Add**.
5. Enter the names of all of the following users:
    a. The services (runtime) user.

    > **NOTE**
    > If the services user is not added to the Reliability Engineers group, some features will not function as expected.

    b. The install user.
    c. All the users who will need to use the APM Configuration Excel Plugin and the Asset Performance Management Configuration.
6. Click **OK**.

    The user names appear in the group list.
7. Click **OK** to close the Reliability Engineers Properties dialog box.

3.3.7 Configure the Anti-Virus Software

If you have McAfee anti-virus software on the server, it must be configured to not block the sending of bulk emails.

1. **Start** > **All Apps** > **VirusScan Console**

    The VirusScan Console window appears.
2. From the Task list, double-click **Access Protection**.

    The Access Protection Properties dialog box appears.
3. Clear the **Prevent Mass Mailing Worms from Sending Mail rule** check box.
4. Click **OK**.

3.3.8 Import the Default Asset Templates

(Optional) Follow the below steps to install the standard model library:

1. Open the following location on the installation media:

    [Installation Media]:\StandardLibrary\Model Library
2. Copy the **AMExport_XML** file.
3. Navigate to the folder in the system with the Model Tools installed, and paste the **AMExport_XML** file:

    [InstallDrive]:\Program Files (x86)\Honeywell\MES\AssetManager\Tools\AMImportExport

4. Run the **AMImportModel** file.
5. Click the **Source** drop-down list and select **Asset Manager Export file**.

   The Asset Template information is displayed in the **Available Models for Import** section.
6. Select the **Select All** check box to import the Asset Template information.

   OR

   From the **Select** column, select the check box beside the asset type that you want to import asset template information.
7. Click **Copy**.

   The selected **Asset Template** information is saved in the Asset Performance Management database.
8. Restart the **AMApplicationServer** service.
9. The imported asset template information is available on the Asset Performance Management user interface after you restart the service.

3.4 On the Web Server

This section contains the procedure for installing the Asset Performance Management Web Server.

> **NOTE**
> This chapter describes a 4-server split installation. If you are installing in a 3-server or 2-server deployment, some steps will differ from those described.

> **NOTE**
> - Ensure you do not install the Asset Performance Management server on the FDM server node.
> - During the installation, TLS 1.2 is enabled on the server.

3.4.1 Install All Microsoft Updates

On all servers:

1. Run Microsoft Updates to ensure all current updates are installed.
2. Restart the server.

3.4.2 Check Services User Permissions

Follow the below steps in the **Database Server** before you begin the installation:

- Ensure you have a **Services User** that does not require login permissions to the Asset Performance Management computer.

> **ATTENTION**
> Do not use the installation (current) user as the Services User. If the selected Services User has logon permissions, they will be removed as a part of the Asset Performance Management installation.
>
> Once the installation is complete, the Services User will no longer be able to log on to the computer locally or remotely.

3.4.3 Stop Services for Honeywell Products

If you are installing or upgrading a system with other Honeywell products installed, before beginning the Asset Performance Management installation or upgrade, stop all services related to the installed Honeywell products.

For example, if Honeywell Uniformance KPI is installed on the system, stop the following services:

- Honeywell Intuition Calculation Service
- Honeywell Intuition Event Processing Service
- Honeywell Uniformance KPI Processing Service.

Once the services are stopped, perform an **IISRESET**.

3.4.4 Prerequisites on Windows 2017/2019

If you are installing Windows 2017/2019, you should install the ARR files.

1. Copy the following folder to a location on the Web Server:

   <installation media>\Installer\Prerequisites\ARR 3.0\Win2016\ARR_TargetFolder
2. On the Web Server, run the **WebPlatformInstaller_amd64_en-US.msi** file found in the following folder:

   <installation media>\Installer\Prerequisites\ARR 3.0\Win2016
3. When the installation is complete, run the following command:
   ```
   C:\Program Files\Microsoft\Web Platform Installer\WebPICMD.exe" /Install
   /Products:"ARRv3_0"/XML:"[Copied_ARR_
   TargetFolder]\feeds\latest\webproductlist.xml
   ```
4. Restart **IIS**.

3.4.5 Installing the Asset Performance Management Web Server

> **NOTE**
> This chapter describes a 4-server split installation. If you are installing in a 3-server or 2-server deployment, some steps will differ from those described.

> **NOTE**
> When entering a server name, use the computer name, not the **Fully Qualified Domain Name (FQDN)**, or IP address for all servers except the DAS Server, unless otherwise mentioned. (The DAS server requires the use of the FQDN.)

1. Perform an **IISRESET**.
2. On the installation media of the new database server, right-click **setup.exe**, and then select **Run as Administrator**.
3. In the RabbitMQ prompt, click **No**.
4. On the Welcome page, click **Next**.
5. Read through the license agreement and click **I accept the terms in the license agreement** if you agree to the license terms and then click **Next**.
6. Do all of the following:

a. Ensure the following is selected:
   - **Web components**

b. Clear the following:
   - **Application components**
   - **Database components**
   - **Calculation Components**

c. If you are installing the Client Tools, the following:
   - **Clientcomponents**: the APM Configuration Excel Plugin, Model Tools, and/or PHD Utility.

d. Click **Next**.

7. Select **Asset Analytics** if you want the asset analytics installed and click **Next**.
8. Do all of the following:

   a. (Optional) If you have selected the Client components, but only want to install a subset, clear any of the following that do not apply:
      - **Bulk Configuration**: The APM Configuration Excel Plugin.
      - **Model Tools**: The APM Configuration Tool and Model Tool.

      > NOTE
      > If you need to import models after installation, Model Tools must be selected.

      - **PHD Utility**: The PHD components.

   b. (Optional) To install to a non-default location, in the **Installation Path** area, click **Browse** and then select the new location.
   c. (Optional) To change the MES Data Files location, in the **Data Files Path** area, click **Browse** and then select the location in which to save MES Data Files.
   d. Click **Next**.

9. In the **Intuition Environment Setup,** correct any issues, and then click **Exit**. (If the Exit button does not appear with the success message, close the browser window to continue.)

   > ATTENTION
   > You must select the **WebComponents** checkbox, enter the **Application Server** name and then click **Configure** button in the **Certificate Configuration** tab even if the system passed all the validation and is ready for installation.

   For example, if Certificate Configuration is flagged, do all of the following:

   a. Beside **Certificate Configuration**, click **Repair**.
   b. Select **Web Components** and enter the Application server name.
   c. Click **Configure**.
   d. In the **Success** dialog box, click **OK**.
   e. Click the **Home** tab and click **Repair all**.

10. Under the **CMMS Alert** section, the path where the CMMS XML files are saved is displayed by default. If you want to change the default path, perform the following steps:
    - Click **Browse**.
    - The **Browse for Folder** page appears.
    - Select the path where you want save the CMMS XML files.
    - Click **OK**.
11. In the **Email Alert** area, in the **Server Name** box, type the SMTP server name, which you made a note of in the Before you Begin section.
12. Click **Next**.
13. Do all of the following:
    a. In the **SQL Server Name** box, type or select the name of the SQL server instance (including instance name if using a named instance) used by Asset Performance Management.
    b. In the **Login Information for Application Services** area, in the **Runtime Domain\User Name** box, type the user name for the Asset Performance Management services user, including the domain name.
    c. In the **Runtime Account Password** box, type the password for the Asset Performance Management services user.

> **NOTE**
> Ensure that the below unsupported characters are not entered in the password.
> - $ Dollar sign
> - space
> - " Quote
> - ' Single quote
> - ` Back quote.

    d. (Optional) If you are using an existing EPHD installation, in the **EPHD Server Details** area, in the **EPHD Server Name** box, type the database server name.
    e. Enable the SQL cluster check box only if u have an SQL cluster.
    If SQL cluster is enabled, Enter the Domain User Group Name. The domain user group name should be in the domain\UserGroupName format. For example, I4-dc\ UASServiceGroup

> **NOTE**
> UserGroupName is not a Windows Group.

    f. In the Service Account Permissions area, select **I accept the service account must be denied the ability to log on to the desktop to continue**

> NOTE
> The installation cannot continue if the service account has login permissions to the desktop. This check box must be selected for the installation to succeed.
>
> Once this change is made, the service account can no longer log on to the computer (locally or remotely) and cannot be used to run programs that interact with the desktop, such as Internet Explorer.

g. Click **Next**.

14. Do all of the following:
    a. In the **Website selection** box, select the website you are using for Asset Performance Management.
    b. In the **net.tcp Binding Information** box type the net.tcp port number.
    c. If the server uses HTTP, clear the **Enable HTTPs based secure installation**. If it uses HTTPS, leave this box selected.
    d. In the **Calculation Server Name** box, type the name of the main calculation server.
    e. Click **Next**.

15. Do all of the following:
    a. In the **Application Server Host Name**, type the name of the Application Server.
    b. In the **Application Server Port Number**, type the port number used on the Application Server.
    c. In the **Application Server Protocol Name**, select the protocol (http or https) used on the Application Server.
    d. In the **Application Server NET.TCP Port Number**, type the net.tcp port number used on the Application Server.
    e. Click **Next**.

16. Click **Install** to start the installation.
17. A message appears prompting you to restart the computer. Click **OK** to restart.

> ATTENTION
> Ensure all other applications or folders are closed before confirming the system restart, as the restart will happen even if other applications have unsaved changes.

18. Once the restart is complete, ensure the Asset Performance Management services user is given the following permissions to the **<InstallDrive>:\ProgramData\Microsoft\Crypto\ RSA\MachineKeys** folder on the Primary Server:
    - Read & Execute
    - List Folder contents
    - Read

3.4.6 FDM Web API Component

If you are connecting to FDM, after the installation is complete, you need to install the FDM Web API Component to ensure asset links to FDM device details open as expected.

3.4.7 Access the Asset Links Folder

1. Navigate to the following folder:

   <InstallDrive>:\Program Files (x86)\Honeywell\MES\Core\WWWMESCore\AMInformationFolders
2. Right-click **AMInformationFolders** and select **Properties**.
3. Click the **Security** tab.
4. Click **Edit**.
5. Click **Add**.
6. Enter the Asset Performance Management services user, and click **OK**.
7. Select the user from the list, and in the **Allow** column select **Read** and **Write**.
8. Click **OK**.

3.4.8 Configuring for SSRS Reports

For the Reports to open as expected with a multi-server deployment, the configuration UI settings file (app.xml) on the Web Server must be updated to use the URL of the Database Server.

Do all of the following on the Web Server:

1. Open the **app.xml** configuration file for editing. By default, this is found in the following location:

   [install drive]:\Program Files (x86)\Honeywell\MES\Core\WWWMESCore\ClientBin
2. Locate the **ReportsURL** key and update the value to use the name of the Database Server, as shown:

   <Add Key="ReportsURL" Value="http://[DBServerName:PortNo]/Reports" />

   Replace **DBServerName** with the name of the Database Server and **PortNo** with the Database Server Port Number.
3. Save and close the file.

3.4.9 (Optional) Enabling Historical Data Source in Cam Page

1. Navigate to the below path and **Web.config** file:

   <InstallDir>\Honeywell\MES\Core\WWWMESCore\WebUX\CAM\ Web.config
2. Locate the appsetting **HistorianDataSourceTypeList**
3. Add the value as **UAS-Historian**
4. Perform an **IISReset.**

3.5 On the Calculation Server

This section contains the procedure for installing the Asset Performance Management Calculation Server.

> NOTE
> - Ensure you do not install the Asset Performance Management server on the FDM server node.
> - Ensure the Asset Performance Management Services user has access to the Intuition Core installation folder on the Calculation Server. This can be accomplished by adding the services user to either the Users (recommended) or Administrators Windows groups.

> NOTE
> During the installation, TLS 1.2 is enabled on the server.

3.5.1 Install All Microsoft Updates

On all servers:

1. Run Microsoft Updates to ensure all current updates are installed.
2. Restart the server.

3.5.2 Check Services User Permissions

Follow the below steps in the **Database Server** before you begin the installation:

- Ensure you have a **Services User** that does not require login permissions to the Asset Performance Management computer.

> **ATTENTION**
> Do not use the installation (current) user as the Services User. If the selected Services User has logon permissions, they will be removed as a part of the Asset Performance Management installation.
>
> Once the installation is complete, the Services User will no longer be able to log on to the computer locally or remotely.

3.5.3 Stop Services for Honeywell Products

If you are installing or upgrading a system with other Honeywell products installed, before beginning the Asset Performance Management installation or upgrade, stop all services related to the installed Honeywell products.

For example, if Honeywell Uniformance KPI is installed on the system, stop the following services:

- Honeywell Intuition Calculation Service
- Honeywell Intuition Event Processing Service
- Honeywell Uniformance KPI Processing Service.

Once the services are stopped, perform an **IISRESET**.

3.5.4 Installing the Asset Performance Management Calculation Server

> **NOTE**
> When entering a server name, use the computer name, not the Fully Qualified Domain Name (FQDN), or IP address for all servers except the DAS Server, unless otherwise mentioned. (The DAS server requires use of the FQDN.)

1. Perform an **IISRESET**.
2. On the installation media of the new database server, right-click **setup.exe**, and then select **Run as Administrator**.
3. If RabbitMQ is not already installed, a prompt appears. In the RabbitMQ prompt, click **Yes**.
4. On the Welcome page, click **Next**.
5. Read through the license agreement and click **I accept the terms in the license agreement** if you agree to the license terms and then click **Next**.
6. Do all of the following:

a. Ensure the following is selected:
   - **Calculation Components**

b. Clear the following:
   - **Web components**
   - **Application components**
   - **Database components**
   - **Reports**: Configure the SQL Server Reporting Service for HAS Reports.

c. Click **Next**.

7. In the **Intuition Environment Setup,** correct any issues, and then click **Exit**. (If the Exit button does not appear with the success message, close the browser window to continue.)

> ATTENTION
> You must click **Configure** button in the **Certificate Configuration** tab even if the system passed all the validation and is ready for installation.

   For example, if Certificate Configuration is flagged, do all of the following:
   a. Beside **Certificate Configuration**, click **Repair**.
   b. Click **Configure**.
   c. In the **Success** dialog box, click **OK**.
   d. Click the **Home** tab and click **Repair all**.

8. Select **Asset Analytics** if you want the asset analytics installed and click **Next**.
9. Do all of the following:
   a. (Optional) If you do not want to install EPHD, clear the **Embedded PHD** check box.
   b. (Optional) To install to a non-default location, in the **Installation Path** area, click **Browse** and then select the new location.
   c. (Optional) To change the MES Data Files location, in the **Data Files Path** area, click **Browse** and then select the location in which to save MES Data Files.
   d. Click **Next**.

10. Do all of the following:
   a. In the **DAS Configuration details** area, select **Configure DAS**.
   b. In the **DAS Configuration** area, in the **Server Name** box, type the *fully qualified domain name* of the DAS server.
   c. In the **DAS Configuration** area, in the **Port Number** box, type the port number on which DAS is installed.
   d. Click **Next**.

11. Do all of the following:
   a. In the **SQL Server Name** box, type or select the name of the SQL server instance (including instance name if using a named instance) used by Asset Performance Management.
   b. In the **Login Information for Application Services** area, in the **Runtime Domain\User Name** box, type the user name for the Asset Performance Management services user, including the domain name.
   c. In the **Runtime Account Password** box, type the password for the Asset Performance Management

services user.

> **NOTE**
> Ensure that the below unsupported characters are not entered in the password:
> - $ Dollar sign
> - space
> - " Quote
> - ' Single quote
> - ` Back quote.

d. (Optional) If you are using an existing EPHD installation, in the **EPHD Server Details** area, in the **EPHD Server Name** box, type the database server name.

e. Enable the SQL cluster check box only if u have an SQL cluster.

   If SQL cluster is enabled, enter the **Domain User Group Name**.

   The domain user group name should be in the domain\UserGroupName format. For example: l4-dc\UASServiceGroup

> **NOTE**
> UserGroupName is not a Windows Group.

f. In the Service Account Permissions area, select **I accept the service account must be denied the ability to logon to the desktop** to continue.

> **NOTE**
> The installation cannot continue if the service account has login permissions to the desktop. This check box must be selected for the installation to succeed.
>
> Once this change is made, the service account can no longer log on to the computer (locally or remotely) and cannot be used to run programs that interact with the desktop, such as Internet Explorer.

g. Click **Next**.

12. Do all of the following:

    a. In the **Website selection** box, select the website you are using for Asset Performance Management.

    b. In the **net.tcp Binding Information** box type the net.tcp port number.

    c. If the server uses HTTP, clear the **Enable HTTPs based secure installation**. If it uses HTTPS, leave this box selected.

    d. Click **Next**.

13. Do all of the following:

a. In the **Web Server Host Name** box, type the name of the Web Server.

> **ATTENTION**
> If you are using https, enter the fully qualified domain name of the Web Server.

b. In the **Web Server HTTP Port No** box, type the port number used on the Web Server.
c. In the **Web Server Protocol Name** box, select the protocol (**http** or **https**) used on the Web Server.
d. Click **Next**.

14. Click **Install** to start the installation.
15. If you selected to install the APM Configuration Excel Plugin, when the APM Excel Add-in Setup page appears, click **Accept**.
16. Click **Finish**.
17. A message appears prompting you to restart the computer. Click **OK** to restart.

    Ensure all other applications or folders are closed before confirming the system restart, as the restart will happen even if other applications have unsaved changes.
18. Once the restart is complete, ensure the Asset Performance Management services user is given the following permissions to the **<InstallDrive>:\ProgramData\Microsoft\Crypto\RSA\MachineKeys** folder on the Primary Server:
    - Read & Execute
    - List Folder contents
    - Read
19. If **Analytics** is selected to install in the calculation server node (in case of split server scenario)

    a. Navigate to the following location:

    C:\Program Files (x86)\Honeywell\MES\AssetManager\CalcHost\PRE

    b. Open the **Settings.py**.
    c. Search for **HOST_NAME**.
    d. Update the fully qualified domain name (FQDN) of the application Server.

3.5.5 Add Users to the Reliability Engineers Group

Once the Application Server is installed, you need to ensure the services and install users are added to the Windows Reliability Engineers Group.

1. **Start** > **Server Manager**.
2. On the server tree, choose **Configuration** > **Local Users and Groups** > **Groups**.
3. Right-click the **Reliability Engineers** group, and click **Add to Group**.
4. In the **Reliability Engineers Properties** dialog box, click **Add**.
5. Enter the names of all of the following users:

a. The services (runtime) user.

> **NOTE**
> If the services user is not added to the Reliability Engineers group, some features will not function as expected.

b. The install user.

c. All the users who will need to use the APM Configuration Excel Plugin and the Asset Performance Management Configuration.

6. Click **OK**.

   The user names appear in the group list.

7. Click **OK** to close the Reliability Engineers Properties dialog box.

3.5.6 Ensure the EPHD Services are Running

If you have installed EPHD, ensure the following services are running under the Asset Performance Management services user on the Calculation Server (they should be started by default):

- Uniformance API Server
- Uniformance DataBase Server
- Uniformance DB Security Service
- Uniformance PHD Server
- Uniformance PHD Startup
- Uniformance RDI Server
- Uniformance Remote API Server

3.5.7 Refresh EPHD

After installing Asset Performance Management and updating the EPHD settings, the EPHD system is left in an Initializing state. To bring the state back to active, run the following commands:

```
phdctl stop /y

phdctl stop /y dbsecurity

phdctl stop /y udbserver

phdctl start cold
```

3.6 Split Server Post-Install

Once all servers have been installed complete the following post-installation tasks.

3.6.1 Certificate Exchange

The certificates on the split server computer must be configured. Perform the export and import procedures on the Application Server, Web Server, and all Calculation Servers.

To export the certificates

1. Open the Console Root (**Start** > **Search** > **mmc**).
2. Do all of the following to add the Certificates Snap-in:
   a. On the **File** menu, select **Add/Remove Snap-in**.
   b. Double-click **Certificates**.
   c. Select **Computer Account**, and then click **Next**.
   d. Click **Finish**.
   e. Click **OK**.
3. On the tree, expand **Certificates**, and select **IntuitionSign**.
4. In the certificate list, right-click the certificate, point to **All Tasks**, and then select **Export**.
5. Complete the steps in the Certificate Export Wizard, and save the certificate in a location that is accessible or can be transferred to the other servers.

To import the certificates

1. In the Console Root, on the tree, expand **Certificates**, and select **IntuitionVerify**.
2. In the **Actions** sidebar, click **More Actions**, point to **All Tasks**, and select **Import**.
3. Complete the Import Certificate Wizard for all of the following:
   - On the Application Server: add the certificates exported from the Web Server and Calculation Server.
   - On the Web Server: add the certificates exported from the Application Server and Calculation Server.
   - On the Calculation Server: add the certificates exported from the Application Server and Web Server.

> ATTENTION
> If any of the SSL certificates used for https binding has expired, the exchange mentioned above should be performed after you renew the SSL certificate and re-run the preinstall checker on all the nodes.
>
> For more information, refer to **Intuition Certificate Renewal** in Asset Performance Management Troubleshooting Guide.

3.6.2 Post installation for InfluxDB

Follow company policy and get a certificate from trusted authorities and then follow the below steps:

> NOTE
> The following steps are applicable in the database server for multi-server.

1. Ensure that you have ssl certificate in .pfx format.
2. Mount the ISO file.
3. Navigate to the following location:
   <Installation Media>\Utilities\ConfigureCertificates\
4. Use **Run as Administrator** to open PowerShell.
5. Run the following command:
   .\Configure-InfluxDBCertificate.ps1

6. The following inputs prompted by the script.
    a. Get the ssl certificate in .pfx format and provide the file location of ssl certificate.

    > **NOTE**
    > SSl certificate should be domain trusted certificate or CA signed certificate and this certification is required to enable HTTPS for InfluxDB.

    b. Enter the password of ssl certificate.

In case the InfluxDB is installed as **http** and **pRdGpInfluxSelection** key value is other than **New**, then the certificate configuration will not ask for a path. To resolve this issue, perform the steps below:

1. Navigate to the following configuration files folder:

   <Install Location>\Honeywell\Asset Performance Management Installer\Install\Configuration files\
2. Open **UI.xml**.
3. Update **PChkboxHttpsInfluxDB** key value to **true**.
4. From the **Configure-InfluxDBCertificate.ps1** file, copy **Influxdbselection -eq** key value to **pRdGpInfluxSelection**.
5. Save **UI.xml**.
6. Run **.\Configure-InfluxDBCertificate.ps1** as mentioned above.

> **ATTENTION**
> Once the InfluxDB certificate is installed, revert the **pRdGpInfluxSelection** key value to the original value in the **UI.xml** file.

Follow the below steps if Asset CX and Asset analytics is not working because of the influx certificate renewal or database migration from one machine to another even after installing the patch (R531.1-FF001) and migrating database from the 520.2 or any earlier version to 531:

1. Check whether the encryption key is available by running the below query in SQL Server.

   ```
   Select * from
   [Honeywell.MES.AssetTask.DataModel.AssetTaskDataModel].dbo.ExternalDataSourceEnc
   ryptionKey
   ```
2. If the key is unavailable, execute the below PowerShell script in administrator in the application server and then in the web server.

   <installation media>\Utilities\EncryptData\EncryptData.ps1
3. Enter the below information:
    a. Service account password
    b. Influxdb user (usually service account name without domain name)
    c. Password of influxdb user (usually, service account password)
4. Perform **IISRESET**.

3.6.3 Post install Step for EPHD with Named SQL Instance

1. In the calculation server, navigate to the below folder and open the AMhost application configuration file.

   <Install location>\ Programfilesx86\Honeywell\Assetmanager\appserver\bin\bin2

2. Ensure SQL_ PHDAPP and SQL_PHDCFG connection strings point to the database server without the instance name.
3. In the application server, navigate to the below folder and open **web.config.**

   <Install Location>\Programfiles x86\HOneywell\ Assetmanager\webserver\data

4. Ensure **SQL_PHDAPP** and **SQL_PHDCFG** connection strings point to the database server without the instance name.
5. In the calculation server and in the Uniformance PHD server using the PHD system console, ensure the database server name points to the database server without the instance name, and in the Uniformance Database Server, ensure that the database data source points to the database server name with the instance name.
6. Run the below commands in the command prompt as administrator:

```
phdctl stop /y
phdctl stop /y
dbsecurity phdctl stop /y
udbserver phdctl start cold
```

3.6.4 Creating an Index

Follow the below steps in the database server to create an index:

1. Navigate to the following folder:

   <Installed Location>:\Installer\Components\UAS\Installers\AssetCX\Scripts\BulkConfig_All_ViewIndexs.sql

2. Run **BulkConfig_All_ViewIndexs.sql** in **SQL Server Management Studio (SSMS)**.

CHAPTER

4 SINGLE SERVER INSTALLATION

NOTE

1. When entering a server name, use the computer name or IP address for all servers except the DAS Server. (The DAS server requires a fully qualified domain name.)
2. Asset Performance Management installs the Data Access Service (DAS) and creates the default instance and connections.

ATTENTION

1. If there is an installation failure, you need to perform password encryption before resuming the installation, for more information refer to Password Encryption.
2. Do not click **Next** twice in the installation dialog manager window as this will lead to skipping of successive inputs screens.
3. Installation of RabbitMQ in a custom location is not possible. By default, RabbitMQ will be installed in the C drive.
4. Ensure you clear the browser cache after installation for the UX changes to reflect.

ATTENTION

If the system prompts for a reboot during installation, click **No** and proceed with Installation.

If you want to proceed with the reboot, click **Yes**, and perform the following steps

1. Mount the media once the reboot is completed.
2. Perform password encryption, for more information refer to Password Encryption.
3. Restart the installation.

   The installation will resume from the point before reboot.

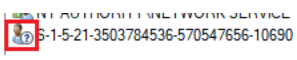
4.1 Single Server Pre-install

Ensure the following pre-requisite conditions are met during the installation of Asset Performance Management:

1. Ensure all Windows updates are completed and the system is restarted before starting the installation.
2. Disable automatic Windows updates in the system during the installation of the Asset Performance Management.
3. You must click **Configure** button in the Certificate Configuration tab even if the system passed all the validation and is ready for installation.

4.1.1 Verify you have the Administrator Privileges on the Server

1. On the Server computer desktop, right-click **Computer** and select **Manage**.
   The **Server Manager** screen appears.
2. On the left pane, expand **Configuration**.
3. Expand **Local Users and Groups**.
4. Click the **Groups** folder.
   The available user groups are displayed on the right pane.
5. Double-click **Administrators**.
   The **Administrator Properties** dialog box appears listing the users who are a part of the administrator user group.

> **NOTE**
> Ensure the list of users part of administrator user group do not contain any invalid entry as highlighted in the below image.
>
> 
>

4.1.2 Verify User Account Control is Set to Low

1. Open the **Control Panel**.
2. Click **System and Security**.
3. Click **Change User Account Control settings**.
4. Slide the setting down to **Never Notify**.
5. Click **OK**.

> **NOTE**
> This setting can be changed to a higher level once all installations are complete.

4.1.3 Verify SQL Server

> **NOTE**
> If SQL Failover Clustered instances are used, when SQL FCI performs a failover, there is a small downtime of the SQL instance. This can disrupt communication between the AM Service and SQL Server, requiring a restart of the AM Service. (This can be automated for minimal disruption.)

4.1.4 Verify the SQL Collation Value

> **NOTE**
> Default SQL collation is supported only in the prerequisite.

1. Open **SQL Server Management Studio**.
2. Enter the details for the database instance to be used by Asset Performance Management, and click **Connect**.
3. Right-click the database instance, and click **Properties**.
4. On the **General** page, ensure the **Server Collation** value is **SQL_Latin1_General_CP1_CI_ AS**.

4.1.5 Verify the SQL Server Service is Running Under "Local System" Account or Under an Account with Administrator Privileges

1. **Start** > **All Apps** > **Services**.
   The Services window appears.
2. From the list, right-click **SQL Server** and select **Properties**.
   The **SQL Server Properties** dialog box appears.
3. Click the **Log On** tab.
4. Click the **Local System** account option.
   (OR)
5. Click the **This account** option and enter the account with administrator privileges.
6. Click **Apply** and then click **OK**.
7. Restart the service.

4.1.6 Verify you have System Administrator Privileges for the Database on which the Asset Performance Management Database will Reside

1. **Start** > **All Apps** > **SQL Server Management Studio**.
   The **Microsoft SQL Server Management Studio** window appears.
2. On the Object Explorer pane, expand **Security > Logins**.
3. Right-click the user on which the setup is running and select **Properties**.
   The **Login Properties** dialog box appears.
4. On the left pane, click **Server Roles**.
   A list of server roles is displayed on the right pane. Verify if the check box beside the **sysadmin** role is selected.

> ATTENTION
> If you restart the Asset Performance Management server, SQL server agent service stops automatically. This is because the SQL server agent service is in **Manual** startup type by default.
>
> This should be changed to an **Automatic** startup type for the SQL server agent service to run automatically even after restarting the server.

Perform the following steps to change the startup type to **Automatic**:

1. **Start > All Apps > Services**
   The **Services** window appears.
2. From the list, right-click **SQL Server Agent** and select **Properties**.
   The **SQL Server Agent Properties** dialog box appears.
3. On the **General** tab, from the **Startup Type** drop-down list, select **Automatic**.
4. Click **OK** and close the services window.

> NOTE
> The SQL Server Agent is required for the following reasons:
>
> - To run the SQL job every 2 hours
> - To delete the records in the HistorisedProcessedAttributes table - If the attribute count exceeds 120 for each attribute, the latest 120 records are retained and the rest are deleted.
> - You must have knowledge of SQL and SQL jobs if you want to modify the number of records retained. We recommend not changing the number of records for performance optimization.

4.1.7 Verify the Web Site Exists in IIS

1. **Start** > **All Apps** > **Internet Information Services (IIS) Manager**

   Internet Information Services (IIS) Manager appears.
2. On the tree, expand the server node, expand the Sites node, and confirm the website you want to use for Asset Performance Management is listed.
3. If the website is not listed, create a new website.

   If you want to use the default website, but it is not listed, create a new website, and then rename it to **Default Web Site**.

> NOTE
> Ensure you have the Hostname box empty in https bindings for the IIS website.

4.1.8 Install all Microsoft Updates

On all servers:

1. Run Microsoft Updates to ensure all current updates are installed.
2. Restart the server.

4.1.9 Stop Services for Honeywell Products

If you are installing or upgrading a system with other Honeywell products installed, before beginning the Asset Performance Management installation or upgrade, stop all services related to the installed Honeywell products.

For example, if Honeywell Uniformance KPI is installed on the system, stop the following services:

- Honeywell Intuition Calculation Service
- Honeywell Intuition Event Processing Service
- Honeywell Uniformance KPI Processing Service.

Once the services are stopped, perform an **IISRESET**.

4.1.10 Install Microsoft SQL Server 2012 Native Client

1. On the installation media, Navigate to the following folder:

   installer/prerequisites/sqlncli.msi
2. Right-click **sqlncli.msi** and click **install**.
3. Follow the on-screen instructions to complete the installation.

4.1.11 Check Services User Permissions

Follow the below steps in the **Database Server** before you begin the installation:

- Ensure you have a **Services User** that does not require login permissions to the Asset Performance Management computer.

> ATTENTION
> Do not use the installation (current) user as the Services User. If the selected Services User has logon permissions, they will be removed as a part of the Asset Performance Management installation.
>
> Once the installation is complete, the Services User will no longer be able to log on to the computer locally or remotely.

> NOTE
> When installing SQL Server, if NT SERVICE\ALL SERVICES no longer has LogOnAsAService permissions, the installation cannot complete as expected.
>
> To ensure the SQL Server installation succeeds, NT SERVICE\ALL SERVICES must have LogOnAsAService permissions in either the local or domain group policy.

4.2 Single Server Installation

This section explains the procedure for installing the Asset Performance Management server.

> ATTENTION
> When there is an installation failure or reboot, you need to perform password encryption before resuming the installation, refer to Password Encryption.

> NOTE
> Ensure you do not install the Asset Performance Management server on the FDM server node.

1. Perform an iisreset.
2. On the installation media of the new database server, right-click **setup.exe**, and then select **Run as Administrator**.
3. If RabbitMQ is not already installed, a prompt appears. In the RabbitMQ prompt, click **Yes**.
4. On the Welcome page, click **Next**.
5. Read through the license agreement and click **I accept the terms in the license agreement** if you agree to the license terms and then click **Next**.
6. Do all of the following:
   a. Ensure all of the following are selected:
      - **Web components**
      - **Application components**
      - **Database components**
      - **Calculation Components**

   b. Clear any of the following optional features that are not required:

- **Reports**: Configure the SQL Server Reporting Service for HAS Reports.
- **Clientcomponents**: the APM Configuration Excel Plugin, Model Tools, and/or PHD Utility.

> NOTE
> If you need to import models after installation, **Client Components** must be selected.

c. If your system requires secure communication between networks, select **Communicate from your business network to your application network**, and then select **Primary Server**.

d. Click **Next**.

- Select **Asset Analytics** check box to install Asset Analytics.

> NOTE
> By selecting the Asset Analytics feature along with Database components, the installer will create an InfluxDB user with Runtime account credentials provided during Asset Performance Management installation. InfluxDB user name will not be prefixed with the domain name.

- Select **Enable HTTPS** for InfluxDB

> NOTE
> You need to have a domain trusted certificate or CA signed certificate to configure HTTPS for InfluxDB. You will need the certification during post-installation. check box

- Select **Create InfluxDB Instance**

7. Click **Next**.
8. Required features are selected by default. Choose any of the following optional features if required, and then click **Next**:
   - **Embedded PHD**: Used when a full PHD server is not available.

> NOTE
> Use of EPHD with Asset Performance Management requires familiarity with PHD.
>
> EPHD does not support SQL Clustering.

   - (If the **Client components** are selected) Asset Performance Management Tools:
     - **Bulk Configuration**: The APM Configuration Excel Plugin.
     - **Model Tools**: The APM Configuration Tool and Model Tool.

> NOTE
> If you need to import models after installation, **Model Tools** must be selected.

     - **PHD Utility**: The PHD components.

9. In the **Intuition Environment Setup,** correct any issues, and then click **Exit**. (If the Exit button does not appear with the success message, close the browser window to continue.)

   For example, if Certificate Configuration is flagged, do all of the following: a. Beside **Certificate Configuration**, click **Repair**.

   a. Select **Web Components** and enter the Application server name.
   b. Click **Configure**.
   c. In the **Success** dialog box, click **OK**.
   d. Click the **Home** tab and click **Repair all**.

10. Perform the following steps:

    a. Under the **CMMS Alert** section, the path where the CMMS XML files are saved is displayed by default. If you want to change the default path, perform the following steps:
       - Click **Browse**.
       - The **Browse for Folder** page appears.
       - Select the path where you want to save the CMMS XML files.
       - Click **OK**.

    b. In the **Email Alert** area, in the **SMTP Server Name** box, type the SMTP server name.
    c. In the **DAS Configuration** area, select **Configure DAS**.
    d. In the **Server Name** box, type the *fully qualified domain name* of the DAS server.
    e. In the **Port Number** box, type the port number on which DAS is installed.
    f. Click **Next**.

11. Do one of the following:

    - To install SQL Server along with Asset Performance Management, select **Install SQL**.
    - To use an existing SQL server, in the **SQL Server Name** box, type or select the name of the SQL server instance (including instance name if using a named instance) used by Asset Performance Management.
    - In the **SQL Server Name** box, type or select the name of the SQL server instance (including instance name if using a named instance) used by Asset Performance Management.

12. Do all of the following:

    a. In the **Login Information for Application Services** area, in the **Runtime Domain\User Name** box, type the user name for the Asset Performance Management services user, including the domain name.
    b. In the **Runtime Account Password** box, type the password for the Asset Performance Management services user.
    c. (Optional) If you are using an existing EPHD installation, in the **EPHD Server Details** area, in the **EPHD Server Name** box, type the database server name.
    d. In the SQL Cluster Support area, select Enable SQL Cluster to enable SQL Cluster deployment.
    e. Enter the Domain User Group Name. The domain user group name should be in the domain\UserGroupName format. For example: l4-dc\UASServiceGroup.
    f. In the Service Account Permissions area, select **I accept the service account must be denied the ability to logon to the desktop to continue**.

> NOTE
> The installation cannot continue if the service account has login permissions to the desktop. This check box must be selected for the installation to succeed.
>
> Once this change is made, the service account can no longer log on to the computer (locally or remotely) and cannot be used to run programs that interact with the desktop, such as Internet Explorer.

g. Click **Next**.

13. Do all of the following:
    a. In the **Website selection** box, select the website you are using for Asset Performance Management.
    b. In the **net.tcp Binding Information** box type the net.tcp port number.
    c. If the server uses HTTP, clear the **Enable HTTPs based secure installation**. If it uses HTTPS, leave this box selected.
    d. Click **Next**.

14. On the Email Dispatcher page, do all of the following:
    a. In the **Sender's Email ID**, type the email address that will appear in the From field of emails sent by the system.
    b. In the **Default Email ID**, type the email address that will receive a notification if an email delivery fails.
    c. In the **SMTP Server Name** box, type the name of the SMTP server.
    d. You can also enable SSL, by clicking the **Enable SSL** check box and provide the following information
        a. "Server Name of SMTP server.
        b. "Port Number of the SMTP server.
        c. "Secure Socket Option - Enter socket option. For more details refer to Configuring Secure Socket Option.

15. Click **Install** to start the installation.
16. If you selected Install SQL, when the installer reaches the Microsoft SQL Server step, a prompt to insert the SQL Server installation media appears. Do all of the following:
    a. Put the provided SQL Server installation media into the listed location/drive.

> NOTE
> If mounting the installation media, ensure it is done so the media remains mounted to the same drive after a server restart.

    b. Click **Yes**.
       The Microsoft SQL Server installation begins.
    c. When the SQL Server installation is complete, a prompt appears to reboot the computer.
    d. If it was removed, return the Asset Performance Management installation media to the installation location/drive.
    e. Click **Yes** to reboot.
    f. Click **Yes**.

17. A message appears prompting you to restart the computer. Click **OK** to restart.

> **ATTENTION**
> Ensure all other applications or folders are closed before confirming the system restart, as the restart will happen even if other applications have unsaved changes.

18. Once the restart is complete, ensure the Asset Performance Management services user is given the following permissions to the **<InstallDrive>:\ProgramData\Microsoft\Crypto\RSA\MachineKeys** folder on the Primary Server:
    - Read & Execute
    - List Folder contents
    - Read
19. When the installation is complete, ensure all required steps in Single Server Post Install have been completed.

> **NOTE**
> To use the UniSim Design (USD) model for monitoring the process or asset with Asset Performance Management, you must also install UniSim Design on the Primary Server. Refer to UniSim Design Suite for instructions.

4.2.1 FDM Web API Component

If you are connecting to FDM, after the installation is complete, you need to install the FDM Web API Component to ensure asset links to FDM device details open as expected.

4.3 Single Server Post Install

4.3.1 Post Installation Steps for InfluxDB

Follow company policy and get a certificate from trusted authorities and then follow the below steps:

> **NOTE**
> The following steps are applicable in the database server for multi-server.

1. Ensure that you have ssl certificate in .pfx format.
2. Mount the ISO file.
3. Navigate to the following location:
   <Installation Media>\Utilities\ConfigureCertificates\
4. Use **Run as Administrator** to open PowerShell.
5. Run the following command:
   .\Configure-InfluxDBCertificate.ps1
6. The following inputs prompted by the script.

a. Get the ssl certificate in .pfx format and provide the file location of ssl certificate.

> **NOTE**
> SSl certificate should be domain trusted certificate or CA signed certificate and this certification is required to enable HTTPS for InfluxDB.

b. Enter the password of ssl certificate.

In case the InfluxDB is installed as **http** and **pRdGpInfluxSelection** key value is other than **New**, then the certificate configuration will not ask for a path. To resolve this issue, perform the steps below:

1. Navigate to the following configuration files folder:
   <Install Location>\Honeywell\Asset Performance Management Installer\Install\Configuration files\
2. Open **UI.xml**.
3. Update **PChkboxHttpsInfluxDB** key value to **true**.
4. From the **Configure-InfluxDBCertificate.ps1** file, copy **Influxdbselection -eq** key value to **pRdGpInfluxSelection**.
5. Save **UI.xml**.
6. Run **.\Configure-InfluxDBCertificate.ps1** as mentioned above.

> **ATTENTION**
> Once the InfluxDB certificate is installed, revert the **pRdGpInfluxSelection** key value to the original value in the **UI.xml** file.

4.3.2 Complete Reports Installation

1. Navigate to the following location:
   <InstallDrive>:\Program Files (x86)\Honeywell\MES\AssetManager\Tools\AMReports
2. Copy the **AMLocalizedAssembly.dll** file.
3. Paste it to the following location:
   <InstallDrive>:\Program Files\Microsoft SQL Server Reporting Services\SSRS\ReportServer\bin
4. Navigate to the below path and open **RS Scripter Load All Items.cmd** in notepad.
5. Update **SET REPORTSERVER** is report server fully qualified domain name and **SET RC** value with the file path to **RS.exe** in the install location.

   Example:

```
::Script Variables
SET LOGFILE="RS Scripter Load Log.txt"
SET SCRIPTLOCATION=
SET BACKUPLOCATION=
SET REPORTSERVER=https://REPORTINGSERVERFQDN:443/ReportServer
SET RS="C"\Program Files\Microsoft SQL Server Reporting Services\Shared Tools\RS.exe"
SET TIMEOUT=60
```

6. Run the following file:

```
<InstallDrive>:\Program Files
(x86)\Honeywell\MES\AssetManager\Tools\AMReports\PublishReports\RS Scripter Load
All Items.cmd
```

4.3.3 Application Key Configuration for Reports

By default, HAS application is configured as **http**report configuration.

For HTTPS installation, and report configuration to work from the HAS application, you should modify the **Honeywell.UAS.reportsURL** key in the **System**> **System Configuration**> **Application**as shown below.

- Double-click the **Value**fieldto update the value for https configuration:

> **NOTE**
> You can change http to https, and provide the DBServer name along with the port number.
>
> example: https:// DBServer FQDN name:Port number/Reports/

4.3.4 Configure the Reports Data Sources

1. Open the SQL Server Reporting Services page:
   http://[ServerName]/reports/
   where [ServerName] is the name or IP address of the server on which Reporting Services is installed.
2. On the reporting services **Home** page, in the toolbar, click **Tiles**, and then select **Show hidden items**.
3. In the folders list, select **Data Sources**.
4. Select **Assets**.
5. On the **Properties** page, do all of the following:
   a. Scroll to the **Credentials** section.
   b. Ensure **Using the following credentials** are selected.
   c. In the **User name** box, type the name installation user.
   d. In the **Password** box, type the password of the installation user.
   e. Click **Apply**.

4.3.5 Post Install Step for EPHD with Named SQL Instance

1. In the calculation server, navigate to the below folder and open the AMhost application configuration file.
   <Install location>\ Programfilesx86\Honeywell\Assetmanager\appserver\bin\bin2
2. Ensure SQL_ PHDAPP and SQL_PHDCFG connection strings point to the database server without the instance name.
3. In the application server, navigate to the below folder and open **web.config.**
   <Install Location>\Programfiles x86\HOneywell\ Assetmanager\webserver\data
4. Ensure **SQL_PHDAPP** and **SQL_PHDCFG** connection strings point to the database server without the instance name.
5. In the calculation server and in the Uniformance PHD server using the PHD system console, ensure the database

server name points to the database server without the instance name, and in the Uniformance Database Server, ensure that the database data source points to the database server name with the instance name.

6. Run the below commands in the command prompt as administrator:

```
phdctl stop /y
phdctl stop /y
dbsecurity phdctl stop /y
udbserver phdctl start cold
```

4.3.6 Add Group or User Access to SQL Reports

1. Open the **SQL Server Reporting Services** page:
   http://[ServerName]/reports/
   where [ServerName] is the name or IP address of the server on which **Reporting Services** is installed.
2. On the reporting services **Home** page, click **Manage Folder**.
3. Click **Add group or user**.
4. Do all of the following:
   a. In the **Group or user name** box, type the name of the group or user who requires access to view the reports.
      For example, we recommend adding the default Reliability Engineer, Maintenance Engineer, and Maintenance Planner groups.
   b. In the **Role** list, select **Browser**.
   c. Click **OK**.
      The user or group appears on the **Security** page for the home folder.

4.3.7 Add Required Users to the Windows Reliability Engineers Group

Once the Application Server is installed, you need to ensure the services and install users are added to the Windows Reliability Engineers Group.

1. **Start** > **Server Manager**.
2. On the server tree, choose **Configuration** > **Local Users and Groups** > **Groups**.
3. Right-click the **Reliability Engineers** group, and click **Add to Group**.
4. In the **Reliability Engineers Properties** dialog box, click **Add**.
5. Enter the names of all of the following users:
   a. The services (runtime) user.

      > **NOTE**
      > If the services user is not added to the Reliability Engineers group, some features will not function as expected.

   b. The install user.
   c. All the users who will need to use the APM Configuration Excel Plugin and the Asset Performance

6. Click **OK**.

The user names appear in the group list.

7. Click **OK** to close the Reliability Engineers Properties dialog box.

4.3.8 Configure the Anti-Virus Software Not to Block Sending Bulk Emails

If you have McAfee anti-virus software on the server, it must be configured to not block the sending of bulk emails.

1. **Start** > **All Apps** > **VirusScan Console**

   The VirusScan Console window appears.
2. From the Task list, double-click **Access Protection**.

   The Access Protection Properties dialog box appears.
3. Clear the **Prevent Mass Mailing Worms from Sending Mail rule** check box.
4. Click **OK**.

4.3.9 (Optional) Import the Default Asset Templates

(Optional) Follow the below steps to install the standard model library:

1. Open the following location on the installation media:

   [Installation Media]:\StandardLibrary\Model Library
2. Copy the **AMExport_XML** file.
3. Navigate to the folder in the system with the Model Tools installed, and paste the **AMExport_XML** file:

   [InstallDrive]:\Program Files (x86)\Honeywell\MES\AssetManager\Tools\AMImportExport
4. Run the **AMImportModel** file.
5. Click the **Source** drop-down list and select **Asset Manager Export file**.

   The Asset Template information is displayed in the **Available Models for Import** section.
6. Select the **Select All** check box to import the Asset Template information.

   OR

   From the **Select** column, select the check box beside the asset type that you want to import asset template information.
7. Click **Copy**.

   The selected **Asset Template** information is saved in the Asset Performance Management database.
8. Restart the **AMApplicationServer** service.
9. The imported asset template information is available on the Asset Performance Management user interface after you restart the service.

4.3.10 Give the Services User Access to the Asset Links Folder

1. Navigate to the following folder:

   <InstallDrive>:\Program Files (x86)\Honeywell\MES\Core\WWWMESCore\AMInformationFolders
2. Right-click **AMInformationFolders** and select **Properties**.
3. Click the **Security** tab.
4. Click **Edit**.

5. Click **Add**.
6. Enter the Asset Performance Management services user, and click **OK**.
7. Select the user from the list, and in the **Allow** column select **Read** and **Write**.
8. Click **OK**.

4.3.11 Refresh EPHD

After installing Asset Performance Management and updating the EPHD settings, the EPHD system is left in an Initializing state. To bring the state back to active, run the following commands:

```
phdctl stop /y

phdctl stop /y dbsecurity

phdctl stop /y udbserver

phdctl start cold
```

4.3.12 InfluxDB Configuration

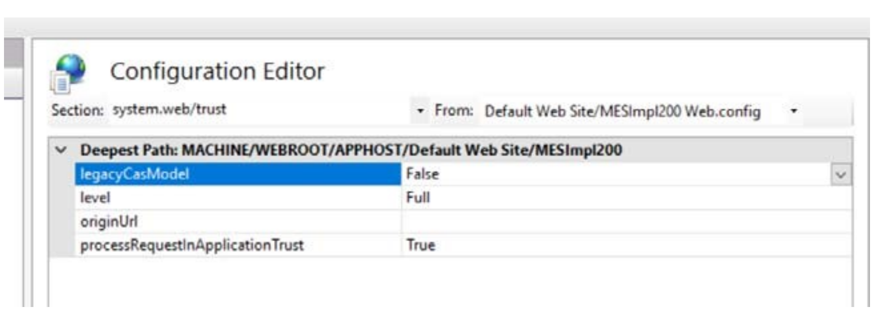
For the InfluxDB Configuration there are two instances such as **Existing InfluxDB Instance** or **Create InfluxDB Instance**.

1. If a New instance is selected as post-installation:
   a. Disabling anonymous authentication.
   b. Creating default user and password as that of the runtime user (without domain) and password.
   c. Updating all the necessary json files and db with these details.

2. If the Existing instance is selected as post-installation:
   a. Post-installation ensures that anonymous authentication is disabled for **influxDB** and configures the default user name and password
   b. Navigate and run the post-install script available at the following location in **Application Server**.
      <Install Media>\Utilities\UpdateInfluxDBUser\UpdateInfluxDBUser.ps1
   c. The above script will prompt for InfluxDB user name and password and updates in .json files and Database.

4.3.13 Trust Level for MES and MESImp200 in IIS

Ensure the trust level for **MES** and **MESImp200** is set to Full.



CHAPTER

5 ADDING ADDITIONAL CALCULATION SERVER

5.1 Installing an Additional Calculation Server

Any Asset Performance Management deployment can install additional Calculation Servers to distribute the calculations. This can help improve calculation performance on systems with a large number of assets.

You need one main calculation server before adding additional calculation servers. This can be the original calculation server in a split server installation or the server with the calculation components installed in an alternate deployment.

> **ATTENTION**
> If there is an installation failure, you need to perform password encryption before resuming the installation, for more information refer to Password Encryption.

> **NOTE**
> - Ensure the Asset Performance Management services user has access to the Intuition Core installation folder on the Calculation Server. This can be accomplished by adding the services user to the Users (recommended) or Administrators Windows groups.
> - Ensure TLS 1.2 is enabled on the server during installation.

> **ATTENTION**
> If the system prompts for a reboot during installation, click **No** and proceed with Installation.
>
> If you want to proceed with the reboot, click **Yes**, and perform the following steps
>
> 1. Mount the media once the reboot is completed.
> 2. Perform password encryption, for more information refer to Password Encryption.
> 3. Restart the installation.
>
>   The installation will resume from the point before reboot.

5.1.1 Update Firewall Settings

Before adding additional calculation servers, ensure the firewall on the main calculation server is configured to open the ports for RabbitMQ. By default, RabbitMQ uses port **5671** and port **5672**.

5.1.2 Install All Microsoft Updates

On all servers:

1. Run Microsoft Updates to ensure all current updates are installed.
2. Restart the server.

5.1.3 Check Services User Permissions

Follow the below steps in the **Database Server** before you begin the installation:

- Ensure you have a **Services User** that does not require login permissions to the Asset Performance Management computer.

> ATTENTION
> Do not use the installation (current) user as the Services User. If the selected Services User has logon permissions, they will be removed as a part of the Asset Performance Management installation.
>
> Once the installation is complete, the Services User will no longer be able to log on to the computer locally or remotely.

5.1.4 Installing the Asset Performance Management Calculation Server

> NOTE
> When entering a server name, use the computer name, not the Fully Qualified Domain Name (FQDN), or IP address for all servers except the DAS Server, unless otherwise mentioned. (The DAS server requires the use of the FQDN.)

1. Perform an **IISRESET**.
2. On the installation media of the new database server, right-click **setup.exe**, and then select **Run as Administrator**.
3. If RabbitMQ is not already installed, a prompt appears. In the RabbitMQ prompt, click **Yes**.
4. On the Welcome page, click **Next**.
5. Read through the license agreement and click **I accept the terms in the license agreement** if you agree to the license terms and then click **Next**.
6. Do all of the following:
   a. Ensure the following is selected:
      - **Calculation Components**

   b. Clear the following:
      - **Web components**
      - **Application components**
      - **Database components**
      - **Reports**: Configure the SQL Server Reporting Service for HAS Reports.

   c. Click **Next**.
7. In the **Intuition Environment Setup,** do all of the following:
   a. Beside **Certificate Configuration**, click **Repair**.
   b. Select **Application Components**.
   c. Click **Configure**.
   d. In the **Success** dialog box, click **OK**.
   e. Click the **Home** tab and click **Repair all**.
   f. Click **Exit**. (If the Exit button does not appear with the success message, close the browser window to continue.)

8. Do all of the following:
   a. Clear the **Embedded PHD** check box.
   b. Select Asset Analytics if you want the Asset Analytics installed and click **Next**.
   c. (Optional) To install to a non-default location, in the **Installation Path** area, click **Browse** and then select the new location.
   d. (Optional) To change the MES Data Files location, in the **Data Files Path** area, click **Browse** and then select the location in which to save MES Data Files.
   e. Click **Next**.
9. Do all of the following:
   a. In the **DAS Configuration details** area, select **Configure DAS**.
   b. In the **DAS Configuration** area, in the **Server Name** box, type the *fully qualified domain name* of the DAS server.
   c. In the **DAS Configuration** area, in the **Port Number** box, type the port number on which DAS is installed.
   d. Click **Next**.
   e. In the **Login Information for Application Services** area, in the **Runtime Domain\User Name** box, type the user name for the Asset Performance Management services user, including the domain name.
   f. In the **Runtime Account Password** box, type the password for the Asset Performance Management services user.

      NOTE
      Ensure that the below unsupported characters are not entered in the password. $ Dollar sign, space, " Quote, ' Single quote, ` Back quote.

   g. Enable the SQL cluster check box only if u have an SQL cluster. If SQL cluster is enabled, Enter the Domain User Group Name.

      The domain user group name should be in the domain\UserGroupName format. For example: I4-dc\UASServiceGroup
   h. In the Service Account Permissions area, select **I accept the service account must be denied the ability to logon to the desktop to continue**.

      NOTE
      The installation cannot continue if the service account has login permissions to the desktop. This check box must be selected for the installation to succeed.

      Once this change is made, the service account can no longer log on to the computer (locally or remotely) and cannot be used to run programs that interact with the desktop, such as Internet Explorer.

   i. Click **Next**.
   j. In the **Website selection** box, select the website you are using for Asset Performance Management.
   k. In the **net.tcp Binding Information** box type the net.tcp port number.
   l. If the server uses HTTP, clear the **Enable HTTPs based secure installation**. If it uses HTTPS, leave this box selected.
   m. Click **Next**.

10. Do all of the following:
    a. In the **Web Server Host Name** box, type the name of the Web Server.

    > **ATTENTION**
    > If you are using https, enter the fully qualified domain name of the Web Server.

    b. In the **Web Server HTTP Port No** box, type the port number used on the Web Server.
    c. In the **Web Server Server Protocol Name** box, select the protocol (**http** or **https**) used on the Web Server.
    d. Click **Next**.
11. Click **Install** to start the installation.
12. If you selected to install the APM Configuration Excel Plugin, when the APM Excel Add-in Setup page appears, click **Accept**.
13. Click **Finish**.
14. A message appears prompting you to restart the computer. Click **OK** to restart.

    > **ATTENTION**
    > Ensure all other applications or folders are closed before confirming the system restart, as the restart will happen even if other applications have unsaved changes.

15. Once the restart is complete, ensure the Asset Performance Management services user is given the following permissions to the **<InstallDrive>:\ProgramData\Microsoft\Crypto\RSA\MachineKeys** folder on the Primary Server:
    - Read & Execute
    - List Folder contents
    - Read

5.2 Exchange the Calculation Server Certificates

Export and import the certificates from the Application Server to the new additional Calculation Server:

- Export the certificate from the Application Server and import it on the additional calculation server.
- Export the certificate from the addition calculation server and import it on the Application Server.

To export the certificate

1. Open the Console Root (**Start** > **Search** > **MMC**).
2. Do all of the following to add the Certificates Snap-in:
    a. On the **File** menu, select **Add/Remove Snap-in**.
    b. Double-click **Certificates**.
    c. Select **Computer Account**, and then click **Next**.
    d. Click **Finish**.
    e. Click **OK**.
3. On the tree, expand **Certificates**, and select **IntuitionSign**.

4. In the certificate list, right-click the **[ServerName].Intuition.Honeywell.com** certificate, point to All Tasks, and then select **Export**.
5. Complete the steps in the Certificate Export Wizard, and save the certificate in a location that is accessible or can be transferred to the other servers.

To import the certificate

1. In the Console Root, on the tree, expand **Certificates**, and select **IntuitionVerify**.
2. In the **Actions** sidebar, click **More Actions**, point to **All Tasks**, and select **Import**.
3. Complete the Import Certificate Wizard for the **[ServerName].Intuition.Honeywell.com** certificate.

> NOTE
> If any of the SSL certificates used for https binding has expired, the exchange mentioned above should be performed after you renew the SSL certificate and re-run the preinstall checker on all the nodes.
>
> For more information, refer to **Intuition Certificate Renewal** in Asset Performance Management Troubleshooting Guide.

5.3 Configuring the RabbitMQ Certificate for Primary and Additional Calculation Servers

Once the calculation servers are installed, the configuration must be configured to allow connections between the calculation servers.

The steps to change the configuration depend on whether your system uses Trusted Certificates (recommended) or self-signed certificates.

5.3.1 Updating Configuration for Servers with Trusted Certificates

Updating Configuration for Primary Calculation Server with Trusted Certificate

1. Stop the following services:
    - AMApplicationServer
    - Honeywell APM Calculation Host
    - Honeywell APM Event Translator services
2. Close the Services window.
3. In a command prompt, run the following command after updating it to include the details for your system:

   [Install Dir]\AssetManager\Tools\InstallConfiguration\Honeywell.UAS.Configuration.exe -r -p [UAS service account password] - s [ServerCertificateLocation] – rabbitserverpassword [ServerCertificatePassword] -c [ClientCertificateLocation] --rabbitclientpassword [ClientCertificatePassword]

   where,

   [ServiceAccountPassword] is the password of the Asset Performance Management services user.

   [ClientCertificateLocation] is the folder location of the Trusted Certificate on the additional calculation server.

   [ServerCertificateLocation] is the folder location of the Trusted Certificate on the primary calculation server.

   [ClientPassword] is the RabbitMQ password for the additional calculation server.
4. Open SQL Server Management Studio and perform the following steps:

a. Expand the **Personal** folder and right-click the certificate in the right pane and then select **All Tasks > Manage Private Keys**.

   Private Keys dialog box appears.

b. In the dialog box ensure the Service User account has read access.

5. Start the following services:
   - AMApplicationServer
   - Honeywell APM Calculation Host

Updating Configuration for Additional Calculation Server with Trusted Certificate

1. Stop the following services:
   - AMApplicationServer
   - Honeywell APM Calculation Host
   - Honeywell APM Event Translator services
2. Close the Services window.
3. In a command prompt, run the following command after updating it to include the details for your system:

   [Install Dir]\AssetManager\Tools\InstallConfiguration\Honeywell.UAS.Configuration.exe -r -p [ServiceAccountPassword] -c [ClientCertificateLocation] --rabbitclientpassword [ClientPassword]

   where,

   [ServiceAccountPassword] is the password of the Asset Performance Management services user.

   [ClientCertificateLocation] is the folder location of the Trusted Certificate on the additional calculation server.

   [ClientPassword] is the RabbitMQ password for the additional calculation server.
4. Start the following services:
   - AMApplicationServer
   - Honeywell APM Calculation Host

Install the RabbitMQ Management Console

On the main calculation server, install the RabbitMQ management console.

1. Navigate to the following location and run the below management command:

   [Install Location]:\Program Files\RabbitMQ Server\rabbitmq_server-3.8.1\sbin\rabbitmq-plugins enable rabbitmq_management
2. Restart the RabbitMQ service in the **Primary Calculation Server**.

Add the Additional Calculation Server User Permissions

On the main calculation server, use the RabbitMQ management console to give the additional calculation server the required user permissions.

1. Open the RabbitMQ Management Console. By default, this is found at the following location:
   http://[CalculationServerName]:15672
2. Enter the username and password of the Asset Performance Management services user.

> **NOTE**
> Enter the service user name without the domain name.

3. Select the **Admin** tab.
4. Do all of the following to add the additional calculation server's user:
   a. Expand **Add a user**.
   b. In the **Username** box, type the following (using the fully qualified domain name of the additional calculation server):
      O=client,CN=[AdditionalCalcServerFQDN]
   c. From the dropdown list select **No password**.
   d. Click **Add User**.
5. Select the new user.
   The user's details page opens.
6. Under Permissions, do all of the following:
   a. From the Virtual host list, select **APM**.
   b. In the **Configure regexp** box, leave the default setting (.*).
   c. In the **Write regexp** box, leave the default setting (.*).
   d. In the **Read regexp** box, leave the default setting (.*).
   e. Click **Set topic permissions**.
7. Under Topic permissions, do all of the following:
   a. From the **Virtual Host** list, select **APM**.
   b. From the **Exchange** list, leave the default setting **(AMQP default)**.
   c. In the **Write regexp** box, leave the default setting (.*).
   d. In the **Read regexp** box, leave the default setting (.*).
   e. Click **Set topic permissions**.

Restart Services

1. In the primary calculation server, restart the services in the following order:
   - Restart the RabbitMQ service.
   - Stop and start the AMapplication service
   - Stop and start the Honeywell APM calculation host.
   - Stop and start the APM Event Translator
2. In the additional calculation server, restart the services in the following order:
   - AMApplicationServer
   - Honeywell APM Calculation Host

The following services can then be **stopped or disabled** in the additional calculation server if they are running:

- RabbitMQ
- Honeywell APM Event Translator Service

Confirm the Connection

On the main calculation server, use the RabbitMQ management console to give the additional calculation server the required user permissions.

1. Open the RabbitMQ Management Console.
2. Select the **Connections** tab.
   Connections with the additional calculation server's user are listed if the connection is established.

5.3.2 Updating Configuration for Servers with Self-Signed Certificates

Install the RabbitMQ Management Console

On the main calculation server, install the RabbitMQ management console.

1. Navigate to the following location and run the below management command:

   <Install Location>:\Program Files (x86)\Honeywell\MES\AssetManager\Tools\InstallConfiguration\Honeywell.UAS.Configuration.exe -r -m -p [ServiceUserPassword]

Update the Certificate Configuration on the Main Calculation Server

1. On the additional calculation server, navigate to the following folder:
   [installationdrive]:\ProgramData\RabbitMQ\cert
2. Open the rabbitmq_ca file in Notepad and copy the contents to a location accessible by the main calculation server.
3. On the main calculation server, navigate to the following folder:
   [installationdrive]:\ProgramData\RabbitMQ\cert
4. Open the rabbitmq_ca file in Notepad, scroll to the bottom, and paste the certificate content from the additional calculation server beneath the existing content.
5. Save and close the file.

Add the Additional Calculation Server User Permissions

On the main calculation server, use the RabbitMQ management console to give the additional calculation server the required user permissions.

1. Open the RabbitMQ Management Console. By default, this is found at the following location:

   https://[CalculationServerName]:15672
2. Enter the user name and password of the Asset Performance Management services user.
3. Select the **Admin** tab.
4. Do all of the following to add the additional calculation server's user:
   a. Expand **Add a user**.
   b. In the **Username** box, type the following (using the fully qualified domain name of the additional calculation server):

      O=client,CN=[AdditionalCalcServerFQDN]

   c. From the dropdown-list select **No password**.
   d. Click **Add user**.

5. Select the new user.

   The user's details page opens.
6. Under Permissions, do all of the following:
   a. From the Virtual host list, select **APM**.
   b. In the **Configure regexp** box, leave the default setting (.*).
   c. In the **Write regexp** box, leave the default setting (.*).
   d. In the **Read regexp** box, leave the default setting (.*).
   e. Click **Set topic permissions**.

7. Under Topic permissions, do all of the following:
   a. From the **Virtual Host** list, select **APM**.
   b. From the **Exchange** list, leave the default setting **(AMQP default)**.
   c. In the **Write regexp** box, leave the default setting (.*).
   d. In the **Read regexp** box, leave the default setting (.*).
   e. Click **Set topic permissions**.

Restart Services

On the additional calculation server, restart the services in the following order:

- AMApplicationServer
- Honeywell APM Calculation Host

The following services can then be stopped on the additional calculation server if they are running:

- RabbitMQ
- Honeywell UAS Event Translator Service

Confirm the Connection

On the main calculation server, use the RabbitMQ management console to give the additional calculation server the required user permissions.

1. Open the RabbitMQ Management Console.
2. Select the **Connections** tab.
   Connections with the additional calculation server's user are listed if the connection is established.

5.4 Distribute Calculations

Once the calculations servers are installed and connected, you can use the Asset Performance Management System Console to distribute calculations.

1. Open **Asset Performance Management**, and on the Contents tree choose **System Console**.
   The **System Console** page appears.
2. In the **Distribute Calculations** section, click Distribute.
   A status message is displayed as the calculations are distributed.

5.5 Appendix: Post installation of Additional Calculation Servers

Below are the post-installation steps of additional calculation servers

1. Run the below command from the command prompt using run as administrator mode
   `<Install Location>:\Installer\Components\EPHD\EPHD\Install\UnfAutoRun.cmd`
2. Navigate to Install Asset Performance Management and Install Uniformance PHD Client interfaces (x 64)

CHAPTER

6 PRIMARY AND SHADOW SERVER INSTALLATION

ATTENTION

- Primary shadow server is only supported for single server architecture.
- Uniformance Insight is not supported with primary shadow server.

6.1 Primary Server Installation

NOTE

- Ensure you do not install the Asset Performance Management server on the FDM server node.
- The Primary Server must be located on a Windows domain.
- Ensure TLS 1.2 is enabled on the server during the installation.
- Asset Performance Management installs the Data Access Service (DAS) and creates the default instance and connections.
- When entering a server name, use the computer name or IP address for all servers except the DAS Server. (The DAS server requires the use of a fully qualified domain name.)
- You must click **Configure** button in the Certificate Configuration tab even if the system passed all the validation and is ready for installation.

ATTENTION

If there is an installation failure, you need to perform password encryption before resuming the installation, for more information refer to Password Encryption.

ATTENTION

During installation, if the system prompts for a reboot, click **No** and proceed with Installation.

If you want to proceed with the reboot, click **Yes** , and perform the following steps

1. Mount the media once the reboot is completed.
2. Perform password encryption, for more information refer to Password Encryption.
3. Restart the installation.

   The installation will resume from the point before reboot.

6.1.1 Verify you have the administrator privileges on the server

1. On the Server computer desktop, right-click **Computer** and select **Manage**.
   The **Server Manager** screen appears.
2. On the left pane, expand **Configuration**.
3. Expand **Local Users and Groups**.
4. Click the **Groups** folder.
   The available user groups are displayed on the right pane.
5. Double-click **Administrators**.
   The **Administrator Properties** dialog box appears listing the users who are a part of the administrator user group.

6.1.2 Verify User Account Control is Set to Low

1. Open the **Control Panel**.
2. Click **System and Security**.
3. Click **Change User Account Control settings**.
4. Slide the setting down to **Never Notify**.
5. Click **OK**.

> NOTE
> This setting can be changed to a higher level once all installations are complete.

6.1.3 Verify SQL Server

> NOTE
> If SQL Failover Clustered instances are used, when SQL FCI performs a failover, there is a small downtime of the SQL instance. This can disrupt communication between the AM Service and SQL Server, requiring a restart of the AM Service. (This can be automated for minimal disruption.)

6.1.4 Verify the SQL Collation Value

> NOTE
> Default SQL collation is supported only in the prerequisite.

1. Open **SQL Server Management Studio**.
2. Enter the details for the database instance to be used by Asset Performance Management, and click **Connect**.
3. Right-click the database instance, and click **Properties**.
4. On the **General** page, ensure the **Server Collation** value is **SQL_Latin1_General_CP1_CI_ AS**.

6.1.5 Verify the SQL Server Service is Running Under "Local System" Account or Under an Account with Administrator Privileges

1. **Start** > **All Apps** > **Services**.

   The Services window appears.

2. From the list, right-click **SQL Server** and select **Properties**.

   The **SQL Server Properties** dialog box appears.

3. Click the **Log On** tab.

4. Click the **Local System** account option.

   (OR)

5. Click the **This account** option and enter the account with administrator privileges.

6. Click **Apply** and then click **OK**.

7. Restart the service.

6.1.6 Verify You have System Administrator Privileges for the Database on Which the Asset Performance Management Database Will Reside

1. **Start** > **All Apps** > **SQL Server Management Studio**.

   The **Microsoft SQL Server Management Studio** window appears.

2. On the Object Explorer pane, expand **Security > Logins**.

3. Right-click the user on which the setup is running and select **Properties**.

   The **Login Properties** dialog box appears.

4. On the left pane, click **Server Roles**.

   A list of server roles is displayed on the right pane. Verify if the check box beside the **sysadmin** role is selected.

> **ATTENTION**
> If you restart the Asset Performance Management server, SQL server agent service stops automatically. This is because the SQL server agent service is in **Manual** startup type by default.
>
> This should be changed to an **Automatic** startup type for the SQL server agent service to run automatically even after restarting the server.

Perform the following steps to change the startup type to **Automatic**:

1. **Start > All Apps > Services**

   The **Services** window appears.

2. From the list, right-click **SQL Server Agent** and select **Properties**.

   The **SQL Server Agent Properties** dialog box appears.

3. On the **General** tab, from the **Startup Type** drop-down list, select **Automatic**.

4. Click **OK** and close the services window.

NOTE
The SQL Server Agent is required for the following reasons:

- To run the SQL job every 2 hours
- To delete the records in the HistorisedProcessedAttributes table - If the attribute count exceeds 120 for each attribute, the latest 120 records are retained and the rest are deleted.
- You must have knowledge of SQL and SQL jobs if you want to modify the number of records retained. We recommend not changing the number of records for performance optimization.

6.1.7 Verify the Web Site Exists in IIS

1. **Start** > **All Apps** > **Internet Information Services (IIS) Manager**

   Internet Information Services (IIS) Manager appears.
2. On the tree, expand the server node, expand the Sites node, and confirm the website you want to use for Asset Performance Management is listed.
3. If the website is not listed, create a new website.

   If you want to use the default website, but it is not listed, create a new website, and then rename it to **Default Web Site**.

NOTE
Ensure you have the Hostname box empty in https bindings for the IIS website.

6.1.8 Install all Microsoft Updates

On all servers:

1. Run Microsoft Updates to ensure all current updates are installed.
2. Restart the server.

6.1.9 Stop Services for Honeywell Products

If you are installing or upgrading a system with other Honeywell products installed, before beginning the Asset Performance Management installation or upgrade, stop all services related to the installed Honeywell products.

For example, if Honeywell Uniformance KPI is installed on the system, stop the following services:

- Honeywell Intuition Calculation Service
- Honeywell Intuition Event Processing Service
- Honeywell Uniformance KPI Processing Service.

Once the services are stopped, perform an **IISRESET**.

6.1.10 Check Services User Permissions

Follow the below steps in the **Database Server** before you begin the installation:

- Ensure you have a **Services User** that does not require login permissions to the Asset Performance Management computer.

> ATTENTION
> Do not use the installation (current) user as the Services User. If the selected Services User has logon permissions, they will be removed as a part of the Asset Performance Management installation.
>
> Once the installation is complete, the Services User will no longer be able to log on to the computer locally or remotely.

> NOTE
> When installing SQL Server, if NT SERVICE\ALL SERVICES no longer has LogOnAsAService permissions, the installation cannot complete as expected. To ensure the SQL Server installation succeeds, NT SERVICE\ALL SERVICES must have LogOnAsAService permissions in either the local or domain group policy.

- Confirm Port Number for the Primary Server

  Determine the port that will be used for the Primary server, as the same port must be used on the Shadow server to communicate with the Primary Server.

> ATTENTION
> Changing port numbers after installation is not recommended, as it can result in issues during future migrations to new versions of Asset Performance Management.

6.1.11 Installing a Primary Server

This section explains the procedure for installing a Asset Performance Management Primary Server.

1. Perform an iisreset.
2. On the installation media of the new data base server, right-click **setup.exe**, and then select **Run as Administrator**.
3. If RabbitMQ is not already installed, a prompt appears. In the RabbitMQ prompt, click **Yes**.
4. On the Welcome page, click **Next**.
5. Read through the license agreement and click **I accept the terms in the license agreement** if you agree to the license terms and then click **Next**.
6. Do all of the following:
   a. Ensure all of the following are selected:
      - **Web components**
      - **Application components**
      - **Database components**
      - **Calculation Components**

   b. Clear any of the following optional features that are not required:

- **Reports**: Configure the SQL Server Reporting Service for APM Reports.
- **Clientcomponents**: the APM Configuration Excel Plugin, Model Tools, and/or PHD Utility.

> NOTE
> If you need to import models after installation, **Client Components** must be selected.

c. If your system requires secure communication between networks, select **Communicate from your business network to your application network**, and then select **Primary Server**.

d. Click Next.

e. Select **Asset Analytics** check box to install Asset Analytics.

> NOTE
> By selecting Asset Analytics feature along with Database components, installer will create InfluxDB user with Runtime account credentials provided during Asset Performance Management installation. InfluxDB user name will not be prefixed with domain name.

f. Select **Enable HTTPS for influxDB** check box.

> NOTE
> You need to have domain trusted certificate or CA signed certificate to configure HTTPS for influxDB. You will need the certification during post installation.

g. Select **Create InfluxDB Instance**.

> NOTE
> During the database server installation, if you select Existing InfluxDB Instance you need to follow the below post installation steps
>
> a. During Post installation, ensure the anonymous authentication is disabled for InfluxDB and configure the default user name and password.
>
> b. Navigate and run the post install script available at the following location in Application Server.
>
> <ISO>\Utilities\UpdateInfluxDBUser\UpdateInfluxDBUser.ps1
>
> c. The above script will prompt for InfluxDB user name and password.
>
> d. Perform an iisreset.

7. Required features are selected by default. Choose any of the following optional features if required, and then click **Next**:
   - **Embedded PHD**: Used when a full PHD server is not available.

     Use of EPHD with Asset Performance Management require familiarity with PHD.

     EPHD does not support SQL Clustering
   - (If the **Client components** are selected) Asset Performance Management Tools:

- **Bulk Configuration**: The APM Configuration Excel Plugin.
- **Model Tools**: The APM Configuration Tool and Model Tool.

> NOTE
> If you need to import models after installation, **Model Tools** must be selected.

- **PHD Utility**: The PHD components.

8. In the **Intuition Environment Setup,** correct any issues, and then click **Exit**. (If the Exit button does not appear with the success message, close the browser window to continue.)

   For example, if Certificate Configuration is flagged, do all of the following:

   a. Beside **Certificate Configuration**, click **Repair**.
   b. Select **Web Components** and enter the Application server name.
   c. Click **Configure**.
   d. In the **Success** dialog box, click **OK**.
   e. Click the **Home** tab and click **Repair all**.

9. Perform the following steps:

   a. Under the **CMMS Alert** section, the path where the CMMS XML files are saved is displayed by default. If you want to change the default path, perform the following steps:
      - Click **Browse**.
      - The **Browse for Folder** page appears.
      - Select the path where you want save the CMMS XML files.
      - Click **OK**.

   b. In **Email Alert** area, in the **SMTP Server Name** box, type the SMTP server name.
   c. In the **DAS Configuration** area, select **Configure DAS**.
   d. In the **Server Name** box, type the *fully qualified domain name* of the DAS server.
   e. In the **Port Number** box, type the port number on which DAS is installed.
   f. Click **Next**.

10. Do one of the following:

    a. To install SQL Server along with Asset Performance Management, select **Install SQL**.
    b. To use an existing SQL server, in the **SQL Server Name** box, type or select the name of the SQL server instance (including instance name if using a named instance) used by Asset Performance Management.
    c. In the **SQL Server Name** box, type or select the name of the SQL server instance (including instance name if using a named instance) used by Asset Performance Management.

11. Do all of the following:

    a. In the **Login Information for Application Services** area, in the **Runtime Domain\User Name** box, type the user name for the Asset Performance Management services user, including the domain name.
    b. In the **Runtime Account Password** box, type the password for the Asset Performance Management services user.

> **NOTE**
> Ensure that the below unsupported characters are not entered in the password. $ Dollar sign, space, " Quote, ' Single quote, ` Back quote.

c. (Optional) If you are using an existing EPHD installation, in the **EPHD Server Details** area, in the **EPHD Server Name** box, type the database server name.

d. Enable the SQL cluster check box only if u have SQL cluster.

   If SQL cluster enabled, Enter the Domain User Group Name. The domain user group name should be in the domain\UserGroupName format. For example: I4-dc\ UASServiceGroup.

e. In the Service Account Permissions area, select **I accept the service account must be denied the ability to logon to the desktop to continue**.

> **NOTE**
> The installation cannot continue if the service account has logon permissions to the desktop. This check box must be selected for the installation to succeed.
>
> Once this change is made, the service account can no longer log on to the computer (locally or remotely) and cannot be used to run programs that interact with the desktop, such as or Internet Explorer.

f. Click **Next**.

12. Do all of the following:
    a. In the **Website selection** box, select the web site you are using for Asset Performance Management.
    b. In the **net.tcp Binding Information** box type the net.tcp port number.
    c. If the server uses HTTP, clear the **Enable HTTPs based secure installation**. If it uses HTTPS, leave this box selected.
    d. Click **Next**.

13. Do all of the following:
    a. In the **Sender's Email ID**, type the email address that will appear in the From field of emails sent by the system.
    b. In the **Default Email ID**, type the email address that will receive a notification if an email delivery fails.
    c. In the **SMTP Server Name** box, type the name of the SMTP server.
    d. You can also enable SSL, by clicking the **Enable SSL** check box and provide the following information
       - Server Name of SMTP server.
       - Port Number of the SMTP server.
       - Secure Socket Option - Enter socket option. For more details refer Configuring Secure Socket Option

14. Click **Install** to start the installation.

15. If you selected Install SQL, when the installer reaches the Microsoft SQL Server step, a prompt to insert the SQL Server installation media appears. Do all of the following:

a. Put the provided SQL Server installation media into the listed location/drive.

> **NOTE**
> If mounting the installation media, ensure it is done so the media remains mounted to the same drive after a server restart.

b. Click **Yes**.

The Microsoft SQL Server installation begins.

c. When the SQL Server installation is complete, a prompt appears to reboot the computer.

d. If it was removed, return the Asset Performance Management installation media to the installation location/drive.

e. Click **Yes** to reboot.

f. Click **Yes**.

16. A message appears prompting you to restart the computer. Click **OK** to restart.

> **ATTENTION**
> Ensure all other applications or folders are closed before confirming the system restart, as the restart will happen even if other applications have unsaved changes.

17. Once the restart is complete, ensure the Asset Performance Management services user is given the following permissions to the [InstallDrive]:\ProgramData\Microsoft\Crypto\RSA\MachineKeys folder on the Primary Server:
    - Read & Execute
    - List Folder contents
    - Read

> **NOTE**
> To use the UniSim Design (USD) model for monitoring the process or asset with Asset Performance Management, you must also install UniSim Design on the Primary Server. See “UniSim Design Suite” on page 237 for instructions.

6.1.12 FDM Web API Component

If you are connecting to FDM, after the installation is complete, you need to install the FDM Web API Component to ensure asset links to FDM device details open as expected.

6.1.13 Enable Windows Features on the Primary Server

1. On the Primary Server, in the **Server Manager**, on the Quick Start, click **Add roles and features**.

   The **Add Features Wizard** appears.
2. On the **Before Your Begin** page, click **Next**.
3. On the **Select Installation Type** page, select **Role-based on feature-based installation**, and click **Next**.
4. On the sidebar, select **Roles**.
5. Select the following feature:
    - **Web Server** > **Security** > **Client Certificate Mapping Authentication**

6. On the sidebar, select **Features**.
7. Select the following feature:
    - **Remote Server Administration Tool** > **Role Administration Tools** > **AD DS and AD LDS Tools** > **AD DS Tools** > **AD DS Snap Ins and Command-Line Tools**
8. Click **Next**.

    The **Confirm Installation Selections** page appears.
9. Click **Install** to start the installation.
10. When the installation completes, click **Close** to close the wizard.

6.1.14 Complete Reports Installation

1. Navigate to the following location:

    <InstallDrive>:\Program Files (x86)\Honeywell\MES\AssetManager\Tools\AMReports
2. Copy the **AMLocalizedAssembly.dll** file.
3. Paste it to the following location:

    <InstallDrive>:\Program Files\Microsoft SQL Server Reporting Services\SSRS\ReportServer\bin
4. Navigate to the below path and open **RS Scripter Load All Items.cmd** in notepad.
5. Update **SET REPORTSERVER** is report server fully qualified domain name and **SET RC** value with the file path to **RS.exe** in the install location.

    Example:

```
::Script Variables
SET LOGFILE="RS Scripter Load Log.txt"
SET SCRIPTLOCATION=
SET BACKUPLOCATION=
SET REPORTSERVER=https://REPORTINGSERVERFQDN:443/ReportServer
SET RS="C"\Program Files\Microsoft SQL Server Reporting Services\Shared Tools\RS.exe"
SET TIMEOUT=60
```

6. Run the following file:

```
<InstallDrive>:\Program Files
(x86)\Honeywell\MES\AssetManager\Tools\AMReports\PublishReports\RS Scripter Load
All Items.cmd
```

6.1.15 Application Key Configuration for Reports

By default, HAS application is configured as **http**report configuration.

For HTTPS installation, and report configuration to work from the HAS application, you should modify the **Honeywell.UAS.reportsURL** key in the **System**> **System Configuration**> **Application**as shown below.

- Double-click the **Value**fieldto update the value for https configuration:

> **NOTE**
> You can change http to https, and provide the DBServer name along with the port number.
>
> example: https:// DBServer FQDN name:Port number/Reports/

6.1.16 Configure the Reports Data Sources

1. Open the SQL Server Reporting Services page:
   http://[ServerName]/reports/
   where [ServerName] is the name or IP address of the server on which Reporting Services is installed.
2. On the reporting services **Home** page, in the toolbar, click **Tiles**, and then select **Show hidden items**.
3. In the folders list, select **Data Sources**.
4. Select **Assets**.
5. On the **Properties** page, do all of the following:
   a. Scroll to the **Credentials** section.
   b. Ensure **Using the following credentials** are selected.
   c. In the **User name** box, type the name installation user.
   d. In the **Password** box, type the password of the installation user.
   e. Click **Apply**.

6.1.17 Add Group or User Access to SQL Reports

1. Open the **SQL Server Reporting Services** page:
   http://[ServerName]/reports/
   where [ServerName] is the name or IP address of the server on which **Reporting Services** is installed.
2. On the reporting services **Home** page, click **Manage Folder**.
3. Click **Add group or user**.
4. Do all of the following:
   a. In the **Group or user name** box, type the name of the group or user who requires access to view the reports.
   For example, we recommend adding the default Reliability Engineer, Maintenance Engineer, and Maintenance Planner groups.
   b. In the **Role** list, select **Browser**.
   c. Click **OK**.
   The user or group appears on the **Security** page for the home folder.

6.1.18 Add Required Users to the Windows Reliability Engineers group

1. **Start** > **Server Manager**.
2. On the server tree, choose **Configuration** > **Local Users and Groups** > **Groups**.
3. Right-click the **Reliability Engineers** group, and click **Add to Group**.
4. In the **Reliability Engineers Properties** dialog box, click **Add**.

5. Enter the names of all of the following users:
   a. The services (runtime) user.

> **ATTENTION**
> If the service user is not added to the reliability engineers group, some features will not function as expected.

   b. The install user.
   c. All the users who will need to use the APM Configuration Excel Plugin and the Asset Performance Management Configuration.
6. Click **OK**.The user names appear in the group list.
7. Click **OK** to close the **Reliability Engineers Properties** dialog box.

6.1.19 Configure the Anti-Virus Software Not to Block Sending Bulk Emails

If you have McAfee anti-virus software on the server, it must be configured to not block the sending of bulk emails.

1. **Start** > **All Apps** > **VirusScan Console**

   The VirusScan Console window appears.
2. From the Task list, double-click **Access Protection**.

   The Access Protection Properties dialog box appears.
3. Clear the **Prevent Mass Mailing Worms from Sending Mail rule** check box.
4. Click **OK**.

6.1.20 (Optional) Import the Default Asset Templates

(Optional) Follow the below steps to install the standard model library:

1. Open the following location on the installation media:

   [Installation Media]:\StandardLibrary\Model Library
2. Copy the **AMExport_XML** file.
3. Navigate to the folder in the system with the Model Tools installed, and paste the **AMExport_XML** file:

   [InstallDrive]:\Program Files (x86)\Honeywell\MES\AssetManager\Tools\AMImportExport
4. Run the **AMImportModel** file.
5. Click the **Source** drop-down list and select **Asset Manager Export file**.

   The Asset Template information is displayed in the **Available Models for Import** section.
6. Select the **Select All** check box to import the Asset Template information.

   OR

   From the **Select** column, select the check box beside the asset type that you want to import asset template information.
7. Click **Copy**.

   The selected **Asset Template** information is saved in the Asset Performance Management database.
8. Restart the **AMApplicationServer** service.
9. The imported asset template information is available on the Asset Performance Management user interface after you restart the service.

6.1.21 Give the Services User Access to the Asset Links Folder

1. Navigate to the following folder:
   <InstallDrive>:\Program Files (x86)\Honeywell\MES\Core\WWWMESCore\AMInformationFolders
2. Right-click **AMInformationFolders** and select **Properties**.
3. Click the **Security** tab.
4. Click **Edit**.
5. Click **Add**.
6. Enter the Asset Performance Management services user, and click **OK**.
7. Select the user from the list, and in the **Allow** column select **Read** and **Write**.
8. Click **OK**.

6.1.22 Refresh EPHD

After installing Asset Performance Management and updating the EPHD settings, the EPHD system is left in an Initializing state. To bring the state back to active, run the following commands:

```
phdctl stop /y

phdctl stop /y dbsecurity

phdctl stop /y udbserver

phdctl start cold
```

6.2 Shadow Server Installation

The section explains the procedure for installing the Asset Performance Management Shadow server.

NOTE

- Ensure you do not install the Asset Performance Management server on the FDM server node.
- When entering a server name, use the computer name or IP address for all servers except the DAS Server. (The DAS server requires a fully qualified domain name.)
- Asset Performance Management installs the Data Access Service (DAS) and creates the default instance and connections.
- The Shadow Server's UTC time should match the Primary Server's UTC time. If the clocks do not match, communication between the two may fail.
- You must click **Configure** button in the Certificate Configuration tab even if the system passed all the validation and is ready for installation.

ATTENTION

- If there is an installation failure, you need to perform password encryption before resuming the installation, for more information refer to Password Encryption .
- A separate license is not required for the Shadow Asset Performance Management server.

ATTENTION
If the system prompts for a reboot during installation, click **No** and proceed with Installation.

If you want to proceed with the reboot, click **Yes**, and perform the following steps

1. Mount the media once the reboot is completed.
2. Perform password encryption, for more information refer to Password Encryption .
3. Restart the installation.
   The installation will resume from the point before reboot.

6.2.1 Verify You have the Administrator Privileges on the Server

1. On the Server computer desktop, right-click **Computer** and select **Manage**.
   The **Server Manager** screen appears.
2. On the left pane, expand **Configuration**.
3. Expand **Local Users and Groups**.
4. Click the **Groups** folder.
   The available user groups are displayed on the right pane.
5. Double-click **Administrators**.
   The **Administrator Properties** dialog box appears listing the users who are a part of the administrator user group.

6.2.2 Verify User Account Control is Set to Low

1. Open the **Control Panel**.
2. Click **System and Security**.
3. Click **Change User Account Control settings**.
4. Slide the setting down to **Never Notify**.
5. Click **OK**.

NOTE
This setting can be changed to a higher level once all installations are complete.

6.2.3 Verify the SQL Collation Value

NOTE
Default SQL collation is supported only in the prerequisite.

1. Open **SQL Server Management Studio**.
2. Enter the details for the database instance to be used by Asset Performance Management, and click **Connect**.
3. Right-click the database instance, and click **Properties**.
4. On the **General** page, ensure the **Server Collation** value is **SQL_Latin1_General_CP1_CI_ AS**.

6.2.4 Verify the SQL Server Service is Running Under "Local System" Account or Under an Account with Administrator Privileges

1. **Start** > **All Apps** > **Services**.

   The Services window appears.
2. From the list, right-click **SQL Server** and select **Properties**.

   The **SQL Server Properties** dialog box appears.
3. Click the **Log On** tab.
4. Click the **Local System** account option.

   (OR)
5. Click the **This account** option and enter the account with administrator privileges.
6. Click **Apply** and then click **OK**.
7. Restart the service.

6.2.5 Verify You have System Administrator Privileges for the Database on Which the Asset Performance Management Database Will Reside

1. **Start** > **All Apps** > **SQL Server Management Studio**.

   The **Microsoft SQL Server Management Studio** window appears.
2. On the Object Explorer pane, expand **Security > Logins**.
3. Right-click the user on which the setup is running and select **Properties**.

   The **Login Properties** dialog box appears.
4. On the left pane, click **Server Roles**.

   A list of server roles is displayed on the right pane. Verify if the check box beside the **sysadmin** role is selected.

> ATTENTION
>
> If you restart the Asset Performance Management server, SQL server agent service stops automatically. This is because the SQL server agent service is in **Manual** startup type by default.
>
> This should be changed to an **Automatic** startup type for the SQL server agent service to run automatically even after restarting the server.

Perform the following steps to change the startup type to **Automatic**:

1. **Start > All Apps > Services**

   The **Services** window appears.
2. From the list, right-click **SQL Server Agent** and select **Properties**.

   The **SQL Server Agent Properties** dialog box appears.
3. On the **General** tab, from the **Startup Type** drop-down list, select **Automatic**.
4. Click **OK** and close the services window.

NOTE
The SQL Server Agent is required for the following reasons:

- To run the SQL job every 2 hours
- To delete the records in the HistorisedProcessedAttributes table - If the attribute count exceeds 120 for each attribute, the latest 120 records are retained and the rest are deleted.
- You must have knowledge of SQL and SQL jobs if you want to modify the number of records retained. We recommend not changing the number of records for performance optimization.

6.2.6 Verify the Web Site Exists in IIS

1. **Start** > **All Apps** > **Internet Information Services (IIS) Manager**
   Internet Information Services (IIS) Manager appears.
2. On the tree, expand the server node, expand the Sites node, and confirm the website you want to use for Asset Performance Management is listed.
3. If the website is not listed, create a new website.
   If you want to use the default website, but it is not listed, create a new website, and then rename it to **Default Web Site**.

NOTE
Ensure you have the Hostname box empty in https bindings for the IIS website.

6.2.7 Install all Microsoft Updates

On all servers:

1. Run Microsoft Updates to ensure all current updates are installed.
2. Restart the server.

6.2.8 Ensure TLS 1.2 is Enforced

To enable the required TLS registry keys, do all of the following:

1. Navigate to the following folder:
   <installation media>\\CoreSystem\BatchFiles
2. Run the following file as an administrator:
   EnableStrongCrypto.bat
3. Restart the server.

6.2.9 Stop Services for Honeywell Products

If you are installing or upgrading a system with other Honeywell products installed, before beginning the Asset Performance Management installation or upgrade, stop all services related to the installed Honeywell products.

For example, if Honeywell Uniformance KPI is installed on the system, stop the following services:

- Honeywell Intuition Calculation Service
- Honeywell Intuition Event Processing Service
- Honeywell Uniformance KPI Processing Service.

Once the services are stopped, perform an **IISRESET**.

6.2.10 Check Services User Permissions

Follow the below steps in the **Database Server** before you begin the installation:

- Ensure you have a **Services User** that does not require login permissions to the Asset Performance Management computer.

> **ATTENTION**
> Do not use the installation (current) user as the Services User. If the selected Services User has logon permissions, they will be removed as a part of the Asset Performance Management installation.
>
> Once the installation is complete, the Services User will no longer be able to log on to the computer locally or remotely.

> **NOTE**
> When installing SQL Server, if NT SERVICE\ALL SERVICES no longer has LogOnAsAService permissions, the installation cannot complete as expected. To ensure the SQL Server installation succeeds, NT SERVICE\ALL SERVICES must have LogOnAsAService permissions in either the local or domain group policy.

6.2.11 Open the Port for the Primary Server

Make a note of the port used for the Primary server.

Open the port used for the Primary server across the firewall to allow communication between the Shadow Asset Performance Management server and Primary Asset Performance Management server.

6.2.12 Installing the Shadow Server

1. Perform an iisreset.
2. On the installation media of the new data base server, right-click **setup.exe**, and then select **Run as Administrator**.
3. In the RabbitMQ prompt, click **No**.
4. On the Welcome page, click **Next**.
5. Read through the license agreement and click **I accept the terms in the license agreement** if you agree to the license terms and then click **Next**.
6. Do all of the following:
   a. Clear the following check box:
      - **Calculation Components**

   b. If any of the following optional features are not required (depending on your topology) clear the related check box:
      - **Reports**: Configure the SQL Server Reporting Service for APM Reports.
      - **Clientcomponents**: the APM Configuration Excel Plugin, Model Tools, and/or PHD Utility.

   c. Select **Communicate from your business network to your application network**, and then select **Shadow Server**.
   d. Click **Next**.
7. In the **Primary/Shadow server details** page, do all of the following:

   a. In the **Primary Server Details** area, in the **Server Name** box, type the name of the Primary Server.
   b. In the **Domain Name** box, type the domain of the Primary Server.
   c. Click **Next**.

8. Select Asset Analytics and InfluxDB if user want the Asset Analytics installed and you should choose **Create InfluxDB Instance** and click **Next**.
9. Required features are selected by default. Choose any of the following optional features if required, and then click **Next** :
   a. **Embedded PHD**: Used when a full PHD server is not available.
   b. (If the **Client components** are selected) Asset Performance Management Tools:
      a. **Bulk Configuration**: The APM Configuration Excel Plugin.
      b. **Model Tools**: The APM Configuration Tool and Model Tool.

      > **NOTE**
      > If you need to import models after installation, **Model Tools** must be selected.

      c. **PHD Utility**: The PHD components.

10. Perform the following steps:
    a. Under the **CMMS Alert** section, the path where the CMMS XML files are saved is displayed by default. If you want to change the default path, perform the following steps:
       - Click **Change**.
         The **Change Current Destination** folder page appears.
       - Click the Look in drop-down list and select the path where you want save the CMMS XML files.
       - Click **OK**.

    b. In **Email Alert** area, in the **Server Name** box, if you are sending e-mails on the L4 or L3.5 network, type the SMTP server name, which you made a note of in the Before you Begin section.
    c. Click **Next**.

11. Do all of the following:
    a. In the **Website selection** box, select the web site you are using for Asset Performance Management.
    b. In the **net.tcp Binding Information** box type the net.tcp port number.
    c. If the server uses HTTP, clear the **Enable HTTPs based secure installation**. If it uses HTTPS, leave this box selected.
    d. Click **Next**.

12. Do all of the following:
    a. In the **Sender's Email ID**, type the email address that will appear in the From field of emails sent by the system.
    b. In the **Default Email ID**, type the email address that will receive a notification if an email delivery fails.
    c. In the **SMTP Server Name** box, type the name of the SMTP server.
    d. In the **SMTP Port Number**, type the port number used on the SMTP server.

13. Do one of the following:

- To install SQL Server along with Asset Performance Management, select **Install SQL**.
- To use an existing SQL server, in the **SQL Server Name** box, type or select the name of the SQL server instance (including instance name if using a named instance) used by Asset Performance Management.
- In the **SQL Server Name** box, type or select the name of the SQL server instance (including instance name if using a named instance) used by Asset Performance Management.

14. Do all of the following:
    a. In the **Login Information for Application Services** area, in the **Runtime Domain\User Name** box, type the user name for the Asset Performance Management services user, including the domain name.
    b. In the **Runtime Account Password** box, type the password for the Asset Performance Management services user.

    > NOTE
    > Ensure that the below unsupported characters are not entered in the password. $ Dollar sign, space, " Quote, ' Single quote, ` Back quote.

    c. (Optional) If you are using an existing EPHD installation, in the **EPHD Server Details** area, in the **EPHD Server Name** box, type the database server name.
    d. Enable the SQL cluster check box only if u have SQL cluster.

    If SQL cluster enabled, Enter the Domain User Group Name. The domain user group name should be in the domain\UserGroupName format. For example: l4-dc\ UASServiceGroup
    e. In the Service Account Permissions area, select **I accept the service account must be denied the ability to logon to the desktop to continue**.

    > NOTE
    > The installation cannot continue if the service account has logon permissions to the desktop. This check box must be selected for the installation to succeed.
    >
    > Once this change is made, the service account can no longer log on to the computer (locally or remotely) and cannot be used to run programs that interact with the desktop, such as or Internet Explorer.

    f. Click **Next**.
15. Click **Install** to start the installation.
16. Click **Finish**.
17. A message appears prompting you to restart the computer. Click **OK** to restart.

> ATTENTION
> Ensure all other applications or folders are closed before confirming the system restart, as the restart will happen even if other applications have unsaved changes.

18. Once the server has restarted, ensure the AMApplicationServer service is running.
19. Once the restart is complete, ensure the Asset Performance Management services user is given the following permissions to the [InstallDrive]:\ProgramData\Microsoft\Crypto\RSA\MachineKeys folder on the Primary Server:

- Read & Execute
- List Folder contents
- Read

6.2.13 Complete Reports Installation

1. Navigate to the following location:

   <InstallDrive>:\Program Files (x86)\Honeywell\MES\AssetManager\Tools\AMReports
2. Copy the **AMLocalizedAssembly.dll** file.
3. Paste it to the following location:

   <InstallDrive>:\Program Files\Microsoft SQL Server Reporting Services\SSRS\ReportServer\bin
4. Navigate to the below path and open **RS Scripter Load All Items.cmd** in notepad.
5. Update **SET REPORTSERVER** is report server fully qualified domain name and **SET RC** value with the file path to **RS.exe** in the install location.

   Example:

```
::Script Variables
SET LOGFILE="RS Scripter Load Log.txt"
SET SCRIPTLOCATION=
SET BACKUPLOCATION=
SET REPORTSERVER=https://REPORTINGSERVERFQDN:443/ReportServer
SET RS="C"\Program Files\Microsoft SQL Server Reporting Services\Shared Tools\RS.exe"
SET TIMEOUT=60
```

6. Run the following file:

```
<InstallDrive>:\Program Files
(x86)\Honeywell\MES\AssetManager\Tools\AMReports\PublishReports\RS Scripter Load
All Items.cmd
```

6.2.14 Application Key Configuration for Reports

By default, HAS application is configured as **http**report configuration.

For HTTPS installation, and report configuration to work from the HAS application, you should modify the **Honeywell.UAS.reportsURL** key in the **System**> **System Configuration**> **Application**as shown below.

- Double-click the **Value**fieldto update the value for https configuration:

> **NOTE**
> You can change http to https, and provide the DBServer name along with the port number.
>
> example: https:// DBServer FQDN name:Port number/Reports/

6.2.15 Configure the Reports Data Sources

1. Open the SQL Server Reporting Services page:
   http://[ServerName]/reports/
   where [ServerName] is the name or IP address of the server on which Reporting Services is installed.
2. On the reporting services **Home** page, in the toolbar, click **Tiles**, and then select **Show hidden items**.
3. In the folders list, select **Data Sources**.
4. Select **Assets**.
5. On the **Properties** page, do all of the following:
   a. Scroll to the **Credentials** section.
   b. Ensure **Using the following credentials** are selected.
   c. In the **User name** box, type the name installation user.
   d. In the **Password** box, type the password of the installation user.
   e. Click **Apply**.

6.2.16 Add Group or User Access to SQL Reports

1. Open the **SQL Server Reporting Services** page:
   http://[ServerName]/reports/
   where [ServerName] is the name or IP address of the server on which **Reporting Services** is installed.
2. On the reporting services **Home** page, click **Manage Folder**.
3. Click **Add group or user**.
4. Do all of the following:
   a. In the **Group or user name** box, type the name of the group or user who requires access to view the reports.
   For example, we recommend adding the default Reliability Engineer, Maintenance Engineer, and Maintenance Planner groups.
   b. In the **Role** list, select **Browser**.
   c. Click **OK**.
   The user or group appears on the **Security** page for the home folder.

6.2.17 Add Required Users to the Windows Reliability Engineers Group

1. **Start** > **Server Manager**.
2. On the server tree, choose **Configuration** > **Local Users and Groups** > **Groups**.
3. Right-click the **Reliability Engineers** group, and click **Add to Group**.
4. In the **Reliability Engineers Properties** dialog box, click **Add**.
5. Enter the names of all of the following users:
   a. The services (runtime) user.

> **ATTENTION**
> If the service user is not added to the reliability engineers group, some features will not function as expected.

b. The install user.

c. All the users who will need to use the APM Configuration Excel Plugin and the Asset Performance Management Configuration.

6. Click **OK**.The user names appear in the group list.
7. Click **OK** to close the **Reliability Engineers Properties** dialog box.

6.2.18 Configure the Anti-Virus Software Not to Block Sending Bulk Emails

If you have McAfee anti-virus software on the server, it must be configured to not block the sending of bulk emails.

1. **Start** > **All Apps** > **VirusScan Console**

   The VirusScan Console window appears.
2. From the Task list, double-click **Access Protection**.

   The Access Protection Properties dialog box appears.
3. Clear the **Prevent Mass Mailing Worms from Sending Mail rule** check box.
4. Click **OK**.

6.2.19 (Optional) Import the Default Asset Templates

(Optional) Follow the below steps to install the standard model library:

1. Open the following location on the installation media:

   [Installation Media]:\StandardLibrary\Model Library
2. Copy the **AMExport_XML** file.
3. Navigate to the folder in the system with the Model Tools installed, and paste the **AMExport_XML** file:

   [InstallDrive]:\Program Files (x86)\Honeywell\MES\AssetManager\Tools\AMImportExport
4. Run the **AMImportModel** file.
5. Click the **Source** drop-down list and select **Asset Manager Export file**.

   The Asset Template information is displayed in the **Available Models for Import** section.
6. Select the **Select All** check box to import the Asset Template information.

   OR

   From the **Select** column, select the check box beside the asset type that you want to import asset template information.
7. Click **Copy**.

   The selected **Asset Template** information is saved in the Asset Performance Management database.
8. Restart the **AMApplicationServer** service.
9. The imported asset template information is available on the Asset Performance Management user interface after you restart the service.

6.2.20 Give the Services User Access to the Asset Links Folder

1. Navigate to the following folder:

   <InstallDrive>:\Program Files (x86)\Honeywell\MES\Core\WWWMESCore\AMInformationFolders
2. Right-click **AMInformationFolders** and select **Properties**.
3. Click the **Security** tab.
4. Click **Edit**.

5. Click **Add**.
6. Enter the Asset Performance Management services user, and click **OK**.
7. Select the user from the list, and in the **Allow** column select **Read** and **Write**.
8. Click **OK**.

6.2.21 Refresh EPHD

After installing Asset Performance Management and updating the EPHD settings, the EPHD system is left in an Initializing state. To bring the state back to active, run the following commands:

```
phdctl stop /y

phdctl stop /y dbsecurity

phdctl stop /y udbserver

phdctl start cold
```

6.3 Configuring Security Certificates

To ensure secure communication between the Primary and Shadow Asset Performance Management Servers, you must configure security certificates on both these servers. These certificates allow the Primary Server and the Shadow Server to communicate with each other securely.

This chapter explains how to configure the security certificates on the Primary and Shadow Asset Performance Management Servers.

> **NOTE**
> For additional information on installing security certificates, see
>
> https://www.digicert.com/ssl-certificate-installation-microsoft-iis-8.htm

To secure the communication channel between the Primary and Shadow Asset Performance Management Servers, you must obtain the required security certificate files.

The security certificates can be classified into the following types depending on where they are sourced from:

- Trusted Certificates (recommended): These certificates are provided by a trusted certifying authority.
- Self-generated Certificates: These self-signed certificates are generated using the Shadow Server Configuration Utility.

To complete the Primary and Shadow Server configuration, you must obtain or generate the following three certificate files:

- The Certificate Authority certificate (CA)

  > **NOTE**
  > This is optional if you are using Trusted Certificates.

- The Primary Server certificate
- The Shadow Server certificate

If using trusted certificates, obtain the certificates from a trusted certifying authority.

If using self-generated certificates, generate the certificates using the Shadow Server Configuration Utility.

> **NOTE**
> If you plan to use the same certificates with future upgrades, ensure a copy of the certificates is saved in a secure location after the servers are configured, so they will be available when upgrading.

6.3.1 Example: Configuring the Primary and Shadow Server Certificates with the Shadow Server Configuration Utility

For this example, we will assume a system with the following settings:

- Certificates: Self-generated
- Primary Server Name: PrimaryServer
- Primary Server Port Number: 808
- Certificate Path on Both Servers: C:\Certificates
- Password for All Certificates: HL4nZbrW
- Domain Account User Name: UserName

On this system, to complete the configuration, from the directory with the Shadow Server Configuration Utility, the following commands would be run.

To generate the certificates:

ShadowServerConfiguration generate --authorityCertPass HL4nZbrW --authority "C:\Certificates\CA.pfx" --primaryCertPass HL4nZbrW --primaryCert "C:\Certificates\Primary.pfx"--shadowCertPass HL4nZbrW --shadowCert "C:\Certificates\Shadow.pfx"

To configure the Primary Server:

ShadowServerConfiguration primary --authorityCertPass HL4nZbrW --authority "C:\Certificates\CA.pfx" --primaryCertPass HL4nZbrW --primaryCert "C:\Certificates\Primary.pfx"--shadowCertPass HL4nZbrW --shadowCert "C:\Certificates\Shadow.pfx" --primaryServerPort 808

To map the certificate to the domain account:

ShadowServerConfiguration mapuser --user UserName --shadowCertPass HL4nZbrW --shadowCert "C:\Certificates\Shadow.pfx"

To configure the Shadow Server:

ShadowServerConfiguration shadow --authorityCertPass HL4nZbrW --authority "C:\Certificates\CA.pfx" --primaryCertPass HL4nZbrW --primaryCert "C:\Certificates\Primary.pfx"--primaryServer PrimaryServer --shadowCertPass HL4nZbrW --shadowCert "C:\Certificates\ Shadow.pfx" --primaryServerPort 808

Shadow Server Configuration Utility Commands

Command	Options	Description	Required
primary	Configure a primary server to accept connections from a shadow server.		
	--authorityCertPass	The password used to protect the authority certificate.	
	--authority	The certificate authority certificate personal exchange format file path.	
	--primaryCertPass	The password used to protect the primary certificate.	Yes
	--primaryCert	The path of the primary server certificate personal exchange format file.	Yes
	--primaryServerPort	The Net TCP port of the primary server. Used on the shadow server to communicate with the Primary server.	
	--primaryAppServerPort	The port of the primary application server. This is set when the application role server is deployed on a different machine than the web role server.	
	--shadowCertPass	The password used to protect the shadow certificate.	Yes
	--shadowCert	The path of the shadow server certificate personal exchange format file.	Yes
shadow	Configure a shadow server to connect to a primary server.		
	--authorityCertPass	The password used to protect the authority certificate.	
	--authority	The certificate authority certificate personal exchange format file path.	
	--primaryCertPass	The password used to protect the primary certificate.	Yes
	--primaryCert	The path of the primary server certificate personal exchange format file.	Yes
	--primaryServerPort	The Net TCP port of the primary server. Used on the shadow server to communicate with the Primary server.	
	--shadowCertPass	The password used to protect the shadow certificate.	Yes
	--shadowCert	The path of the shadow server certificate personal exchange format file.	Yes
mapuser	Map a certificate to a domain account. You must be a domain administrator to perform this action.		
	--shadowCertPass	The password used to protect the shadow certificate.	Yes
	--shadowCert	The path of the shadow server certificate personal exchange format file.	Yes
	-user	The Active Directory user to map the shadow certificate with (just the user name, do not domain).	Yes
single	Resets the configuration from either primary or shadow server configuration to a single server configuration.		

Command	Options	Description	Required
generate	Generate self-signed certificates for both primary and shadow servers.		
	NOTE Self-generated certificates created by the Shadow Server Configuration Utility are only good for two years from the generation date.		
	--authorityCertPass	The password used to protect the authority certificate.	
	--authority	The certificate authority certificate personal exchange format file path.	
	--primaryCertPass	The password used to protect the primary certificate.	Yes
	--primaryCert	The path of the primary server certificate personal exchange format file.	Yes
	--shadowCertPass	The password used to protect the shadow certificate.	Yes
	--shadowCert	The path of the shadow server certificate personal exchange format file.	Yes

6.3.2 Enable Windows Features on the Primary Server

1. On the Primary Server, in the **Server Manager**, on the Quick Start, click **Add roles and features**.

   The **Add Features Wizard** appears.
2. On the **Before Your Begin** page, click **Next**.
3. On the **Select Installation Type** page, select **Role-based on feature-based installation**, and click **Next**.
4. On the sidebar, select **Roles**.
5. Select the following feature:
   - **Web Server** > **Security** > **Client Certificate Mapping Authentication**
6. On the sidebar, select **Features**.
7. Select the following feature:
   - **Remote Server Administration Tool** > **Role Administration Tools** > **AD DS and AD LDS Tools** > **AD DS Tools** > **AD DS Snap Ins and Command-Line Tools**
8. Click **Next**.

   The **Confirm Installation Selections** page appears.
9. Click **Install** to start the installation.
10. When the installation completes, click **Close** to close the wizard.

6.3.3 Configure the Certificate Management Console on Primary and Shadow

Perform the following procedure on both the Primary and Shadow servers.

1. Open the certificate management console: open Windows Explorer, and in the address bar, type **MMC**, and then press **Enter**.

   The **Certificate Management Console** appears.
2. Click **File** > **Add/Remove Snap-in**.

The **Add or Remove Snap-ins** dialog box appears.

3. In the **Available Snap-ins** list, select **Certificates**.
4. Click **Add**.

   The **Certificates snap-in** wizard appears.
5. Select **Computer Account** and click **Next**.

   The **Select Computer** page appears.
6. Select **Local Computer** and click **Finish**.

   The snap-in is created and the wizard is closed.

   This new snap-in, named **Certificates**, is listed under **Selected Snap-ins** in the **Add or Remove Snap-ins** dialog box.
7. Click **OK** to close the **Add or Remove Snap-ins** dialog box and return to the certificate management console.

   A new node named **Certificates** is created under **Console Root** in the tree.

6.3.4 (Optional) Generate the Certificates

If you are not using a Trusted Certificate, you need to generate the self-generated certificates.

1. Navigate to the below folder in either server and run the Shadow Server Configuration Utility:

   <installation media>\Utilities\ShadowServerConfiguration
2. Update the following command to use your system details, and then run the command to configure and generate the certificates.

```
[%PathToUtility%]\ShadowServerConfiguration.exe generate --authorityCertPass
[%password%]--authority "[%location to save Trusted Authority Certificate%]" --
primaryCertPass[%password%] --primaryCert "[%location to save Primary Server
Certificate%]" --shadowCertPass[%password%] --shadowCert "[%location to save
Shadow Server Certificate%]
```

6.3.5 Configure the Primary Server Certificate

> **NOTE**
> Step 3 of this procedure requires the use of a domain administrator account.

1. On the Primary Server, run the Shadow Server Configuration Utility.

   This is located in the following folder on the installation media:

   [installation media]\Utilities\ShadowServerConfiguration
2. Update the following command to use your system details, and then run the command to configure the Primary server. (Refer to Shadow Server Configuration Utility Commands for descriptions of the commands and options.)

```
[%PathToUtility%]\ShadowServerConfiguration.exe primary --authorityCertPass
[%password%]
--authority "[%path to Trusted Authority Certificate%]" --primaryCertPass
[%password%]
--primaryCert "[%path to Primary Server Certificate%]" --shadowCertPass
[%password%]
--shadowCert "[%path to Shadow Server Certificate%]" --primaryServerPort [%port
number%]
```

3. As a domain administrator, run the command to map the certificate to a domain account, after updating it to use the

information for your system. (Refer to Shadow Server Configuration Utility Commands for the commands and options.)

```
[%PathToUtility%]\ShadowServerConfiguration.exe mapuser --user "[%user name without domain%]" --shadowCertPass [%password%] --shadowCert "[%path to Shadow Server Certificate%]
```

6.3.6 Configure the Shadow Server Certificate

1. Navigate to the following folder in the shadow server, and run the Shadow Server Configuration Utility.

   <installation media>\Utilities\ShadowServerConfiguration

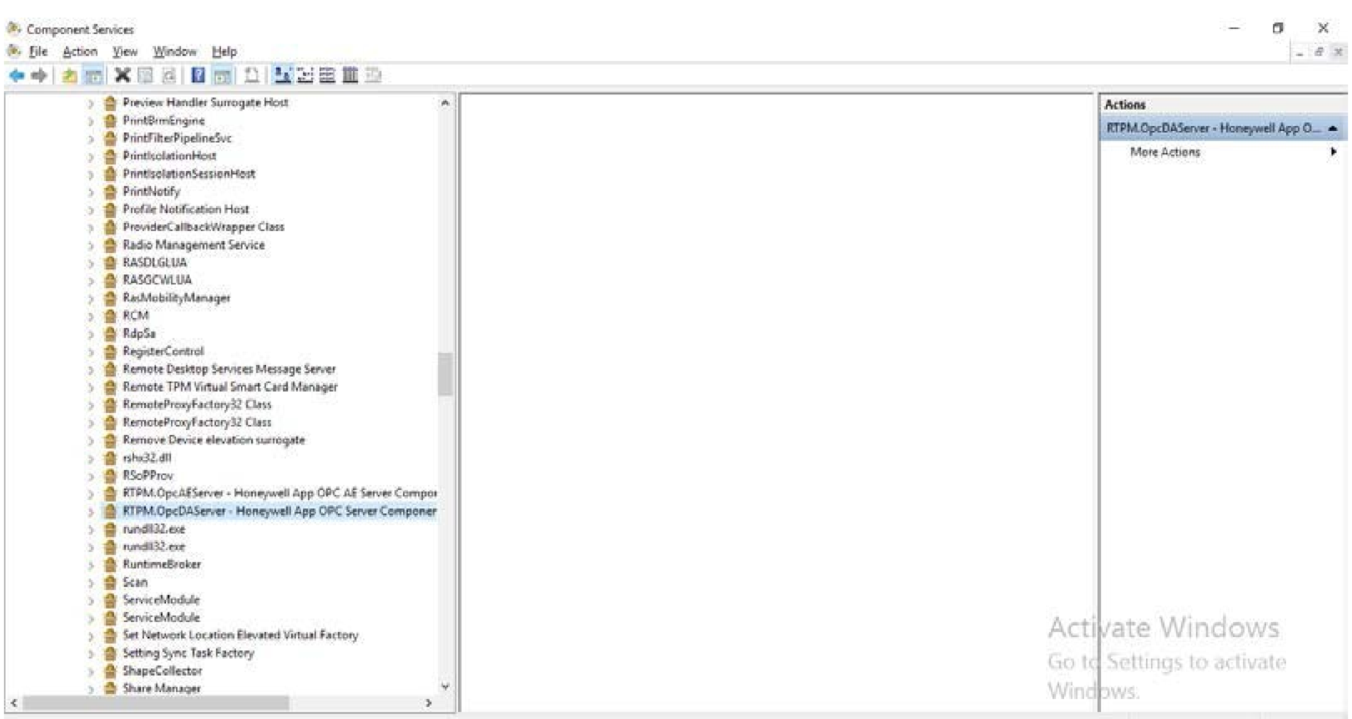
2. Update the following command to use your system details, and then run the command to configure the Shadow server. (Refer to Shadow Server Configuration Utility Commands for the commands and options.)

```
[%PathToUtility%]\ShadowServerConfiguration.exe shadow --authorityCertPass [%password%]

--authority "[%path to Trusted Authority Certificate%]" --primaryCertPass [%password%]

--primaryCert "[%path to Primary Server Certificate%]" --primaryServer "[%Primary Server name

or address%]" --shadowCertPass [%password%] --shadowCert "[%path to Shadow Server

Certificate%]" --primaryServerPort [%port number%]
```

6.3.7 DCOM Configuration Security Recommendations

Choose **Run>DCOMCNFG>Component Services.**

The screen appears as shown below:

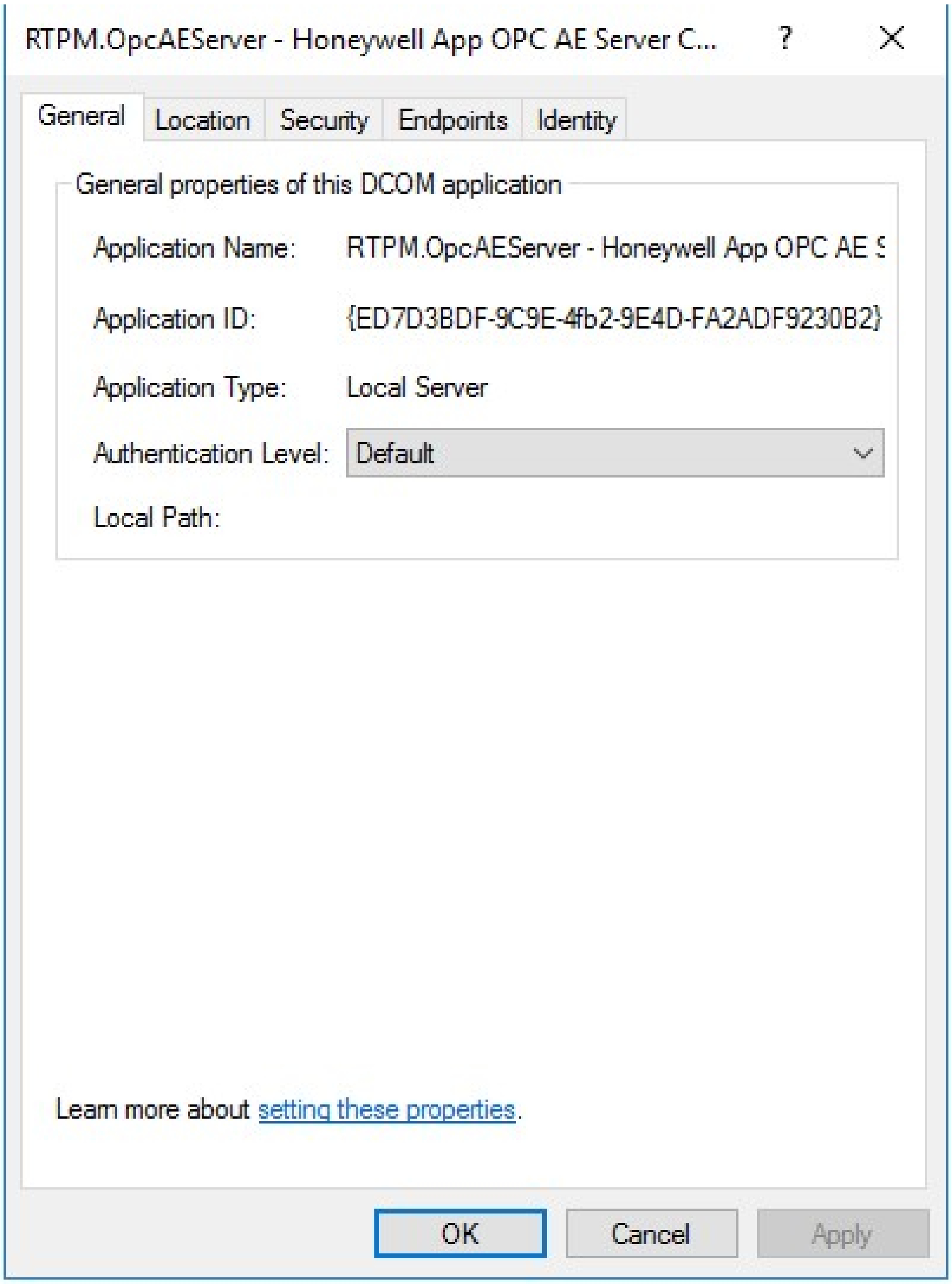


Perform the following steps on the APM DCOM services:

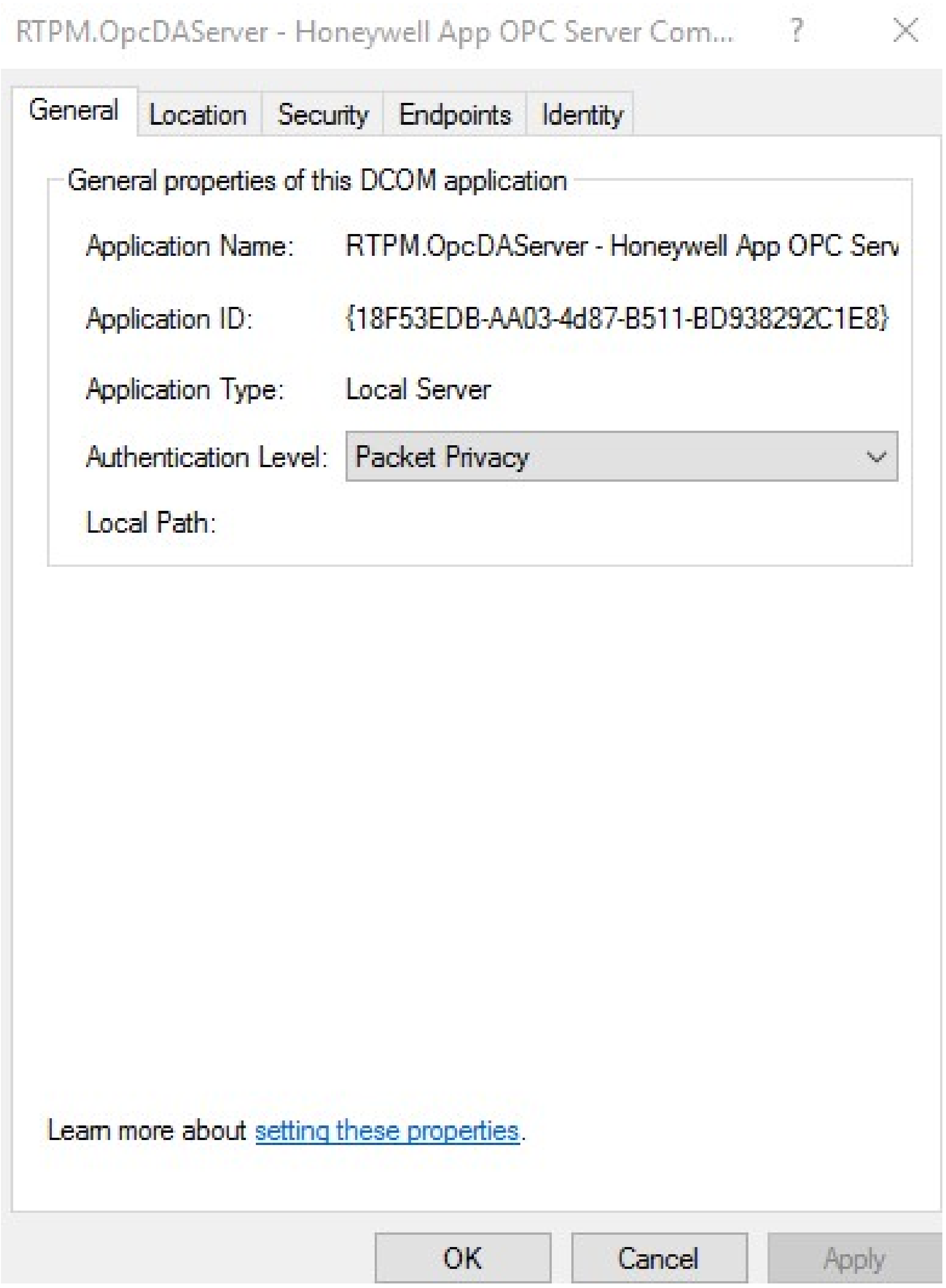
1. RTPM.OpcAEServer - Honeywell App OPC AE Server Component - exe Server.
2. RTPM.OpcDAServer - Honeywell App OPC Server Component - exe Server.
3. Select the service, **right click >Properties >General.**
4. Change Authentication level from **Default to Packet Privacy**.

Example:

- By default, APM DCOM servers use Authentication Level = Default as shown below:



- Change the **Authentication Level to Packet Privacy,** as shown below to provide the communication encryption.



- Perform this for both RTPM.OpcAEServer and RTPM.OpcDAServer .

6.4 Configuring SQL Replication

This chapter describes how to configure SQL replication for the Shadow Server. Replication is a technology used to copy and distribute data from one database to another database. The copied data is synchronized between the two databases to maintain consistency.

The user **AMViewOnly** has permission to perform the **ViewOnly** tasks on the Shadow Asset Performance Management server. You can refer to *Appendix* in the Asset Performance Management *Configuration Guide* to know more about the **ViewOnly** tasks.

> NOTE
> The Asset Performance Management primary server (L3) acts as a Publisher and the Shadow Asset Performance Management server (L3.5) acts as a Subscriber with respect to the SQL replication.

> ATTENTION
> You must be familiar with the basics of SQL Administration concepts, SQL Scripting, and Windows Server administration concepts to configure SQL Replication.

6.4.1 Note Port Number

- Make a note of the port number which is open in the hardware firewall.

6.4.2 Update the Host File in the Publisher and Subscriber

1. Navigate to **C:\WINDOWS\system32\drivers\etc** and open the host file in notepad.
2. Type the server name and IP address of the subscriber in the publisher and the server name and IP address of the publisher in subscriber.
3. Save and close the file.

6.4.3 Create a SQL Login for the Windows User in the Publisher and Subscriber

1. Create a Windows user in the Asset Performance Management server and Shadow

   Asset Performance Management server. Ensure that the username and password are the same on both servers. The Windows user created here is used by the replication service.
2. Open **Microsoft SQL Management Studio**.
3. In the **ObjectExplorer** pane, expand **Security**.
4. Right-click **Login** and select **NewLogin**.

   The **Login-New** dialog box appears. By default, the **General** page is displayed.
5. Provide the following information:
   a. In the **Loginname** box, type the name of the Windows user that you created for the replication service.
   b. Click **Windows Authentication**.
6. From the **Select a page** pane, click **User Mapping** and perform the following steps:
   a. From the **Users mapped to this login** list, select the check box provided next to **Honeywell.MES.AssetTask.DataModel.AssetTaskDataModel**.
   b. From the **Database role membership for** the list, select the **db_owner** check box.
7. Click **OK**.

6.4.4 Configure SQL TCP Ports in the Subscriber

The following configuration steps must be performed only when the SQL replication must use a specific TCP port. We recommend using **55900** SQL TCP port.

NOTE
The certificate used must have the fully qualified domain name of the SQL Server in the Subject. (The fully qualified domain name of the SQL Server can also be used as the certificate name.)

The service account under which SQL Server is running requires permission to read the chosen certificate for the encrypted connection.

1. On the Subscriber's machine, open **SQL Server Configuration Manager**.
2. On the left pane, expand **SQL Server Network Configuration** and select **Protocols for SQL Instance used by Asset Performance Management** .
3. On the right pane, double-click **TCP/IP**.

   The **TCP/IP Properties** dialog box appears.
4. Click the **IP Addresses** tab, expand **IPAll** and in the **TCP Port** box type the port number that is used by replication.
5. Click **OK**.

6.4.5 Configure Encryption in the Subscriber

1. On the Subscriber's machine, open **SQL Server Configuration Manager**.
2. On the left pane, expand **SQL Server Network Configuration** right-click **Protocols for SQL Instance used by Asset Performance Management** , and select **Properties**.
3. Click the **Certificate** tab, and then select the certificate in the list.

   If you do not see the certificate, it must be imported. Refer to Configure the Certificate Management Console on Primary and Shadow for more information.
4. Click the **Flags** tab, and in the **Force Encryption** row select **Yes.**
5. Click **OK**.

6.4.6 Create TCP/IP Alias in the Publisher

1. On the Publisher's machine, open **SQL Server Configuration Manager**.
2. On the left pane, expand **SQL Native Client Configuration**, left-click **Aliases** and select **Alias**.

   The **Alias-New** dialog box appears.
3. Provide the information required for the **Subscriber**:
   a. Alias name: host name of the Subscriber machine.
   b. Port number: TCP port identified for SQL replication.
   c. Server name: host name of the Subscriber machine.
4. Click **OK**.

   The **TCP/IP** status is displayed as **Enabled**.

6.4.7 Configure Distribution in the Publisher

1. On the **Microsoft SQL Server Management Studio** window, right-click **Replication** and select **Configure Distribution**.

   The **Configure Distribution Wizard** page is displayed.
2. Click **Next**.

   The **Distributor** page is displayed.
3. Select the **Publisher acting as its own distributor** option and then click **Next**.

The **Snapshot Folder** page is displayed.

4. Click **Next**.
5. The **Distribution Database** page is displayed.
6. Provide the following information:
    a. In the **Distribution database name** box, type a name for the distribution database.
    b. Type the location where you want to store the distribution database file and database log file and then click **Next**.

       The **Publishers** page is displayed.
7. Select the name of the publisher you configured and then click **Next**.

   The **Wizard Action** page is displayed.
8. Select **Configure Distribution** and then click **Next**.
9. Click **Finish** to close the **Complete the Wizard** page.
10. Provide db_owner role in the distribution database for the user created in the **Create a SQL login for the Windows user in the Publisher and Subscriber** section.
    a. For providing the db_Owner role, browse to **Database > System Database > Distribution > Security > Users** and add the user created in the **Create a SQL login for the Windows user in the Publisher and Subscriber** section. Under **Schemas** and **Roles**, provide DB owner role to the user.

> NOTE
> The user must also be given permission to read and write to the backup directory specified in Create a Subscription in Publisher.

6.4.8 Create a Publication in Publisher

1. Stop AM Application server service running on the Publisher.
2. From the **Object Explorer** pane, right-click **Local Publications** and select **NewPublication**.

   The **New Publication Wizard** page is displayed.
3. Select the **Honeywell.MES.AssetTask.DataModel.AssetTaskDataModel** option for publishing.
4. From the **Publication type** list, select **TransactionalPublication**.
5. From the **Objecttopublish** a list, expand **Table** and select all the tables, except **dbo.ActiveFaultHistories** table for publishing.
6. Click **Next**.

   The **Filter Tables Rows** page is displayed.
7. Do not configure the filters. Click **Next**.
8. On the **Snapshot Agent** page, do not enter any information on when to run the snapshot.
9. Click **Next**.

   The **Agent Security** page is displayed.
10. Provide the security settings for **Snapshot Agent** and **Log Reader Agent**.

ATTENTION

- Ensure that you use the Windows username that you created in the section **Create a SQL login for the Windows user in the Publisher** and Subscriber for the snapshot agent and log reader agent.
- Ensure that you select the by impersonating the process account option in the **Snapshot Agent Security** and **Log Reader Agent Security** dialog boxes to connect to the publisher.

11. Click **Next**.

    The **Wizards Action** page is displayed.
12. Select the **Create the publication** check box and click **Next**.

    The **Complete the Wizard** page is displayed
13. In the **Publicationname** box, type a name for the publication and click **Finish**.

    The **Creating Publication** page indicates the success of the publication.
14. Click **Close** to close the **CreatingPublication** page.

    The configured database is displayed under **LocalPublications**.

6.4.9 Change the Publication Settings in the Publisher

1. On the **Microsoft SQL Server Management Studio**, select the Asset Performance Management database, and click **NewQuery**.
2. In the **SQL Query** pane, type the following script:

```
EXEC sp_changepublication
@publication = N'<PublicationName>',
@property = N'allow_initialize_from_backup',
@value = 'true'
GO
```

   In the above query, replace **<PublicationName>** with the name of the publication.
3. Run the Query
4. Expand **LocalPublication**, right-click the configured database, and select **Properties**.
5. From the **Publication Properties** dialog box, click **Publication Access List**.
6. Click **Add** to add the user created in the **Create a SQL login for the Windows user in the Publisher and Subscriber** section and click **OK**.
7. Click **OK** to close the **Publication Properties** dialog box.

6.4.10 Take a Backup of the Asset Performance Management Server Database

1. From the **Object Explorer** pane, expand **Databases**.
2. Right-click the database **Honeywell.MES.AssetTask.DataModel.AssetTaskDataModel** and choose **Tasks> Backup**.

   The **Backup Database** page is displayed.
3. Provide the required information and then click **OK**.

   A message indicating the success of the database backup is displayed.

6.4.11 Restore the Database on the Shadow Server

1. Open **SQL Server Management Studio**.
2. From the **Object Explorer** pane, expand **Databases**.
3. Right-click the database you created and choose **Tasks> Restore > Database**.

   The **Restore Database** page is displayed.
4. Provide the information of Shadow Asset Performance Management server and then click **OK**.

   A message indicating the success of the database restoration is displayed.

> **NOTE**
> When you restore the database on the Shadow Asset Performance Management server, the details of the Windows user in the **AssetTaskDataModel** database is replaced by the details of the Asset Performance Management server (Publisher) user. Therefore, it is necessary to add the details of the Windows user created in Shadow Asset Performance Management server to **AssetTaskDataModel** database again and provide db_owner permission after the restore.

6.4.12 Disable the CAM Archive SQL Job in Subscriber

1. From the **Object Explorer** pane, expand **SQL Server Agent > Jobs**.
2. Right-click the **CAM_Archive_AuditTrail** job and then select **Disable**.

6.4.13 Create a Subscription in Publisher

> **ATTENTION**
> - Ensure that the Windows user created in the Shadow Asset Performance Management server for replication use has the db_owner role for the **Honeywell.MES.AssetTask.DataModel.AssetTaskDataModel** database of Shadow Asset Performance Management server. For more information on creating a Windows user refer to, Create a SQL login for the Windows user in the Publisher and Subscriber.
> - Ensure the user has access to the backup directory.
> - Ensure that the SQL server agent service is running before you run the script.

1. On the **Microsoft SQL Server Management Studio**, select theAsset Performance Management database, and then click **NewQuery.**
2. On the **SQL Query** pane, type the following script:

```
EXEC sp_addsubscription
@publication = N'<PublicationName>',
@subscriber = N'<SubscriberServerName>',
@destination_db = N'Honeywell.MES.AssetTask.DataModel.AssetTaskDataModel',
@subscription_type = N'Push',
@sync_type = N'initialize with backup',
@backupdevicetype = N'disk',
@backupdevicename = N'<backup file location and name>',
@article = N'all',
@update_mode = N'read only',
@subscriber_type = 0
```

```
EXEC sp_addpushsubscription_agent
@publication = N'<PublicationName>',
@subscriber = N'<SubscriberServerName>',
@subscriber_db = N'Honeywell.MES.AssetTask.DataModel.AssetTaskDataModel',
@job_login = '<domain\user create in step 2>',
@job_password = '<Password of the user created in step 2>',
@subscriber_security_mode = 1,
@subscriber_login = null,
@subscriber_password = null,
@frequency_type = 64,
@frequency_interval = 0,
@frequency_relative_interval = 0,
@frequency_recurrence_factor = 0,
@frequency_subday = 0,
@frequency_subday_interval = 0,
@active_start_time_of_day = 0,
@active_end_time_of_day = 235959,
@active_start_date = 0,
@active_end_date = 99991231,
@enabled_for_syncmgr = N'False',
@dts_package_location = N'Distributor'
GO
```

3. In the above query, replace the following elements.
   - <**PublicationName**> - The name of the Publisher
   - <**SubscriberServerName**> - Server name of the Subscriber
   - <**backup file location and name**> - Name and the location of the database used while creating a backup for the database. For example, "BackupDatabase.bak"
   - <**domain\user create in step 2**> - SQL login for the Windows user created for publisher and subscriber
   - <**Password of the user created in step 2**> - Password of the Windows user created for publisher and subscriber
4. Click **Execute**.

> **ATTENTION**
> Refer to the Asset Performance Management Troubleshooting guide if you encounter any errors while performing replication.

6.4.14 Run the Query

1. Open **Microsoft SQL Management Studio** and run the below query:

```
GO
EXEC sp_configure 'show advanced options', 1;
RECONFIGURE;
GO
```

```
EXEC sp_configure 'max text repl size', -1;
GO
RECONFIGURE; GO
```

2. Click **Execute**.

6.4.15 Copying Asset Links and Asset Icons

When certain links, documents, and icons are uploaded to theAsset Performance Management server after performing the replication, the data does not sync automatically between theAsset Performance Management server database and Shadow Asset Performance Management server databases.

You should manually copy the links, documents, and icons from Asset Performance Management server to the Shadow Asset Performance Management server. The links, documents, and icons must be copied to the following location **<MES installed path>\Honeywell\MES\ Core\WWWMESCore\AMInformationFolders** on the Shadow Asset Performance Management server.

6.4.16 Verifying the SQL Replication

1. On the **Microsoft SQL Server Management Studio** window, expand **Local Subscription**.
2. Right-click the created subscription and select **View Synchronization Status**.

   The **View Synchronization Status** dialog box is displayed. You can view the status of the synchronization.

CHAPTER

7 PRE-INSTALL FOR CLIENTS

This chapter describes how to install the Asset Performance Management prerequisites on the server and the client computers.

7.1 Prerequisite Installation

Installing Dynamic Content Compression Roles

1. **Start** > **Server Manager**.

   The **Server Manager** window is displayed.
2. On the left pane, expand **Roles > Web Server**.

   The **Summary** page is displayed on the right pane.
3. Click the **Add Role Services** link.

   The **Add Role Services** dialog box is displayed.
4. In the **Roles** list, expand **Web Server > Performance** and select the **Dynamic Content Compression** check box.
5. Click **Install**.

   The **Dynamic Content Compression** roles are installed and displayed in the **RolesServices** list.

Install ULM

Prerequisite

1. Stop all Asset Performance Management services.
2. Uninstall the previous version of ULM that was installed along with the Honeywell Intuition Core System installation: **Start** > **Control Panel** > **Programs** > **Programs and Features**.
3. Remove the following components:
   - Honeywell ULM Admin Tools
   - Honeywell ULM License Server
   - ULM License Client

To install ULM 6.6

1. On the installation media, browse to the **ISSetupPrerequisites** folder.
2. To install **Honeywell ULM Admin Tools**, perform the following steps:
   a. Browse to **ULM 6.6 > Admin Tools** and double-click **Honeywell ULM Admin Tools**. The **Welcome** page appears.
   b. Read through the welcome information, details, and warnings and click **Next**. The **CustomerInformation** page appears.
   c. Type the user name and the name of your organization and then click **Next**. The **Destination Folder** page

appears.

d. Click **Next** to install ULM on the default location. The **Ready to Install the Program** page appears.

e. Click **Install**.

f. Click **Finish**.

The **Honeywell ULM Admin Tools** component is installed.

3. To install **Honeywell ULM License Server**, perform the following steps:

 a. Browse to **ULM 6.6 > Server** and double-click **Honeywell ULM License Server**. The **Welcome** page appears.

 b. Read through the welcome information, details, and warnings, and then click **Next**.

 The **Ready to Install the Program** page appears.

 c. Click **Install**.

 d. Click **Finish**.

 The **Honeywell ULM License Server** component is installed.

7.2 Installing Report Server

We recommend installing the report server on the Database Server. For instructions on installing a report server, refer to the MSDN website.

Configuring Report Server for a Default Port

1. **Start** > **All Apps** > **Reporting Services Configuration Manager**.
2. The **Reporting Services Configuration Connection** dialog box appears. Provide the information to configure the report server instance.
 a. In the **Server Name** box, the name of the Asset Performance Managementserver is displayed by default. If the Asset Performance Management server name is not displayed, click **Find** and select the Asset Performance Management server.
 b. The default instance name is displayed in the **Report Server Instance** box. It is recommended not to change the value in this field.
 c. Click **Connect**.
3. The **Report Server Status** page appears.
4. On the left pane, click **Web Service URL**.

 The **Web Service URL** dialog box appears.
5. Do all of the following:
 a. In the **Report Server Web Service Virtual Directory** section, type the virtual directory for the Web Service URL. Ensure that the virtual directory for Web Service URL is **ReportServer**.
 b. In the **TCP Port** box, the default port number where Report Server is installed is displayed as 80 by default.
 c. Click **Apply**.
6. On the left pane, click **Web Portal URL**.

 The **Web Portal URL** page appears.

> **NOTE**
> Ensure that the virtual directory for the Web Portal URL is Reports.

7. Perform the following steps:
    a. Click **Advanced**. By default, the TCP port number is displayed as 80. If the port number is other than 80, click **Edit** to change the port number to 80.
    b. The **Edit a Web Portal HTTP URL** dialog box appears. In the **TCP Port** box, type the default port number as 80.
    c. Click **OK**.
8. On the left pane, click **Database** to configure the report server database.
9. The **Report Server Database** page appears. In the **Current Report Server Database** section, ensure the following:
    - The SQL server name displayed is the name of the server where you want to install Asset Performance Management.
    - Always ensure that the database name is displayed as **ReportServer**. If the database name displayed is different other than **ReportServer** then, click **ChangeDatabase** to change the database.
10. The **Change Database** page appears. Select the appropriate option and then click **Next**.
11. The following page appears. Provide the following information to connect to the database server:
    a. In the **Server Name** box, type the name of the SQL server where the Report Server is installed.
    b. Click the **Authentication Type** drop-down list and select **Current User- Integrated Security**.
    c. Click **Test Connection**.
    A message is displayed if the connection is successful,
    d. Click **Next**.
12. The following page appears. In the **Database Name** box, the name of the database appears by default as **ReportServer**.
    Click **Next** to proceed to the next screen.
13. Under **Report Server Mode**, ensure that you select the **Native Node** option.
14. Click **Next**.
15. The Credentials page appears. Click the **Authentication Type** drop-down list and select **Service Credentials** and then click **Next**.
    The information about the existing report server database is displayed.
16. Click **Next**.
17. After successful configuration of the database, click **Finish** and **Exit**.
    The Report Server is configured to the Report Server database.

Configuring Report Server for a non-default port

If MES is installed on a non-default port, you must make the following changes in the Report Service Configuration Manager page to access reports from Asset Performance Management.

1. **Start** > **All Apps** > **Reporting Services Configuration Manager**.
2. The **Reporting Services Configuration Connection** dialog box appears. In the **ReportServices Configuration Connection** dialog box, provide the information to configure the report server instance.
   a. In the **Server Name** box, the name of the Asset Performance Management server is displayed by default. If the Asset Performance Management server name is not displayed, click **Find** and select the Asset Performance Management server.
   b. By default, MSSQLSERVER instance name is displayed in the Report Server Instance textbox. It is recommended not to change the value in this field.
   c. Click **Connect**.
3. The **Report Server Status** page appears. On the left pane, click **Web Service URL**.
   a. In the **Report Server Web Service Virtual Directory** section, type the virtual directory for the Web Service URL. Ensure that the virtual directory for Web Service URL is **ReportServer**.
   b. By default, the **TCP Port** number is displayed as 80.
   c. Click **Advanced** to add a new port number.
   d. In the **Advanced Multiple Website Configuration** dialog box, click **Add** to add the port number.
      The **Add a Report Server HTTP URL** dialog box appears.
   e. Click the **IP Address** drop-down list and select **All Assigned (Recommended)**.
   f. In the **TCP Port** box, type the non-default website port number. For example, 8001.
   g. Click **OK**.
   h. The IP Address and the port number appear in the **Advanced Multiple Website Configuration** dialog box. Click **OK** and then click **Apply**.
4. On the left pane, click **Report Manager URL**
   The Report Manager URL page appears.

> **NOTE**
> Ensure that the virtual directory for Report Manager URL is Reports.

   a. Click **Advanced** to add the new port number.
   b. In the **Advanced Multiple Website Configuration** dialog box, click **Add** to add the port number.
      The **Edit a Report Manager HTTP URL** dialog box appears.
   c. Click the **IP Address** drop-down list and select **All Assigned**.
   d. In the **TCP Port** box, type the non-default website port number. For example, 8001.
   e. Click **OK**.
   f. The IP Address and the port number appear in the **Advanced Multiple WebsiteConfiguration** dialog box. Click **OK** and then click **Apply**.

Checking if Report Server is Installed

Check if Reporting Services Configuration Manager option is available:

- **Start > All Apps > Reporting Services Configuration Manager.**

If the option is available, the Report server is installed.

Checking if Report Server is Configured

After successful installation of the report server, access the following url **http://[MACHINENAME]/Reports** to open the **Report Manager** home page (where **MACHINENAME** is the computer name where you want to install Asset Performance Management). If an error message is displayed in the Report Manager home page, the Report server is not configured correctly.

Adding a User to the System Administration Role

The prerequisite installer automatically adds the installation user to the System Administrator role in Intuition. This role is required to complete the Asset Performance Management installation.

If the same user cannot be used for both installations, ensure the installation user is added to the System Administrator role before beginning the Asset Performance Management installation.

Add users to the System Administrators role

1. Open the **Intuition Landing page** application from the server: **Start** > **Honeywell** > **Home**), and open the Security page (**System > System Configuration > Security**).

   The **Configure Security** page appears.
2. From the **Permissions** pane, select the System Administrator role.
3. In the **Role Members** pane, in the **Windows User or Group Name** box, type the name of the Asset Performance Management installation user.
4. Click  .

   The user is now associated with the role.

CHAPTER

8 CLIENT INSTALLATION

This chapter describes the procedure for installing different client components in Asset Performance Management.

> **ATTENTION**
> If there is an installation failure, you need to perform password encryption before resuming the installation, for more information refer to Password Encryption.

> **ATTENTION**
> If the system prompts for a reboot during installation, click **No** and proceed with Installation.
>
> If you want to proceed with the reboot, click **Yes**, and perform the following steps
>
> 1. Mount the media once the reboot is completed.
> 2. Perform password encryption, for more information refer to Password Encryption.
> 3. Restart the installation.
>
>   The installation will resume from the point before reboot.

8.1 Verify the Connection to the APM Server

1. If the Asset Performance Management is installed on the default port, type **http://[WebServerFullyQualifiedDomainName]>/MES/Default.aspx** in the URL of the Web Browser. The default port number is **80**.

   OR

   If the Asset Performance Management application is installed on a different port, type **http://[WebServerFullyQualifiedDomainName]:[PortNumber]/MES/Default.aspx** in the URL of the Web Browser.
2. After providing the administrator username and password, verify if the Asset Performance Management home page appears. If it does not appear, it indicates that the Asset Performance Management installation was not successful. Contact technical support for help troubleshooting the installation.

8.2 Installing Model Tools

> **NOTE**
> - Verify if you have administrator privileges on the client
> - Verify if you have system administrator privileges for the database on which the Asset Performance Management database will reside

1. On the installation media, right-click **setup.exe**, and then select **Run as Administrator**.
2. In the RabbitMQ prompt, click **No**.
3. After you have read through the welcome information, click **Next**.
4. Read through the license agreement and click **I accept the terms in the license agreement** if you agree to the license terms. Click **Next**.
5. Do all of the following:
    a. Ensure all the following are cleared:
        - Application components
        - Web components
        - Database components
        - Reports
        - Calculation components

    b. Select **Client Components**.
    c. Click **Next**.
6. Follow the below steps depending on the component to be installed:

<table>
<tr><th>Component Name</th><th>Steps</th></tr>
<tr><td>Model Tool</td><td>NOTE
To use Import Thermo Configuration Tool, it must be installed and run on the Application Server. Other APM Configuration Tools can be used when it is installed on a client computer.

1. Leave Model Tool selected.
2. Clear Bulk Configuration and PHD Utility.
3. Select Asset Analytics and Influxdb if you want the Asset Analytics installed and click Next.
4. (Optional) To install to a non-default location, in the Installation Path area, click Browse and then select the new location.
5. (Optional) To change the MES Data Files location, in the Data Files Path area, click Browse and then select the location in which to save MES Data Files.
6. Click Next.

NOTE
Once the model tool installation is completed, refer to Updating Model Tool Configuration Files.</td></tr>
<tr><td>APM Configuration Excel Plugin</td><td>1. Leave Bulk Configuration selected.
2. Clear Model Tools and PHD Utility.
3. Select Asset Analytics and Influxdb if you want the Asset Analytics installed and click Next.
4. (Optional) To install to a non-default location, in the Installation Path area, click Browse and then select the new location.</td></tr>
</table>

	5. (Optional) To change the MES Data Files location, in the **Data Files Path** area, click **Browse** and then selection the location in which to save MES Data Files. 6. Click **Next**. **NOTE** Ensure you are added to the reliability engineers group in the installed server, refer to Adding User to Reliability Engineers Group on the Server for more information. **ATTENTION** If any Asset Performance Management components are already installed, they must be uninstalled and reinstalled with the APM Configuration Excel Plugin.
PHD Components	1. Leave **PHD Utility** selected. 2. Clear **Bulk Configuration** and **Model Tools**. 3. Select Asset Analytics and Influxdb if you want the Asset Analytics installed and click **Next**. 4. (Optional) To install to a non-default location, in the **Installation Path** area, click **Browse** and then select the new location. 5. (Optional) To change the MES Data Files location, in the **Data Files Path** area, click **Browse** and then select the location in which to save MES Data Files. f.Click **Next**.

7. Do all of the following:
    a. In the **Web Server Host Name** box, type the name of the Web Server.

    > **ATTENTION**
    > If you are using https, enter the fully qualified domain name of the Web Server.

    b. In the **Web Server HTTP Port No** box, type the port number used on the Web Server.
    c. In the **Web Server Server Protocol Name** box, select the protocol (**http** or **https**) used on the Web Server.
    d. Click **Next**.
8. Do all of the following:
    a. In the **Application Server Host Name** box, type the name of the Application Server.
    b. In the **Application Server Port Number** box, type the net.tcp port number used on the Application Server.
    c. From the **Application Server Protocol Name**, select the protocol used on the Application Server (**http** or **https**).
    d. In the **APM DB Server Nam**e box, type the name of the Database Server.
    e. In the **APM Calc Server Name** box, type the name of the Calculation Server.
    f. Click **Next**.
9. Click **Install**.
10. If the **Microsoft Visual Studio Tools for Office Runtime** notification appears, click **Accept** to accept the license agreement and continue with the installation.

11. Click **Finish**.

    A message appears prompting you to restart the computer.
12. Click **Yes** to restart.

> **NOTE**
> To import Thermo definitions from UniSim, UniSim Design software must be installed on the APM client computer. Refer to UniSim Design Suite for instructions.

8.3 Updating Model Tool Configuration Files

Follow the below steps to update the model tool configuration files:

1. To update the APM Configuration tool configuration file, do all of the following:
   a. Open the following file:

      [InstallDrive]:\Program Files (x86)\Honeywell\MES\AssetManager\Tools\SentinelConfiguration\SentinelConfiguration.exe.config
   b. Locate the <appSettings> entry, and in the ServerLocation value, replace localhost with the name of the Asset Performance Management server.

      For a split server, use the name of the Application Server.
   c. Locate the <system.serviceModel><client><endpoint> entry, in the address, replace localhost with the name of the Asset Performance Management server.

      For a split server, use the name of the Application Server.
   d. Save and close the file.

2. To update the Model Tool configuration file, do all of the following:
   a. Open the following file:

      <InstallDrive>:\Program Files (x86)\Honeywell\MES\AssetManager\Tools\AMImportExport\ImportModels.exe.config
   b. Locate the <appSettings> entry, and in the RTData, RTConfigurationEqp, AssetImagesPath values, replace localhost with the name of the Asset Performance Management server.
   c. Locate the <connectionString> entry, and in the AssetTaskDataModelEntities connection string, replace localhost with the name of the Asset Performance Management server.

      For a split server, use the name of the Database Server.

3. Save and close the file.

8.4 Adding User to Reliability Engineers Group on the Server

1. **Start** > **Server Manager**.
2. On the server tree, choose **Configuration** > **Local Users and Groups** > **Groups**.
3. Right-click the **Reliability Engineers** group, and click **Add to Group**.
4. In the **Reliability Engineers Properties** dialog box, click **Add**.

5. Enter the names of all of the following users:
    a. The services (runtime) user.

    > **ATTENTION**
    > If the service user is not added to the Reliability Engineers group, some features will not function as expected.

    b. The install user.
    c. All the users who will need to use the APM Configuration Excel Plugin and the Asset Performance Management Configuration.
6. Click **OK**.

    The user names appear in the group list.
7. Click **OK** to close the Reliability Engineers Properties dialog box.

8.5 Post Installation Considerations for Split Servers

If the client computer is connected to a Asset Performance Management system installed in a split server arrangement, some changes need to be made to some of the configuration files.

To update the client tool configuration files for the split server

1. If any of the following tools are installed, update the listed configuration files to use the name of the Database Server in place of localhost:
    - Model Tool:

      <Install Location>:\Program Files (x86)\Honeywell\MES\AssetManager\Tools\AMImportExport\ImportModels.exe.config
    - APM Configuration Tool:

      <Install Location>:\Program Files (x86)\Honeywell\MES\AssetManager\Tools\SentinelConfiguration\SentinelConfiguration.exe.config

CHAPTER

9 CHANGE SERVER SETTINGS

This section covers the following procedures related to renaming the Asset Performance Management servers.

9.1 Changing the Services (Runtime) Account

This section covers the following procedures related to changing the services user for Asset Performance Management.

9.1.1 Copy the Required Files

1. Copy the contents (files and folder) of the following folder on the installation media:
   <installationmedia>\Installer\RuntimeAccountChangeMaster
2. Do all of the following (in a multi-site installation, perform these steps on both the Application and Web Servers):
   a. Navigate to the following folder:
      <InstallPath>\Honeywell\MES\Tools\RuntimeAccountChange
   b. At that location, create a folder called **UASAccountRename**.
   c. In the UASAccountRename folder, paste the files copied in Step 1.
      The UASAccountRename folder should now contain the following:
      - RunTimeAccountChange folder (and contents)
      - ST_Replace-UserAccount file
      - ST_UserAccount-Replace file

9.1.2 Prerequisites

Following prerequisites are required to complete the renaming procedures:

- The old name of all servers where Asset Performance Management is installed.
- The services account user name and password.
- SQL Server Management Studio is installed on the database server.
- Administrator rights on the Asset Performance Management SQL instance.
- The PowerShell SqlServer module is installed on all servers.
- Permission to create certificates or to access the already created one.

9.1.3 Add New Services User to the MES Services Role

> **ATTENTION**
> The new services account must be added to the MES Services role before it can be used.

1. Open the **Intuition Landing page** application from the server: **Start** > **Honeywell** > **Home**), and open the Security page (**System > System Configuration > Security**).

   The **Configure Security** page appears.
2. From the **Permissions** pane, select the **Services** role.
3. In **Role Members** pane, in the **Windows User or Group Name** box, type the name of the Asset Performance Management installation user.
4. Click .

9.1.4 Change the Services User on the Application Server

1. Add the new services user to the following Windows groups:
   - MESDBUsers
   - Product Administrators
2. Use **Run as Administrator** to open PowerShell.
3. In PowerShell, change the directory to the local UASAccountRename folder created in Copy the Required Files.

   <InstallPath>\Honeywell\MES\Tools\RuntimeAccountChange
4. Run the **ST_Replace-UserAccount.ps1** PowerShell script in the UASAccountChange folder using the following command:

   ```
   .\UASAccountRename\ST_Replace-UserAccount.ps1 OldServicesAccount
   NewServicesAccount
   ```

   where OldServicesAccount is the old services user and NewServicesAccount is the new services user.
5. Run the **Replace-UserAccount.ps1** PowerShell script using the following command:

   ```
   .\Replace-UserAccount.ps1 OldServicesAccount NewServicesAccount
   ```

   where OldServicesAccount is the old services user and NewServicesAccount is the new services user.
6. **Start** > **All Apps** > **Internet Information Services (IIS) Manager**

   The **Internet Information Services (IIS) Manager** window appears.
7. On the left pane, expand the server and select **Application Pool**.

   The Application Pools pane is displayed.
8. Do all of the following to change the update the NotificationConfig4.0 application pool to the new user:
   a. Right-click the NotificationConfig4.0 application pool, and from the menu select **Advanced Settings**.
   b. In the **Advanced Settings** dialog box, in the **Process Model** > **Identity** box, type or select the new runtime user, and then click **OK**.
9. Do all of the following to change the update the **APM DASInstance** application pool to the new user:
   a. Right-click the SentinelDASInstance application pool, and from the menu select **Advanced Settings**.
   b. In the **Advanced Settings** dialog box, in the **Process Model** > **Identity** box, type or select the new runtime user, and then click **OK**.
10. Perform an **IISRESET**.

> ATTENTION
> If the Asset Performance Management home page does not load after changing the services user, see Solution 4 for **Asset Performance Management Home Page Is Not Displayed** in the Asset Performance Management Troubleshooting Guide.

9.1.5 Change the Services User on the Web Server

1. Add the new services user to the following Windows groups:
    - MESDBUsers
    - Product Administrators
2. Use **Run as Administrator** to open PowerShell.
3. In PowerShell, change directory to the local UASAccountRename folder created in Copy the Required Files.

    <InstallPath>\Honeywell\MES\Tools\RuntimeAccountChange
4. Run the **ST_Replace-UserAccount.ps1** PowerShell script in the UASAccountChange folder using the following command:

    ```
    .\UASAccountRename\ST_Replace-UserAccount.ps1 OldServicesAccount
    NewServicesAccount
    ```

    where **OldServicesAccount** is the old services user and NewServicesAccount is the new services user.
5. Run the **Replace-UserAccount.ps1** PowerShell script using the following command:

    ```
    .\Replace-UserAccount.ps1 OldServicesAccount NewServicesAccount
    ```

    where **OldServicesAccount** is the old services user and NewServicesAccount is the new services user.
6. **Start** > **All Apps** > **Internet Information Services (IIS) Manager**

    The **Internet Information Services (IIS) Manager** window appears.
7. On the left pane, expand the server and select **Application Pool**.

    The **Application Pools** pane is displayed.
8. Do all of the following to change the update the **NotificationConfig4.0** application pool to the new user:
    a. Right-click the **NotificationConfig4.0** application pool, and from the menu select **Advanced Settings**.
    b. In the **Advanced Settings** dialog box, in the **Process Model** > **Identity** box, type or select the new runtime user, and then click **OK**.
9. Do all of the following to change the update the **SentinelDASInstance** application pool to the new user:
    a. Right-click the **SentinelDASInstance** application pool, and from the menu select **Advanced Settings**.
    b. In the **Advanced Settings** dialog box, in the **Process Model** > **Identity** box, type or select the new runtime user, and then click **OK**.
10. Perform an **IISRESET**.

> **ATTENTION**
> If the Asset Performance Management home page does not load after changing the services user, refer to Asset Performance Management **Home Page Is Not Displayed** in the Asset Performance Management Troubleshooting Guide.

CHAPTER

10 UNINSTALL ASSET PERFORMANCE MANAGEMENT

This section explains the procedure to uninstall Asset Performance Management server.

10.1 Remove Asset Performance Management from a Server Computer

NOTE
- Ensure that you log on to the computer with administrator privileges.
- Ensure the Asset Performance Management installation media is available in an accessible location.

1. **Start** > **Control Panel** > **Programs** > **Programs and Features.**

   The **Programs and Features** dialog box appears listing the software that is installed on your computer.
2. Right-click and select **Uninstall** for all of the following (when prompted with a confirmation dialog, select **Yes**):
   - **Honeywell Calculation Host Setup**
   - **Honeywell Asset Performance Management Data Source Plugins**
   - **Honeywell Intuition Data Access Service - SentinelDASInstance**
   - **Asset Performance Management Excel Addin**
   - **Asset Performance Management**

     A dialog box appears to confirm the system reboot.
3. Click **OK**.

10.2 Remove Asset Performance Management from a Client Computer

ATTENTION
- Ensure that you log on to the computer with administrator privileges.
- Ensure that you remove the Asset Performance Management client on the client computers that has the Experion Asset Synchronizer or APM Configuration Excel Plugin.

1. **Start** > **Control Panel** > **Programs** > **Programs and Features**

   The **Programs and Features** dialog box appears listing the software that is installed on your computer.
2. Right-click Asset Performance Management and select **Change**.

   The **Welcome** page appears.
3. Click **Next**.

   The **Program Maintenance** page appears.
4. Click **Remove**.

The **Programs and Features** dialog box appears asking you if you want to retain the database.

5. Click **Yes** to retain the database, else **No** to remove the database.

   A confirmation dialog box appears.
6. Click **Yes** to remove Asset Performance Management.

   A dialog box appears to confirm the system reboot.
7. Click **OK**.

   The Asset Performance Management is removed.

10.3 Removing ULM

The uninstallation of the Asset Performance Management server does not uninstall the ULM components. You must manually uninstall the installed version of ULM and its components from your computer.

To remove Honeywell ULM Admin Tools

1. **Start > Control Panel > Programs > Programs and Features**.

   The **Programs and Features** dialog box appears listing the software that is installed on your computer.
2. Right-click **Honeywell ULM Admin Tools** and select **Uninstall**.

   A confirmation dialog box appears.
3. Click **Yes**.

   The **Honeywell ULM Admin Tools** software is removed.

To remove Honeywell ULM License Server

1. **Start > Control Panel > Programs > Programs and Features**.

   The **Programs and Features** dialog box appears listing the software that is installed on your computer.
2. Right-click **Honeywell ULM License Server** and select **Uninstall**.

   A confirmation dialog box appears.
3. Click **Yes**.

   The Honeywell ULM License Server software is removed.

CHAPTER

11 Licensing Licensing Asset Performance Management

This chapter describes the procedure for setting up the licenses in the Asset Performance Management application.

The Asset Performance Management application uses UniSim License Manager (ULM) to generate and validate user licenses.

The ULM software is installed along with the typical installation of Honeywell Intuition Core System. You can also upgrade previous versions to version ULM 6.6. The setup for installing ULM is available on the Asset Performance Management media.

If you encounter any licensing related issues, contact your Honeywell representative.

> **NOTE**
> After you install and configure Asset Performance Management, if you are unable to view the licenses in the Current User tab of the Admin Console Application screen, it is recommended that you upgrade ULM to version 6.24. For more information on how to install ULM, refer to Install ULM.

The following are the characteristics of the user license.

- **Runtime**: A license is required to start the Asset Performance Management application successfully.

  You do not require a license to install the application. However, the license is essential to run and use Asset Performance Management.
- **Subscription-based**: Issued licenses do not typically have a term limit. There may be exceptions where the software is provided for a trial period and the delivered license includes a date of expiry after which it must be renewed for you to continue using Asset Performance Management.
- **Hardware-bound**: A license is issued based on the hardware characteristics of the computer for which the license is intended. Hence, once issued, the license cannot be installed on another computer.

> **NOTE**
> Contact Honeywell to arrange for a new license to be issued.

11.1 Asset Performance Management Licenses

The following are the Asset Performance Management licensing features:

- **Base License:** Master license to run the application.
- **User Calcs:** License for calculations associated with an asset. The license count increases for every asset with custom calculations.
- **Rule Only Model:** License for Fault Symptoms associated with an asset. The license count increases for every asset with fault/symptom rules.

- **Heat Exchanger:** License for the in-built heat exchanger models. The license count increases for every asset with a Heat Exchanger model.
- **Centrifugal Compressor:** License for the in-built centrifugal compressor models. The license count increases for every stage of the Centrifugal Compressor (for example, a 2-stage compressor is 2 licenses).
- **Steam Turbine:** License for the in-built steam turbine models. The license count increases for every asset with a Steam Turbine model.
- **Gas Turbine:** License for the in-built gas turbine models. The license count increases for every asset with a Gas Turbine model.
- **UniSim Link:** License for enabling the UniSim Design (USD) external data source.
- **Standard Library:** License for enabling the Standard Library models (for example, pumps, blowers, motors, and boilers).
- **Thermo Library:** License for the thermo engine calculations.
- **Extended Models**: License for the Custom Functions feature. This allows you to create custom functions (in C# code) for use in calculations.
- **CMMS Notification:** License for CMMS notification feature to generate XML notifications for integration with business systems.

> NOTE
> If migrating from **Experion** synced assets to **Field Device Manager (FDM)** synced assets, you have to procure new calculation licenses for FDM synced assets since each instrumentation asset will consume user calculation as well as rule only model licenses.

11.2 Confirm the ULM Components are Installed

The ULM components are included in the Honeywell Intuition Core System installer setup and are installed automatically when you install the Honeywell Intuition Core System application.

To view the list of installed ULM components, do the following:

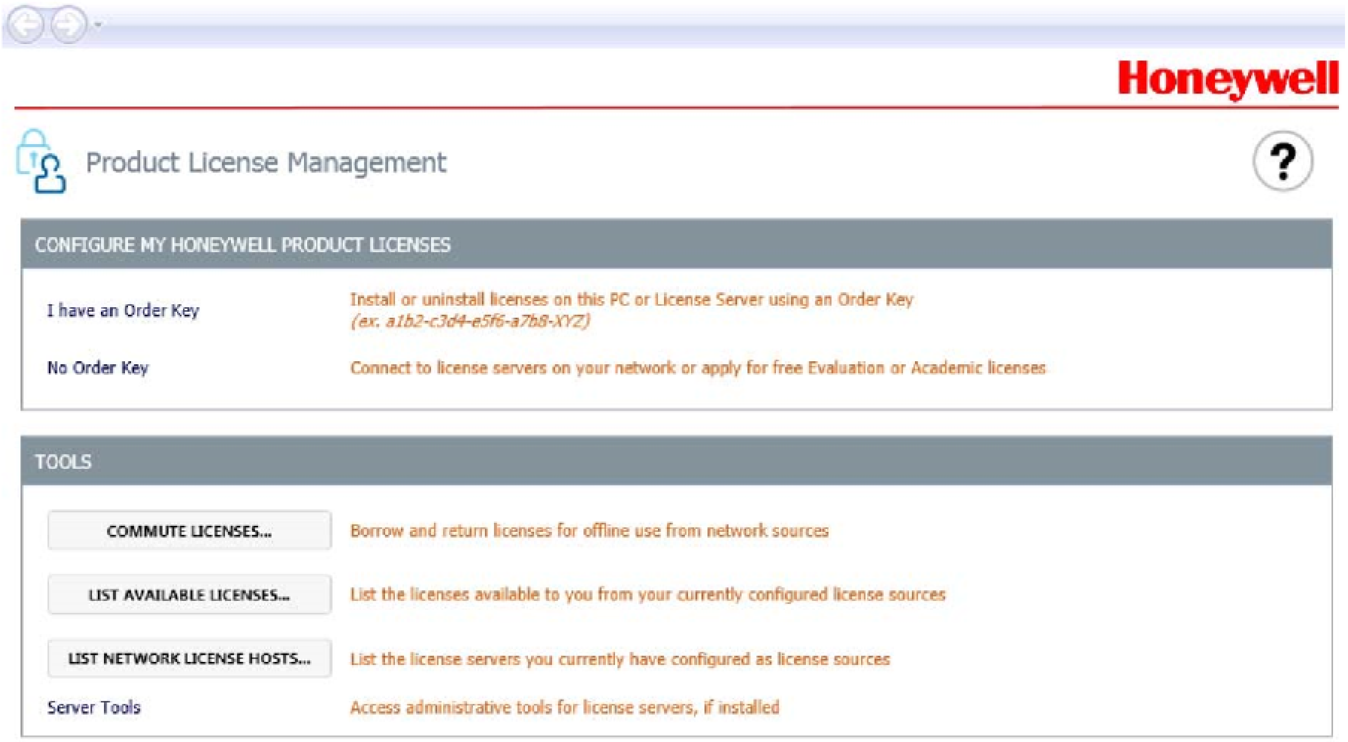
- **Start** > **All Apps** > **ULM Config Wizard**.

11.3 Generate and Mailing the Host ID File

To request a license file through e-mail, you must generate the Host ID file containing the hardware details that identify the computer on which the license is generated.

1. **Start** > **All Apps** > **ULM Config Wizard**.

   The **ULM Configuration Wizard**



2. Click **I have an Order Key**, enter the order key, and then click **Next**.

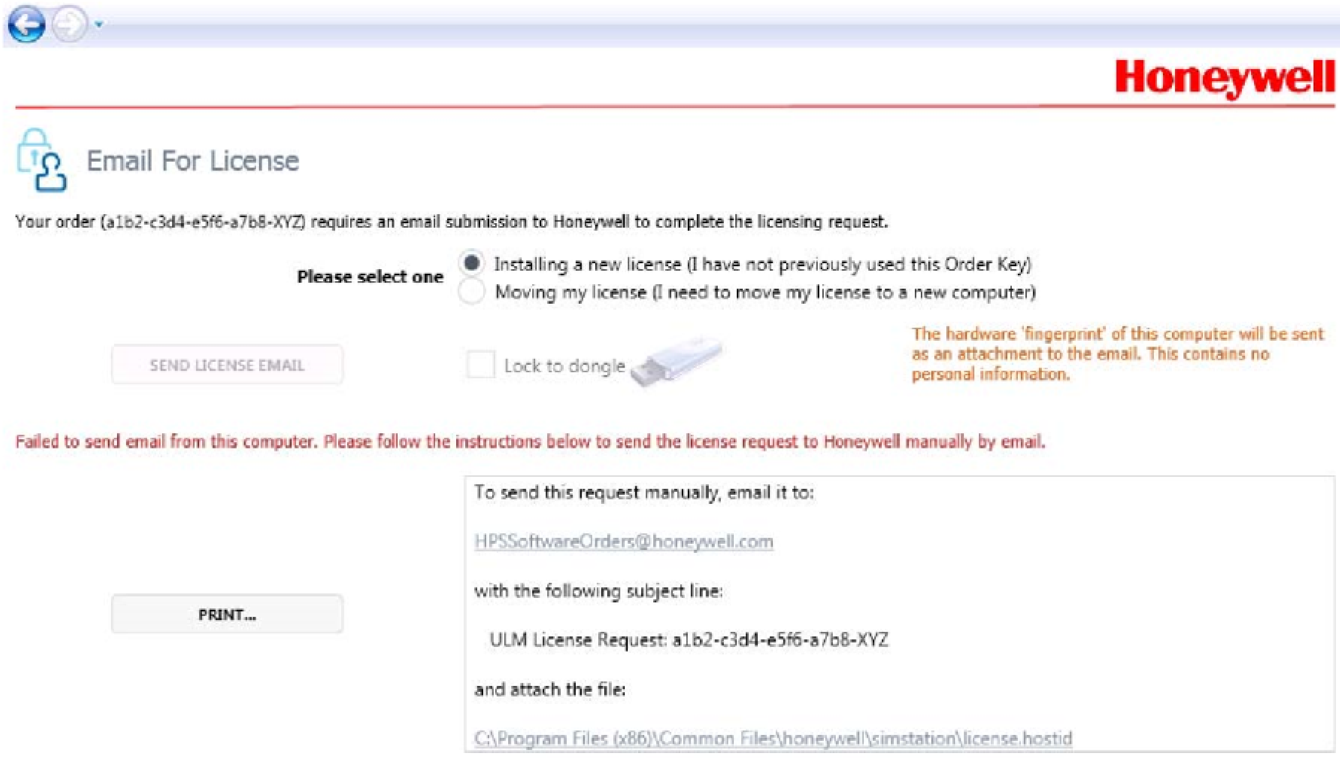
   The **Email For License** page appears.

> **ATTENTION**
> If you do not have the order key, contact Honeywell support.

3. Select one of the following options:
   - Select **Installing a new license** if this order key has never been used.
   - Select **Moving my license** if this order key was previously used on a different computer and needs to be moved.

4. Click **Send License Email**.

   If the ULM Configuration Wizard can send the e-mail, the details of the current computer are automatically sent to the Asset Performance Management Licensing team (HPS Software Orders (AZ16) "ACTHPSLicense@honeywell.com") from your account.



5. If e-mails send fails, the license details are displayed. Manually send an e-mail from a computer with an Internet connection with the following information as shown in the wizard:
    - To: ACTHPSLicense@honeywell.com
    - Subject: ULM License Request: [OrderKey]
    - Attachment: The Host ID file. Click the link shown in the ULM Configuration Wizard. By default, the Host ID file is saved as **[InstallDrive]:\Program Files\Common Files\honeywell\simstation\license.hostid**.

11.4 Install the License

After receiving the e-mail with the Host ID file and the client key, the Asset Performance Management Licensing Team generates a license and sends it back to you through e-mail.

You must copy the license received from the Asset Performance Management Licensing Team onto your server computer. Double-click the license file to install it on your computer.

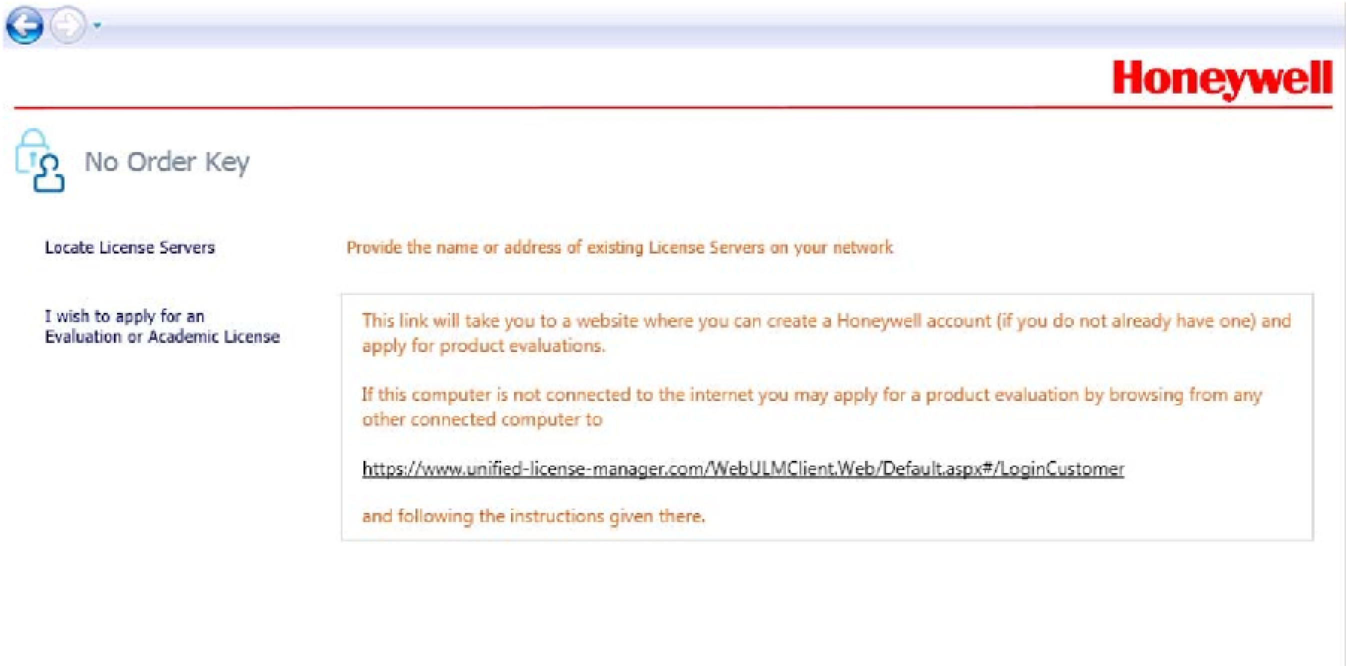
> **ATTENTION**
> Ensure that you install the license received from the Asset Performance Management Licensing Team on the computer from which the Host ID file was generated.

11.5 Request the License From the Network License Server

If your network has installed and configured a ULM Network License Server, you can request a license from the server.

1. **Start** > **All Apps** > **ULM Config Wizard**.

   The ULM Configuration Wizard dialog box appears.
2. Click **No Order Key**, and click **Next**.



3. Click **Locate License Servers**.
4. In the **New Server Name** box, enter the IP Address of the network license server, and then click **Add**.

5. Click **Commit**.

CHAPTER

12 Localization

You can localize the Asset Performance Management application for certain languages or cultures.

A culture indicates a unique combination of language and location. The translation is performed for each culture in a .NET application, the naming convention followed for representing a culture is as given below:

aa-BB

- **aa** indicates the language; for example, "ru" for Russian
- **BB** indicates the location; for example, "RU" for Russia

Asset Performance Management uses the Intuition Localization Tool to assist in the localization process. The Intuition Localization Tool provides both neutral (ru) and locale-specific (ru-RU) cultures as options when adding a new language.

> **ATTENTION**
> To ensure localization functions as expected, create both the locale-specific and neutral localization files for the selected language.

The Asset Performance Management application includes an **English (United States)** version of language resources (DLL and HTM files) installed by default.

> **ATTENTION**
> While Asset Performance Management is capable of displaying **Right-to-Left (RTL)** languages, the user interface will not mirror elements (for example, sidebars, buttons, and forms) and the user experience has not been verified for RTL languages. We recommend you avoid RTL language localization until fully supported.

12.1 Install Intuition Localization Tool

> **ATTENTION**
> Ensure that you install the Intuition Localization Tool on the Asset Performance Management primary server (or the Web Server in a multi-server installation).

1. On the installation media, browse to the **Utilities\Localization Tool R310** folder.
2. Double-click **Honeywell Intuition Localization Tool.msi**
3. Click **Next** in all the install screens and proceed with the installation until the **Final** screen is displayed.
4. Click **Finish**.

12.2 Localization Steps

This section describes the five main tasks that must be performed as a part of localization:

1. **Select Resources**: describes the steps involved in creating a repository and selecting the application resources.
2. **Add a Language**: describes the steps involved in adding a language of user's choice for the application resource.
3. **Localizing Application Resources**: describes the steps involved in translating the versions of language resources.
4. **Deploy Resources**: describes the steps involved in deploying the localized resources.

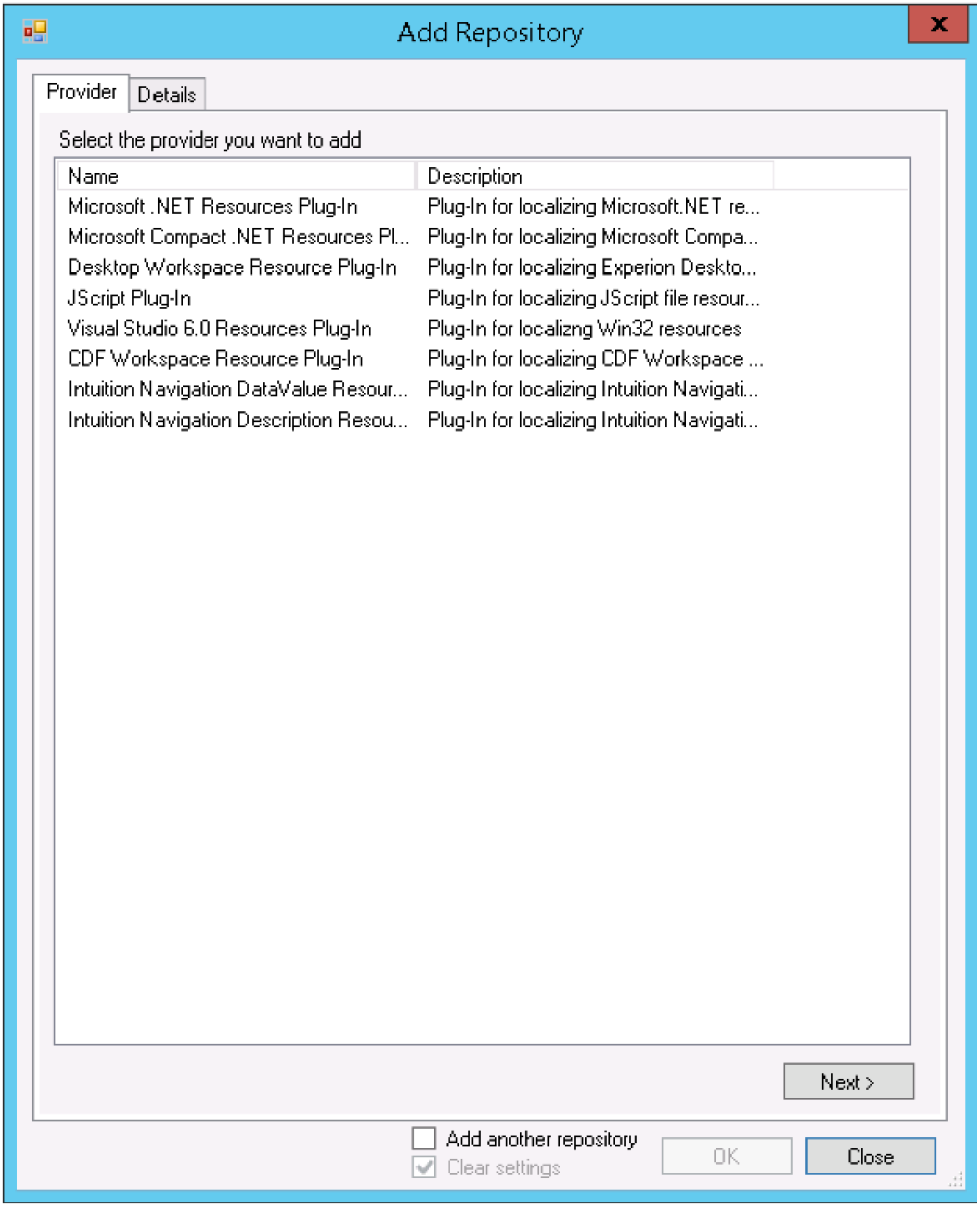
12.3 Select Resources

This section describes how to select application resources for **Microsoft.NET Resources Plug-In** resource type.

Perform the following steps on the Application Server:

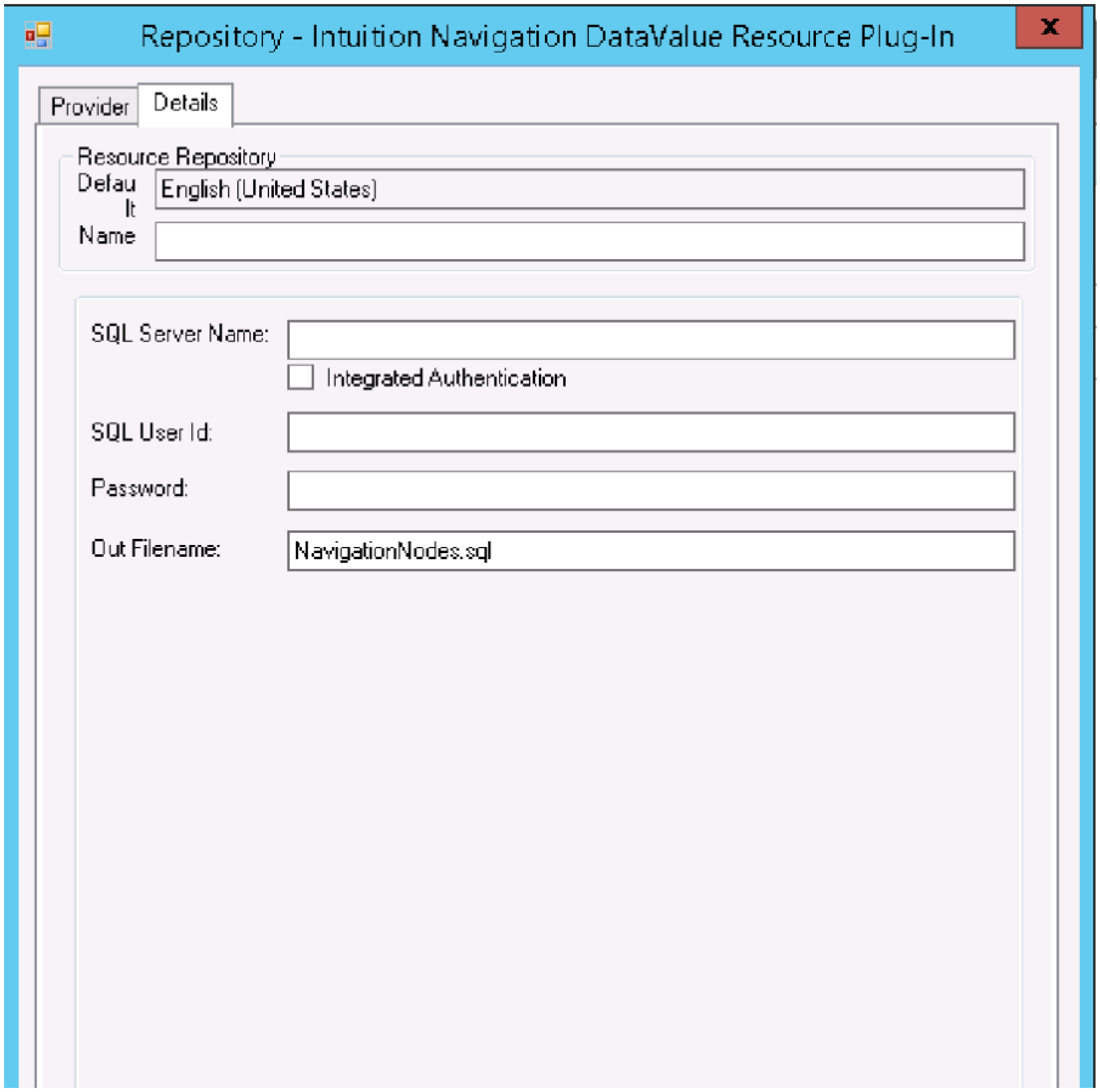
1. **Start** > **Honeywell** > **Localization Tool**.
   The **Localization Tool** is displayed.
2. Choose **File > New**.
   A new node **Untitled** is created.
3. Choose **File > Save** and in the **Save Project** dialog box, type the name of the project and then click **OK**.
4. Right-click the node and select **Add Repository**.

The **Add Repository** dialog box is displayed.



5. (Optional) To localize menus, do all of the following:

a. Select Intuition Navigation DataValue Resource Plug IN.



b. In the **Name box**, type a name for the resource repository. For example NavigationLocalization.

c. In the **SQL Server Name** box, type the machine name where MES database resides.

d. Select **Integrated Authentication**.

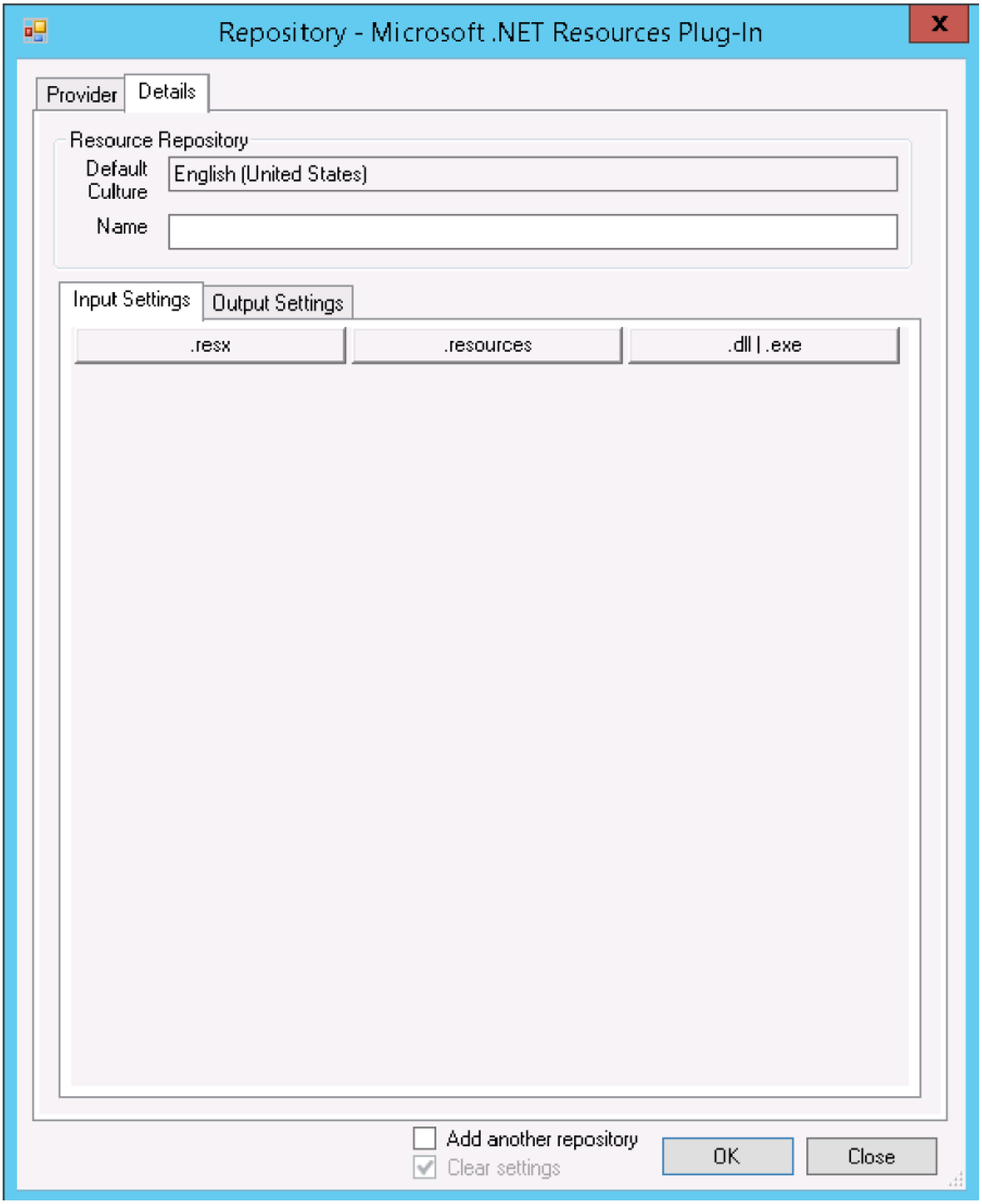
e. Click **OK**.

6. Right-click the node and select **Add Repository**.

   The **Add Repository** dialog box is displayed.

7. Under the **Provider** tab, from the list of Plug-ins, select **Microsoft.NET Resources PlugIn** and then click **Next**.

   The **Details** tab is displayed.

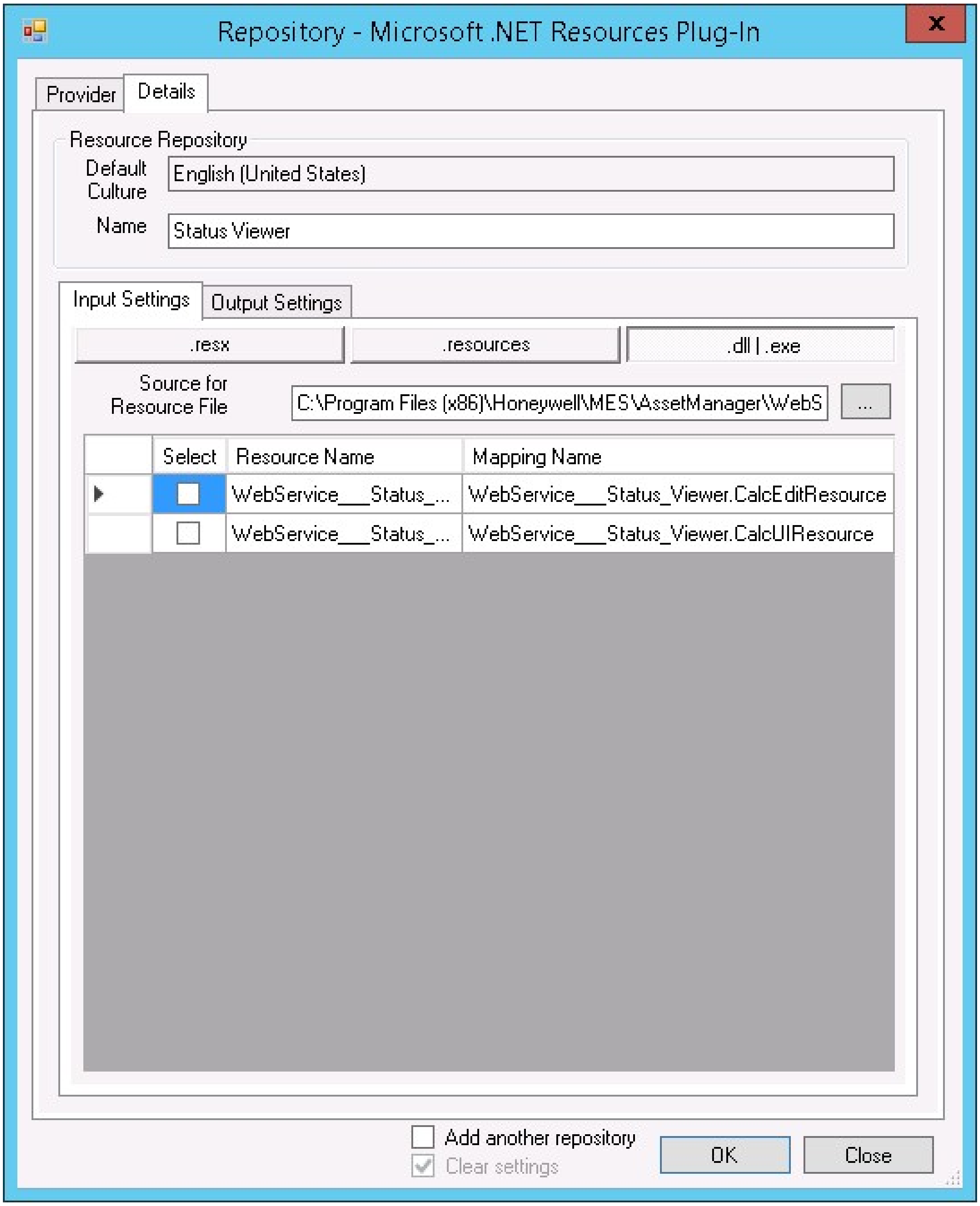


8. Perform the following steps:

a. In the **Default Culture** box, the language **English (United States)** appears by default.

b. In the **Name** box, type a name for the resource repository.

c. Under the **Input Settings** tab, click the **.dll/.exe** button.



9. Do all of the following for each of the files listed in the following table:

a. Click the Browse/Ellipses (...) button by the **Source for Resource File** field, browse to the required folder, select the dll or resource file, and then click **Open**.

   The selected dll is listed in the **Repository - Microsoft.NET Resources Plug-In** dialog box.

b. Select the required resource by selecting the check box provided beside the **Resource Name**.

   To localize the user interface, you must select the **UIReources.resources** for all the listed DLLs.

c. Click the **Output Settings** tab and then click the **.dll** button.

d. Click the browse button and browse to and select the same dll selected on the **Input Settings** tab (folders listed in the resources table above).

e. From the **Resource Name** drop-down list, select the resource. Ensure that you select the same resource name that was selected in the **Input Settings** tab.

f. Click **OK**.

   The application resources added appear on the left pane.

> NOTE
> Ensure you have added a **Resource Repository Name** before clicking **OK**.

The following table provides information about the folders and the files required for localization.

To localize	dll Name	Folder
Reports	AMLocalizedAssembly.dll	[InstallDrive]:\Program Files (x86)\Honeywell\MES\AssetManager\Tools\AMReports\
Performance Curve	Honeywell.AM.WebUI.reources.dll	[InstallDrive]:\Program Files (x86)\Honeywell\MES\AssetManager\WebServer\UI\bin\
Status Viewer	WebService - Status Viewer.dll	[InstallDrive]:\Program Files (x86)\Honeywell\MES\AssetManager\WebServer\Status\bin\
Calculation UI	CalcUI.resources.dll	[InstallDrive]:\Program Files (x86)\Honeywell\MES\AssetManager\WebServer\CalcUI\bin\
	CalcUI.ResourceAssembly.resources.dll	[InstallDrive]:\Program Files (x86)\Honeywell\MES\AssetManager\WebServer\CalcUI\bin\

10. Right-click the node and select **Add Repository**.

    The **Add Repository** dialog box is displayed.

11. Under the **Provider** tab, from the list of Plug-ins, select **Microsoft.NET Resources PlugIn** and then click **Next**.

    The **Details** tab is displayed.

12. Do all the following to add the resource files:

    a. Click the **Input Settings** tab, and then click the **.resx** button.

    b. For all the files shown in the following table, click the browse button in an empty row and navigate to the file.

    c. Click the **Source for Resource File** browse button and browse to the required folder, select the dll or resource file, and then click **Open**.

The following table provides information about the folders and the files required for localization.

To localize	resx Name	Folder
Web User Interface	Resources.resx	[InstallDrive]:\Program Files (x86)\Honeywell\MES\AssetManager\WebServer\WUI\App_GlobalResources\
Event Viewer	FaultViewerResources.resx	[InstallDrive]:\Program Files (x86)\Honeywell\MES\AssetManager\WebServer\FaultViewer\App_GlobalResources\

d. Click the **Output Settings** tab, and then click the **.resx** button.

e. For all the listed .resx files, click the button in the Default column.

f. Click **OK**.

The application resources added appear on the left pane.

12.4 Add a Language

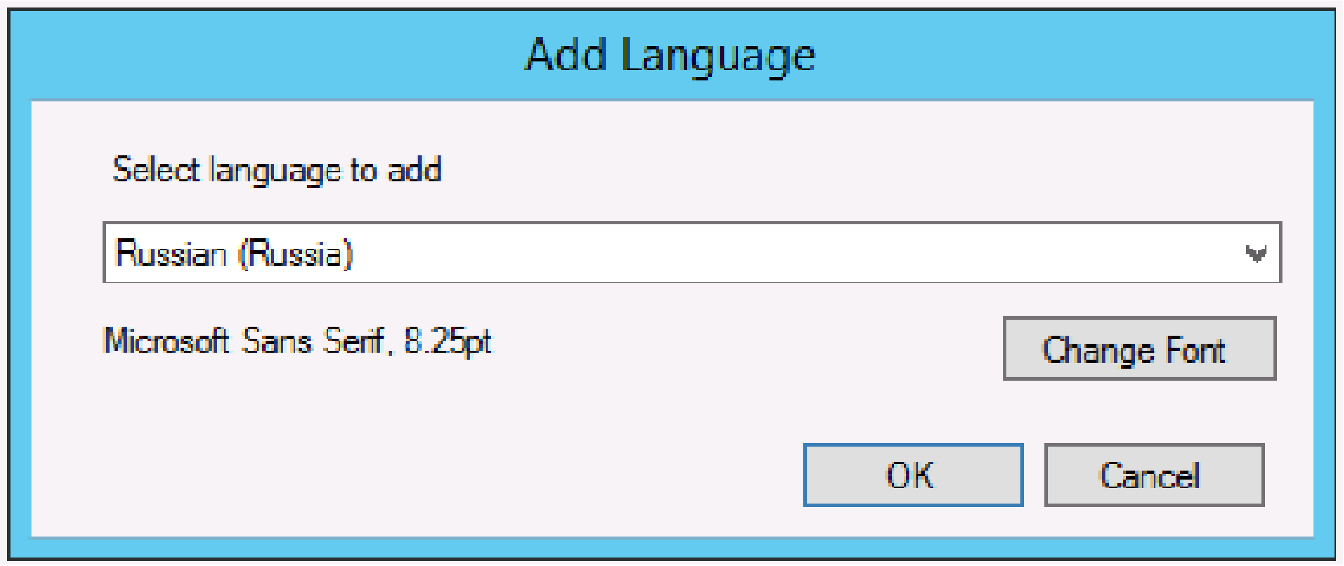
This section describes the procedure to add the language to which you want to translate the application resources.

> **ATTENTION**
> To ensure localization functions as expected, create both the locale-specific and neutral localization files for the selected language, as directed.

The **Russian (Russia)** "ru-RU" language (culture) and **Russian** "ru" neutral language are used as an example. You can use the same procedure for adding additional cultures.

1. On the **Localization Tool** menu bar, choose **Language > Add Language**.

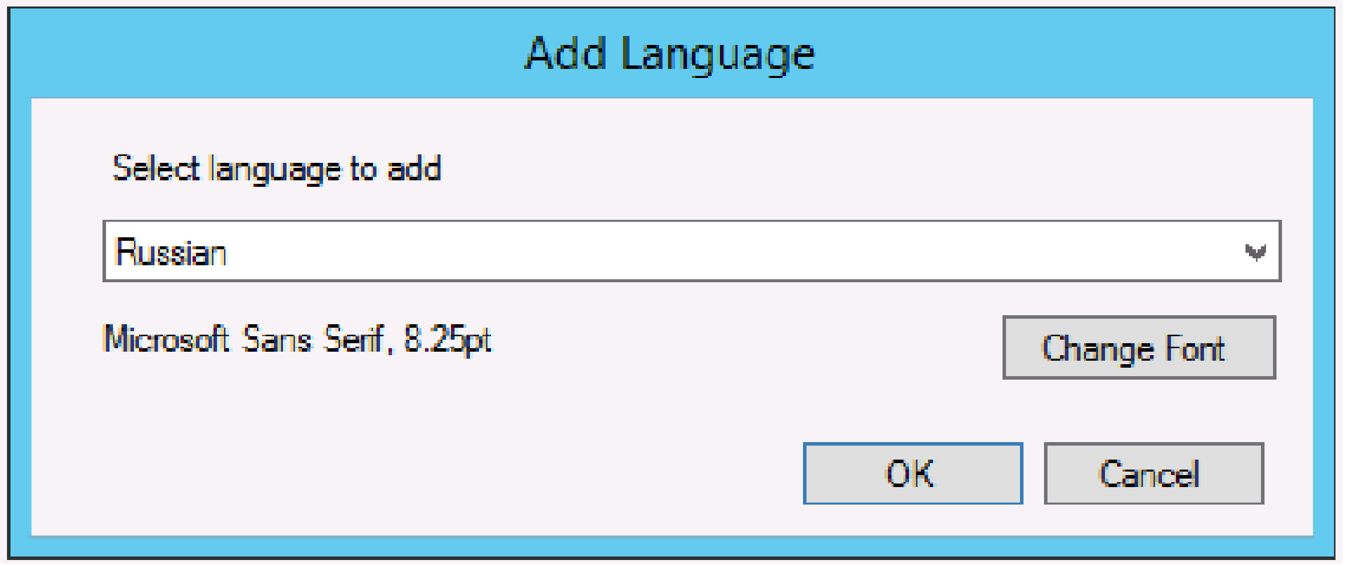
   The **Add Language** dialog box is displayed.



2. From the **Select language to add** drop-down list, select **Russian (Russia)** (localespecific) and then click **OK**.

   The selected language is displayed on the left pane.

3. Choose **Language > Add Language**.



4. From the **Select language to add** drop-down list, select **Russian** (neutral) and then click **OK**.

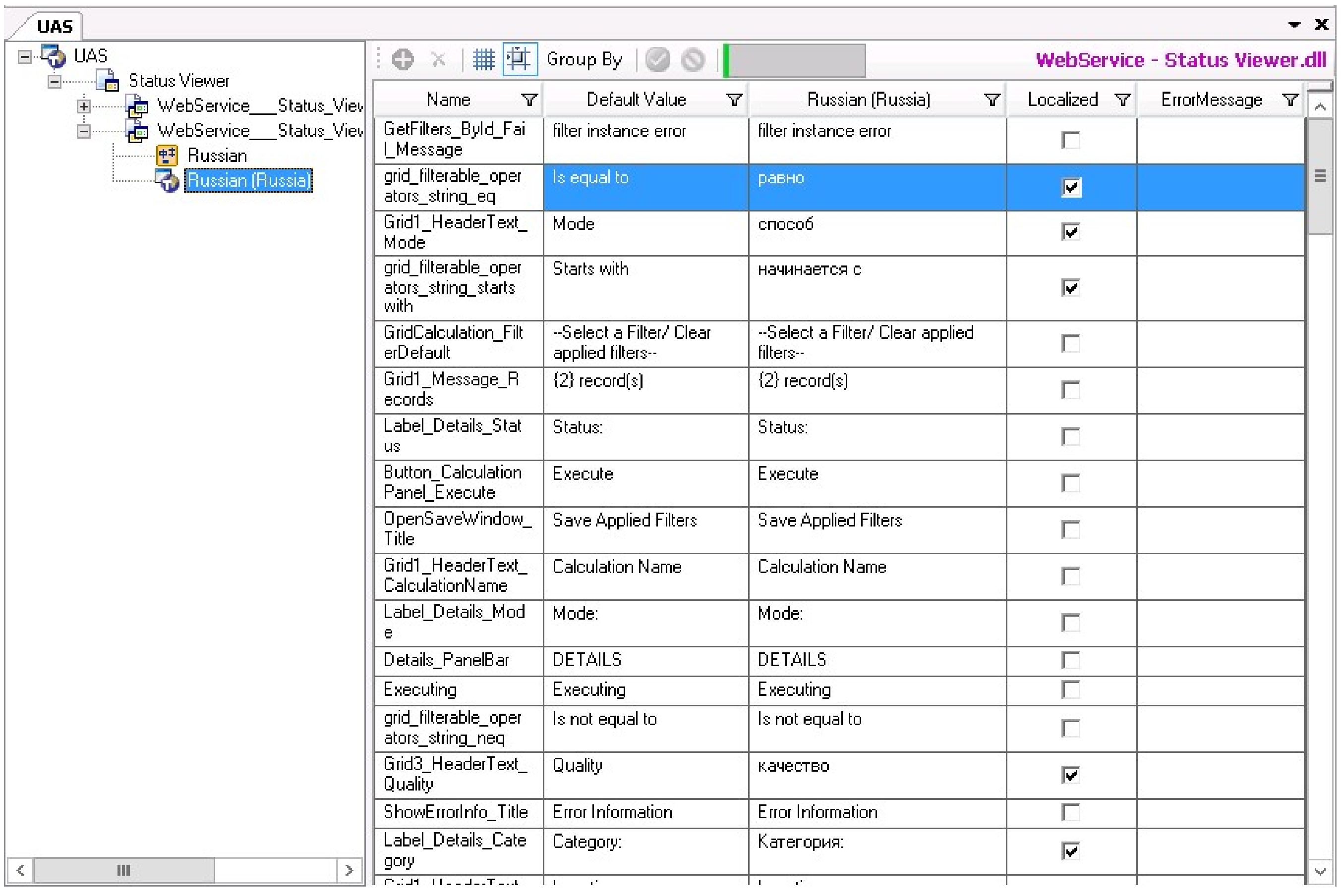
   The selected language is displayed on the left pane.

12.5 Localizing Application Resources

1. On the left pane, expand the application resource and then double-click the language name.

   For example, double-click **Russian (Russia)** to create the Russian locale-specific resources and double-click **Russian** to create the Russian neutral language resources.

   The details of the resource are displayed on the right pane.



2. Click the value on the first row of the language column and translate the value. Translate the default values for all the resource files in the grid to the corresponding values (for example, in Russian).

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To ensure localization functions as expected, create both the locale-specific and neutral resources files. For example, translate both **Russian (Russia)** and **Russian**

> TIP
> - You can use a Translator tool to translate the values. Copy all the values in the Default Value column and paste it on the translator you are using. Translate the values and later copy the translated values back to the Russian column.
> - You can use the Export/Import option and export all the values to an Excel sheet. After you localize the values, you can import them back to the Localization tool.

12.6 Create Signed Translation Packages

1. From the installation media, copy the files listed in the following table to a temporary folder on your computer.

File Name	Source Folder
CAM.ttp	[installationmedia]:\Installer\Components\USC\CAM_Localization
CAM.UI.ttr	[installationmedia]:\Installer\Components\USC\CAM_Localization
CAM.DAL.ttr	[installationmedia]:\Installer\Components\USC\CAM_Localization
Honeywell.ILM.UI.ttp	[installationmedia]:\\Installer\Components\ILM\Localization
Honeywell.ILM.UI.Commo	[installationmedia]:\\Installer\Components\ILM\Localization
MESFoundation	[installationmedia]:\\Installer\Components\CoreSystem\CoreEnv\Localization
MES Foundation.ttp	[installationmedia]:\\Installer\Components\CoreSystem\CoreUI\Localization
IntuitionDAS.ttp	[installationmedia]:\\Installer\Components\DAS\Localization
Timemanagement.ttp	[installationmedia]:\\Installer\Components\TimeManagement\Localization
Controls.ttp	[installationmedia]:\\Installer\Components\IntuitionControls\Localization
EventProcessor.ttp	[installationmedia]:\\Installer\Components\EventProcessor\Localization
EmailDispatcher.ttp	[installationmedia]:\\Installer\Components\EmailNotification\LocalizationProject
Notification.ttp	[installationmedia]:\\Installer\Components\Notification\Localization
NotificationDisplays.ttp	[installationmedia]:\\Installer\Components\Notification\Localization
SPConfigurationDisplay.ttp	SharePointNotification\Localization

2. For each of the files above, do all of the following:
   a. Choose **File** > **Open**.
   b. Click **OK**.

      The resource files are displayed in the left pane.
   c. On the left pane, right-click the main node and select **Reload All Repositories**.
   d. Double-click the **language** under the resource node of a project.

      All the strings in the resource are displayed on the right pane.

e. Translate all resource strings to the required language using the procedure you follow to translate the resource strings.

f. Once you translate the resource, ensure the **Localized** check box is selected to localize the resource string.

3. Choose **File** > **Save** and provide the location where you want to save the updated project on your local computer.

   The project file (ttp file) and the repository files (ttr files) are saved.

4. Send the saved **ttp** and **ttr** files for signing to Honeywell Product Support.
5. The support team provides the resource files as dlls.

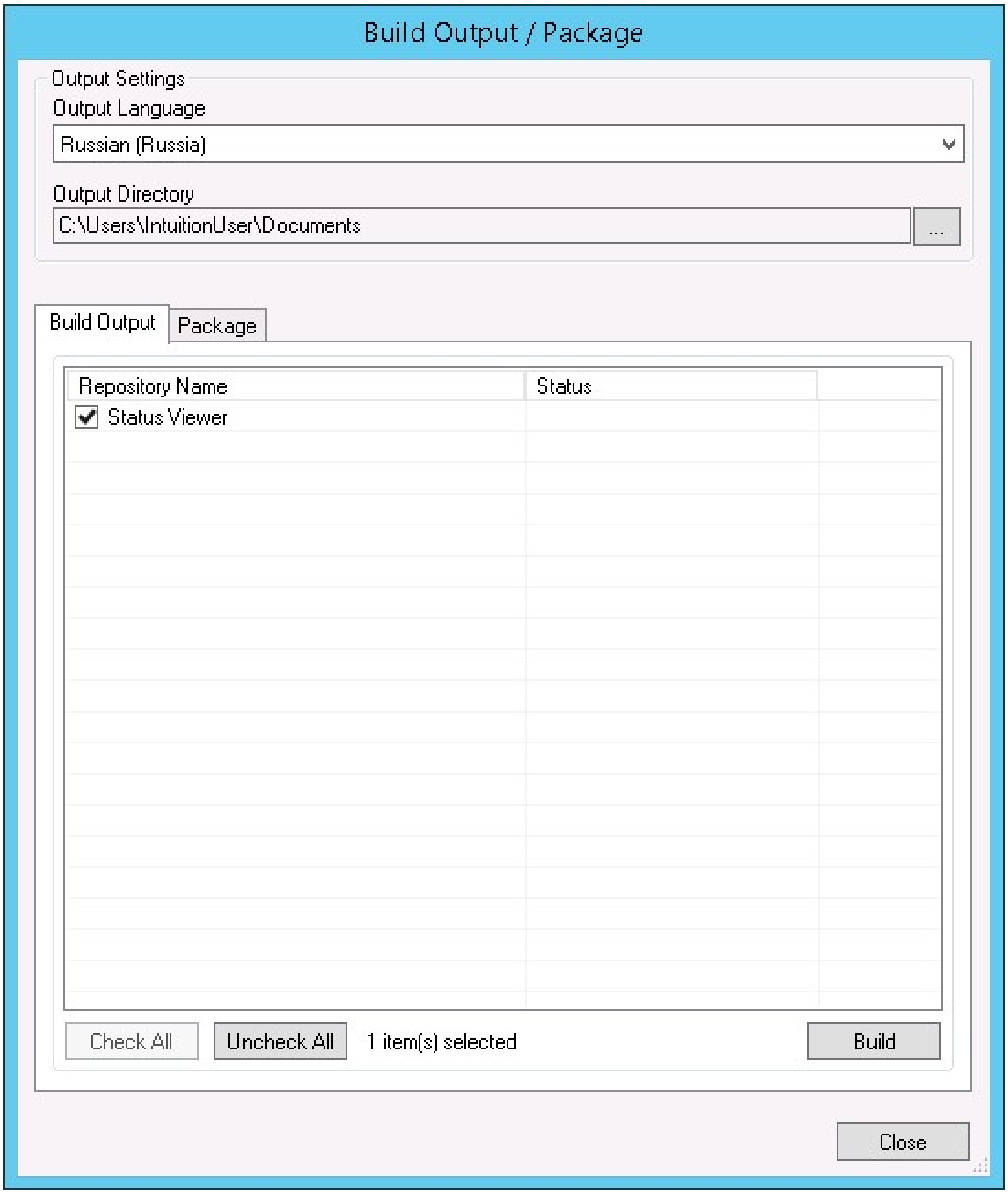
12.7 Deploy Resources

This section describes how to deploy the localized resources.

The **Russian (Russia)** "ru-RU" language (culture) and **Russian** "ru" neutral language are used as an example. You can use the same procedure for adding additional cultures.

1. Choose **File > Build Output/Package**.

   The **Build Output/Package** dialog box appears.



2. Do all of the following to build the locale-specific files:
   a. From the **Output Language** drop-down list, select **Russian (Russia)**.
   b. Click the **Output Directory** browse button and browse to the path where you want to save the localized resource files and then click **OK**.
   c. Under the **Build Output** tab, select the repository that you want to build.

d. Click **Build**.

   A Message is displayed indicating that the files are built successfully. If you have selected "MES Navigation Resource Plug IN" for localizing menus, an output file with an extension ".sql" is created in "ru-RU" folder.

e. Click **OK**.

3. Do all of the following to build the neutral language files:
   a. From the **Output Language** drop-down list, select **Russian**.
   b. Click the **Output Directory** browse button and browse to the path where you want to save the localized resource files and then click **OK**.
   c. Under the **Build Output** tab, select the repository that you want to build.
   d. Click **Build**.

      A message is displayed indicating that the files are built successfully. If you have selected "MES Navigation Resource Plug IN" for localizing menus, an output file with an extension ".sql" is created in "ru" folder.
   e. Click **OK**.
4. Click **Close**.
5. Browse to the path where you have saved the localized resource files. Note that the "ruRU" and "ru" folder is created. Copy the "ru-RU" and "ru" folders and paste them in the following path on the Applications Server.
6. For all of the dll files shown in the following table, do all of the following:

a. Create an "ru-RU" folder in the related folder path.
b. Create an "ru" folder in the related folder path.
c. Copy the files from the saved localized resource folders to both the "ru-RU" and "ru" folders.

File Name	Folder Path
AMLocalizedAssembly.dll	[InstallDrive]:\Program Files (x86)\Honeywell\MES\AssetManager\Tools\AMReports\
Honeywell.AM.WebUI.resources.dll	[InstallDrive]:\Program Files (x86)\Honeywell\MES\AssetManager\WebServer\UI\bin\
WebService - Status Viewer.dll	[InstallDrive]:\Program Files (x86)\Honeywell\MES\AssetManager\WebServer\Status\bin\
CalcUI.resources.dll	[InstallDrive]:\Program Files (x86)\Honeywell\MES\AssetManager\WebServer\CalcUI\bin\
CalcUI.ResourceAssembly.resources.dll	[InstallDrive]:\Program Files (x86)\Honeywell\MES\AssetManager\WebServer\CalcUI\bin\
CAM.UI.dll	[InstallDrive]:\Program Files (x86)\Honeywell\Honeywell\MES\Core\WWWMESCore\WebUX\ CAM\bin\
CAM.DAL.dll	[InstallDrive]:\Program Files (x86)\Honeywell\Honeywell\MES\Core\Service\CommonAssetService\1.0\Bin\
Honeywell.ILM.UI.resources.dll	[Install Drive]:\Program Files (x86)\Honeywell\MES\ILM\WWWMESCore\UI\bin\[LanguageCode]\
Application Resources	[Install Drive]:\Program Files (x86)\Honeywell\MES\ILM\WWWMESCore\UI\App_GlobalResources\
Honeywell.Intuition.Controls.TimeControl.ResourceAssembly.dll	[Install Drive]:\Program Files (x86)\Honeywell\MES\Controls\bin
Honeywell.MES.Core.UX.CE.Shell.ResourceAssembly.resources.dll	[Install Drive]:\Program Files (x86)\Honeywell\MES\Core\wwwMEScore\ClientBin5\
Honeywell.MES.Core.UX.ConfigureModelModule.ResourceAssembly.resources.dll	[Install Drive]:\Program Files (x86)\Honeywell\MES\Core\wwwMEScore\ClientBin5\
Honeywell.MES.Core.UX.ConfigureSecurityModule.ResourceAssembly.resources.dll	[Install Drive]:\Program Files (x86)\Honeywell\MES\Core\wwwMEScore\ClientBin5\
Honeywell.MES.Core.UX.ConfigureSystemCatalogModule.ResourceAssembly.resources.dll	[Install Drive]:\Program Files (x86)\Honeywell\MES\Core\wwwMEScore\ClientBin5\
Honeywell.MES.Core.UX.InfrastructureModule.ResourceAssembly.dll	[Install Drive]:\Program Files (x86)\Honeywell\MES\Core\wwwMEScore\C

File Name	Folder Path
	lientBin5\
Honeywell.MES.Core.UX.NavigationModule.ResourceAssembly.resources.dll	[Install Drive]:\Program Files (x86)\Honeywell\MES\Core\wwwMEScore\ClientBin5\
Honeywell.MES.Core.UX.RootModule.ResourceAssembly.resources.dll	[Install Drive]:\Program Files (x86)\Honeywell\MES\Core\wwwMEScore\ClientBin5\
Honeywell.MES.Core.UX.Shell.ResourceAssembly.resources.dll	[Install Drive]:\Program Files (x86)\Honeywell\MES\Core\wwwMEScore\ClientBin5\
Honeywell.MES.Core.UX.FlowSheetRuntime.ResourceAssembly.dll	[Install Drive]:\Program Files (x86)\Honeywell\MES\Core\wwwMEScore\ClientBin5\
Honeywell.MES.Core.UX.Utilities.ResourceAssembly.dll	[Install Drive]:\Program Files (x86)\Honeywell\MES\Core\wwwMEScore\ClientBin5\
Honeywell.MES.Diagnostics.UX.LogViewer.ResourceAssembly.dll	[Install Drive]:\Program Files (x86)\Honeywell\MES\Core\wwwMEScore\ClientBin5\
Honeywell.MES.Core.Service.DiagnosticsService.Resource.dll	[Install Drive]:\Program Files (x86)\Honeywell\MES\Core\Service\DiagnosticsService\1.0\bin\
Honeywell.MES.Core.UX.Core.Web.ResourceAssembly.dll	[Install Drive]:\Program Files (x86)\Honeywell\MES\WWWMESCommon\bin\
Honeywell.MES.Core.Service.LicenseRuntime.Resource.resources.dll	[Install Drive]:\Program Files (x86)\Honeywell\MES\Core\Service\License Manager\1.0\bin\
Honeywell.MES.Core.Service.Navigation.Resource.resources.dll	[Install Drive]:\Program Files (x86)\
Honeywell.MES.Core.UX.PluginConfigurationModule.ResourceAssembly.dll	[installpath]\Honeywell\MES\Core\ wwwMEScore\ClientBin\<lang>\
Honeywell.IDA.Configuration.Service.Resources.dll	[install path]\Honeywell\MES\DataAccessService\220\1.0\bin\<lang>\
Honeywell.IDA.DataAccessService.Resources.dll	[install path]\Honeywell\MES\DataAccessService\220\1.0\bin\<lang>\
Honeywell.Intuition.DataAccess.DataItemBrowserService.Resources.dll	[install path]\Honeywell\MES\Core\DataAccess\DataItemBrowser\1.0\bin\<lang>\

File Name	Folder Path
Honeywell.IDA.CacheManagerPlugins.InMemory.Resources.dll	[install path]\Honeywell\MES\DataAccessService\220\1.0\bin\<lang>\
Honeywell.IDA.ConfigurationManagerPlugins.ConfigurationManager.Resources.dll	[install path]\Honeywell\MES\DataAccessService\220\1.0\bin\<lang>\
Honeywell.IDA.DataAccessPlugins.IntuitionDataSource.Resources.dll	[install path]\Honeywell\MES\DataAccessService\220\1.0\bin\<lang>\
Honeywell.IDA.DataAccessPlugins.OData.Resources.dll	[install path]\Honeywell\MES\DataAccessService\220\1.0\bin\<lang>\
Honeywell.IDA.DataAccessPlugins.WCFTimeSeries.Resources.dll	[install path]\Honeywell\MES\DataAccessService\220\1.0\bin\<lang>\
Honeywell.IDA.DiagnosticPlugins.PerformanceCounters.Resources.dll	[install path]\Honeywell\MES\DataAccessService\220\1.0\bin\<lang>\
Honeywell.IDA.Interfaces.Resources.dll	[install path]\Honeywell\MES\DataAccessService\220\1.0\bin\<lang>\
Honeywell.IDA.SecurityManagerPlugins.DefaultSecurityManager.Resources.dll	[install path]\Honeywell\MES\DataAccessService\220\1.0\bin\<lang>\
Honeywell.IDA.SubSystems.TimeSeries.ITimeSeriesProvider.Resources.dll	[install path]\Honeywell\MES\DataAccessService\220\1.0\bin\<lang>\
Honeywell.IDA.DataSourcePlugins.DAS.Resources.dll	[install path]\Honeywell\MES\DataAccessService\220\1.0\bin\<lang>\
Honeywell.IDA.DataSourcePlugins.Date.Resources.dll	[install path]\Honeywell\MES\DataAccessService\220\1.0\bin\<lang>\
Honeywell.MES.Service.EventProcessor.EventService.Resource.dll	[install path]\Honeywell\MES\EventProcessor\Service\EventService\1.0\Bin\ <lang>
Honeywell.MES.Core.Service.EmailNotification.Resource.dll	[install path]\Honeywell\MES\Core\Service\EmailNotification\1.0\bin\<lang>
Honeywell.MES.Core.Service.EmailTransformer.Resource.dll	[install path]\Honeywell\MES\Core\Service\EmailTransformer\1.0\bin\<lang>

File Name	Folder Path
Honeywell.MES.Core.Service.EmailNotification.Resource.resources.dll	[install path]\Honeywell\MES\Core\Service\EmailNotification\1.0\bin\<lang>
Honeywell.MES.Core.Service.EmailTransformer.Resource.resources.dll	[install path]\Honeywell\MES\Core\Service\EmailTransformer\1.0\bin\<lang>
Honeywell.MES.Core.Service.NotificationService.Resource.dll	[install path]\Honeywell\MES\Core\Service\NotificationService\1.0\bin\<lang>
Honeywell.MES.Core.DataModel.EventModel.Resource.dll	[install path]\Honeywell\MES\Core\DataModel\EventModel\1.0\bin\<lang>
Honeywell.MES.Service.Diagnostics.Core.Resource.dll	[install path]\Honeywell\MES\Core\Service\NotificationDiagnostics\1.0\bin\<lang>
Honeywell.MES.Service.Core.NotificationLogSrv.Resource.dll	[install path]\Honeywell\MES\Core\Service\NotificationLogService\1.0\bin\<lang>
Honeywell.MES.Core.Service.NotificationType.ServiceExtensions.Resource.dll	[install path]\Honeywell\MES\Core\Service\NotificationType\bin\<lang>
Honeywell.MES.Core.Service.NotificationService.Resource.resources.dll	[install path]\Honeywell\MES\Core\Service\NotificationService\
Honeywell.MES.Core.DataModel.EventModel.Resource.resources.dll	[install path]\Honeywell\MES\Core\DataModel\EventModel\1.0\bin\<lang>
Honeywell.MES.Service.Diagnostics.Core.Resource.resources.dll	[install path]\Honeywell\MES\Core\Service\NotificationDiagnostics\1.0\bin\<lang>
Honeywell.MES.Service.Core.NotificationLogSrv.Resource.resources.dll	[install path]\Honeywell\MES\Core\Service\NotificationLogService\1.0\bin\<lang>
Honeywell.MES.Core.Service.NotificationType.ServiceExtensions.Resource.resources. dll	[install path]\Honeywell\MES\Core\Service\NotificationType\bin\<lang>
Honeywell.MES.NotificationUX.UX.NotificationDisplay.ResourceAssembly.dll	[install path]\Honeywell\MES\Core\WWWMESCore\ClientBin\<lang>
Honeywell.Intuition.Notification.ConfigurationUI.ResourceAssembly.dll	[install path]\Honeywell\MES\Core\UX\SPNotificationConfiguration\bin \<lang>

File Name	Folder Path
Honeywell.MES.Core.Service.SharePointTransformer.Resource.dll	[install path]\Honeywell MES\Core\Service\SharePointTransformer\1.0\bin \<lang>

7. To localize the required files in the Global Assembly Cache (GAC), do all of the following:
   a. Navigate to the following folder to access the **Honeywell.MES.Sample.MESGACInstallUtility.exe** file:

      [installdrive]:\Program Files (x86)\Honeywell\LT\Toolkit

   b. Open a command prompt and execute the following command for all assemblies listed in the following table:

      `[current folder]\Honeywell.MES.Sample.MESGACInstallUtility` **`"Assembly Name"`**

Assembly Name
Honeywell.MES.Core.DataModel.SystemCatalogs.ResourceAssembly.dll
Honeywell.MES.Core.Service.TimeManagement.Resource.dll
Honeywell.MES.Core.Service.ExceptionHandling.Resources.resources.dll
Honeywell.MES.Core.Service.Security.Resources.resources.dll
Honeywell.MES.Core.Service.Resources.resources.dll
Honeywell.Intuition.DataAccess.OData.Router.ResourceAssembly.resources.dll
Honeywell.MES.Core.Service.TimeManagement.Resource.dll

8. Execute the following scripts in SQL Server Management Studio:
   - NavigationNodes.sql

9. If you are localizing reports, copy the "ru-RU" and "ru" folders, along with the dlls, to the following locations:
   - [SQL Reporting Service Installed Path]\Microsoft SQL Server\MSRS[VersionNumber].[InstanceID]\Reporting Services\ReportServer\bin
   - [Microsoft Visual Studio 3.5 framework installed path]Microsoft Visual Studio 9.0\Common7\IDE\PrivateAssemblies

> **ATTENTION**
> Ensure that you do not copy the folder when you build output for the same repository next time. You must only copy the corresponding dll available inside the folder and replace it with the dll available in the path "[InstallDrive]:\Program Files (x86)\Honeywell\MES\Core\WWWMESCore\clientbin5\[Folder]".

10. For the resource files on the web server, create a copy of the localized files with the language code in the file name and then paste them in the original locations, as follows:

To localize	resx localized Name	Folder
Web User	Resources.**ru-RU**.resx	[InstallDrive]:\Program Files

To localize	resx localized Name	Folder
Interface		(x86)\Honeywell\MES\AssetManager\WebServer\WUI\App_GlobalResources\
	Resources.**ru**.resx	[InstallDrive]:\Program Files (x86)\Honeywell\MES\AssetManager\WebServer\WUI\App_GlobalResources\
Event Viewer	FaultViewerResources.**ruRU**.resx	[InstallDrive]:\Program Files (x86)\Honeywell\MES\AssetManager\WebServer\FaultViewer\App_GlobalResources\
	FaultViewerResources.**ru**.resx	[InstallDrive]:\Program Files (x86)\Honeywell\MES\AssetManager\WebServer\FaultViewer\App_GlobalResources\

11. Open the Intuition Home Page: **Start** > **Honeywell** > **Home**.
12. Navigate to the Model Settings (**System** > **System Configuration** > **Application**).
13. On the tree, select **AppConfigs**.
14. Scroll to **Honeywell.MES.Core.UI.Localization.IsLocalized**, double-click the Value, type true, and then press Enter.
15. Open Asset Performance Management.

    The interface should now appear in the localized language.

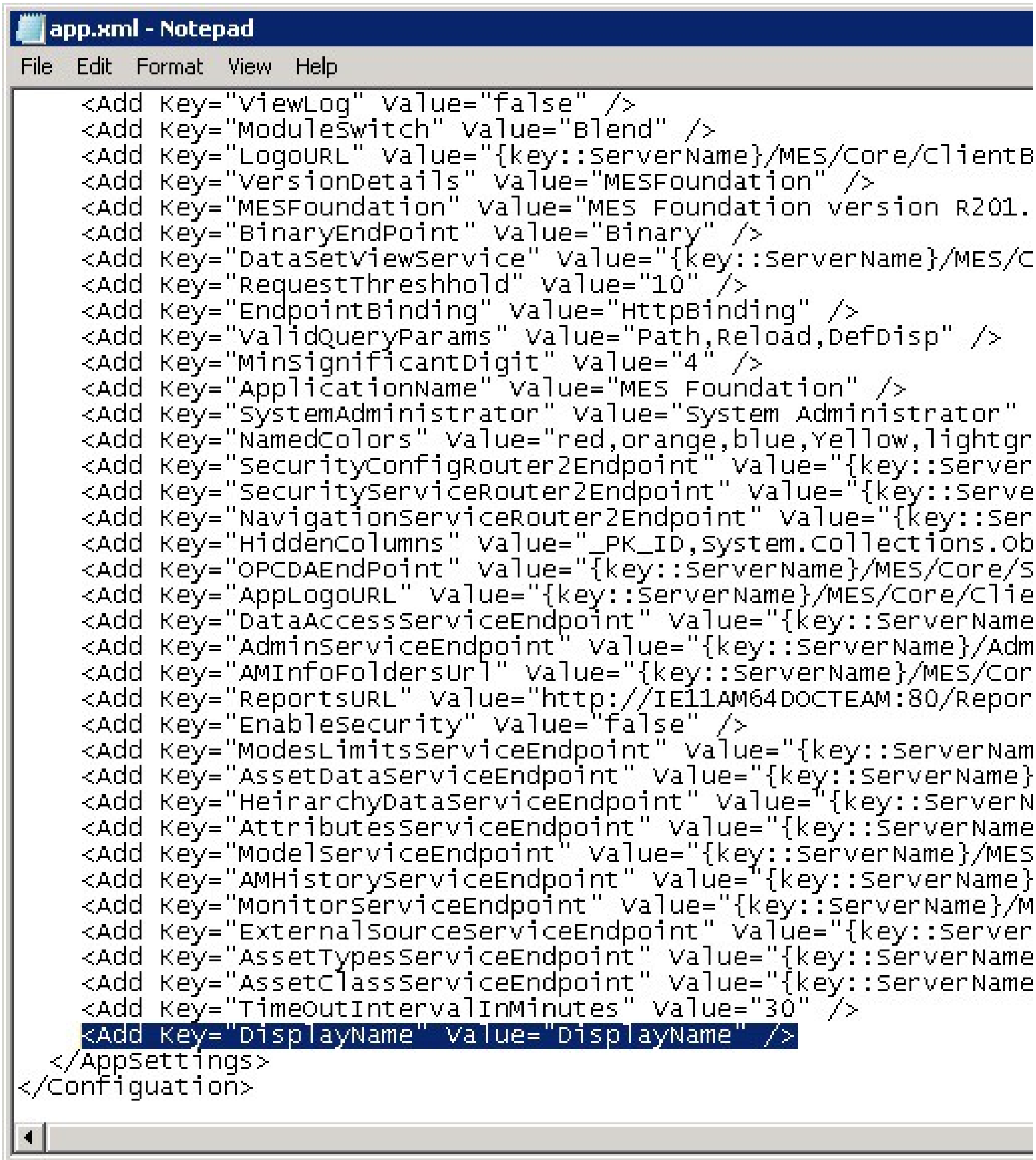
12.8 Update the System to Use Display Name

This section provides more information on how to change the settings of the configuration UI settings file (app.xml) so the Display Name, which can hold localized strings, is used in place of the Name.

> NOTE
> By default, the Asset Performance Management reports are displayed with the display name.

1. Browse to **[MES Installed Path]\Core\WWWMESCore\clientbin5** and open the app.xml file.
2. Browse to "**<Add Key=DisplayName Value=Name />**" key.
3. Change **Value=Name** to **Value = DisplayName**.

    > NOTE
    > Ensure that you always use **Value = Name** at the time of configuration.



4. Click **File > Save** and then close the app.xml file.

APPENDIX

A SETTING UP DAS AND PHD FOR UOM SUPPORT

To add Units of Measure to the attributes in Asset Performance Management, you need to configure DAS and PHD to ensure they have access to the required data sources.

A.1 Configuring the DAS Instance

This section describes how to configure the DAS instance required for UOM Support.

To configure the DAS instance

1. On the Honeywell applications page, choose **System** > **System Configuration** > **Data Access**.
2. In the **Data Access** configuration window, on the **Services** tree, expand **SentinelDASInstance**, and then select **UASDataSource**.
3. Click the green plus icon to add the data source required for UOMs.

   The **Data Source Configuration** dialog box appears.
4. In the **Name** box, type **AMHistorian**.
5. From the **Select Plug-in** drop-down list, select **OPCHDADataSource**.
6. Click **Next**.

   The **External Connection Configuration** dialog box appears.
7. In the **External Connection Configuration** dialog box, do all of the following in the default connection:
   a. In the **Name** box, type **AMHistorian**.
   b. In the **Connection String** box, use the following connection string for the OPCHDA data source:

      Url=opchda://localhost/OPC.PHDServerHDA.1
   c. In the **Description** box, type a description for the connection string.

   > **ATTENTION**
   > The Test Connection feature may not succeed when connected to an Asset Performance Management data source. To check the configuration settings, complete the configuration and security settings, and look for the Asset Performance Management data in the Intuition model browser after performing the **IISRESET** on the Intuition Application Server and Intuition Web Server.

8. Click **Next**.

   The **Security and Cache Configuration** dialog box is displayed.
9. Click **Finish**.
10. On the tree, select **AMHistorian**, and click the **Edit** button.
11. In the **TagSource Name** box, type **AMHistorian**, and click **Save**.

A.2 Configuring PHD for UOMs

Once the DAS instance is created, you should configure PHD to allow it to browse OPC tags.

Do all of the following on the Calculation Server.

To Ensure PHD can Browse the OPC Tags

1. On the Calculation Server, browse to **Apps**> **Uniformance** > **Uniformance System Console**.
2. On the **Uniformance System Console** page, click **Define Server**.
3. In the **Node Name** dialog box, type localhost or the Calculation Server name, and then click **OK**.
4. On the **Control Root** tree, expand **Uniformance System Console**, right-click **localhost** (or the Calculation Server name), and on the shortcut menu, point to **All Tasks**, and then select **Connect**.
5. In the Logon dialog box, click **Trusted Connection**, and then click **OK**.
6. Under localhost, expand **Registry Editor** > **SOFTWARE** > **Honeywell** > **Uniformance**, and then select **OPCServer**.
7. Double-click **BrowseMode**.
8. In the DWORD Value dialog box, in the **Value Data** box type **1**, and then click **OK**.
9. Perform an **IISRESET**.

APPENDIX

B CONFIGURING EPHD

This section contains the procedure for configuring EPHD for use with Asset Performance Management.

> **NOTE**
> - Use of EPHD with Asset Performance Management requires familiarity with PHD.
> - EPHD does not support SQL Clustering.

If the settings are configured as listed, but EPHD is not functioning as expected, please contact Honeywell Product Support (see Support and other contacts).

B.1 Configuring the Database Server for Remote EPHD

If you have installed EPHD, you should update the settings to allow the Calculation Server and the Database Server to communicate.

Follow the below steps in a **Database Server**, if the communication between the calculation server and database server was not allowed during the database server installation or if there is a problem in connection:

1. Open the command prompt as an administrator and run the following file:

   `[Installation Media]\Installer\Components\EPHD\EPHD\Install\UnfAutoRun.cmd`

   The **Uniformance Installation Utility** appears.
2. If not already installed, do all of the following to install the **Uniformance PHD Database**:
   a. On the Uniformance Installation Utility, click the **Install Uniformance** tab.
   b. Expand the **Database System**, double-click or right-click **Uniformance Configuration Database**, and then choose **Install Product**.

      The Uniformance Configuration Database installation wizard appears.
   c. On the Welcome page, click **Next**.

      The License Key Information page appears.
   d. Click **Next**.

      The Destination Folder page appears.
   e. Click **Browse** to select a different install location.
   f. Click **Next**.

      The Uniformance Database Information appears.
   g. In the **SQL Server Instance Name for (local)** box, type the name of the Asset Performance Management database instance (without the server name), and click **Next**.

      For example, if you are installing to a default SQL instance, such as DBServerName, leave the default text as is. However, if you are installing to a named instance, such as DBServerName/Intuition, replace the default text with Intuition.

The Ready to Install Application screen appears.

h. Click **Next**.

i. When the installation is complete, click **Finish**.

3. If not already installed, do all of the following to install the **Uniformance PHD Database Services**:

   a. On the Uniformance Installation Utility, click the **Install Uniformance** tab.

   b. Expand the **Database System**, double-click or right-click **Uniformance Database Services**, and then choose **Install Product**.

      The Uniformance Database Services installation wizard appears.

   c. On the Welcome page, click **Next**.

      The License Key Information page appears.

   d. Click **Next**.

      The Destination Folder page appears.

   e. Click **Browse** to select a different install location.

   f. Click **Next**.

      The Uniformance Database Information appears.

   g. In the **SQL Server Instance Name for (local)** box, type the name of the Asset Performance Management database instance, and then click **Next**. The Ready to Install Application screen appears.

   h. Click **Next**.

   i. When the installation is complete, click **Finish**.

4. Close the Uniformance Installation Utility.

5. Open Windows services, and ensure the Uniformance Database Server services are running under the Asset Performance Management services user:

   - Uniformance Database Server
   - Uniformance DB Security Service
   - Uniformance PHD Startup

B.2 Connect EPHD on the Database Server

Follow the below steps to connect EPHD to the Database Server:

1. Open the Uniformance System Console (**Start** > **Uniformance** > **Uniformance System Console**).

2. Click **Define Server**.

   If this option is not visible, use the menu command: **Action** > **All Tasks** > **Define Server** .

3. In the Node Name field, type **localhost**.

4. Confirm the EPHD Database Settings on the Database Server.

   In the PHD Server Tool, ensure the EPHD server is pointing to the Database Server.

5. Open the Uniformance System Console (**Start** > **Uniformance** > **Uniformance System Console**).

6. On the tree, expand **Uniformance System Console**, right-click **localhost** > **All Tasks**, and then select **Connect**.

7. Log in as the installation account.

8. Right-click **UniformanceDatabase Server**, and select **Properties**.
9. Select the **UDB Server Configuration** tab.
10. In the UDB Server Configuration dialog box, change the Database Data Source to the name of the Database Server.

B.3 Confirm the EPHD PHD Settings on the Database Server

In the PHD Server Tool, ensure the EPHD server is pointing to the Database Server.

1. Open the Uniformance System Console (**Start** > **Uniformance** > **Uniformance System Console**).
2. On the tree, expand **Uniformance System Console**, right-click **localhost** > **All Tasks**, and then select **Connect**.
3. Log in as the installation account.
4. Right-click **Uniformance PHD Server**, and then select **Properties**.
5. Select the **PHDServer Configuration** tab.
6. In the PHD Server Configuration dialog box, in the **DataBase Server Name** enter the name of the Database Server (do not include the instance name).

B.4 Confirm the EPHD Startup Settings on the Database Server

1. Open the Uniformance System Console (**Start** > **Uniformance** > **Uniformance System Console**).
2. On the tree, expand **Uniformance System Console**, right-click **localhost** > **All Tasks**, and then select **Connect**.
3. Log in as the installation account.
4. Right-click **Uniformance PHD Startup**, and select **Properties**.
5. From the **Startup Type list**, ensure **Automatic** is selected.
6. In the **Log On As** area, ensure a user account with access to run PHD is configured.

B.5 Ensure the PHD Services are Running on the Calculation Server

If you have installed EPHD, ensure the following services are running under the Asset Performance Management services user on the Calculation Server (they should be started by default):

- Uniformance API Server
- Uniformance Database Server
- Uniformance DB Security Service
- Uniformance PHD Server
- Uniformance PHD Startup
- Uniformance RDI Server
- Uniformance Remote API Server

> **NOTE**
> Add the service account in the product administrator group in database server and calculation server for split server setup.

B.6 Refresh EPHD

After installing Asset Performance Management and updating the EPHD settings, the EPHD system is left in an Initializing state. To bring the state back to active, run the following commands:

```
phdctl stop /y

phdctl stop /y dbsecurity

phdctl stop /y udbserver

phdctl start cold
```

APPENDIX

C UNISIM DESIGN SUITE

Asset Performance Management monitors asset performance by evaluating the performance in real time from plant data. Asset Performance Management has the capability to perform basic process calculations and leverages UniSim Design for advanced process calculations.

Users could purchase UniSim Design run time license with Asset Performance Management and if a developer license is required then a separate UniSim Design license is required.

Asset Performance Management supports **UniSim DesignR460**, **R480**, and **R492** .

C.1 Installing the UniSim Design Suite

1. Shut down all other programs on your computer before starting the installation process.
2. Insert the UniSim Design media into the DVD drive of your computer.

   If your computer has the autorun feature enabled, the DVD browser should start and you can skip to step 5.
3. From the **Start** menu, select **Run**.
4. In the Run view, type **D:\setup.exe** and click **OK**.

   If your DVD-ROM uses a different drive letter, use the corresponding drive letter.
5. Once the installer starts, you may need to install or update the **.NET Framework** on your computer and a reboot may be required. The installer should automatically restart after this reboot.

   The opening screen appears.
6. Click on **Install Products** in the upper right-hand corner.
7. Click **UniSim Design**.
8. Click **UniSim Design** to begin the installation of the UniSim Design application.

   The installation wizard appears.
9. On the **Welcome** screen, click **Next**.

   The **Customer Information** page appears.
10. Do all of the following:
    a. In the **User Name** field, type the user name.
    b. (Optional) In the **Organization** field, type the organization to which the suite is registered.
    c. Click **Next**.

       The **Destination Folder** page appears.
11. Do all of the following:
    a. If required, click **Change** to change the installation location.
    b. Click **Next**.

       The **Setup Type** page appears.
12. Do all of the following:

a. Select **Complete** to install the default components.

b. Click **Next**.

The **Ready to Install the Program** page appears.

13. Click **Install**.

UniSim software is licensed as part of Asset Performance Management and is only for runtime and if a developer license is required then a separate UniSim Design license is required.

C.2 Third-Party Software Considerations for the UniSim Design Suite

This product may contain or be derived from materials, including software, of third parties. The third-party materials may be subject to licenses, notices, restrictions, and obligations imposed by the licensor. The licenses, notices, restrictions, and obligations, if any, may be found in the materials accompanying the product, in the documents or files accompanying such third-party materials, in a file named Third Party Licenses.pdf on the media containing the product, or at http://www.honeywellprocess.com/thirdpartylicenses.

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Options

The following third-party options are available with the UniSim Design install when purchased in addition to UniSim Design:

- UniSim Amines Option: Amines thermodynamic package allowing simulation of gas and liquid sweetening processes. Technology supplied by Schlumberger (formerly DBR).
- UniSim Multiflash Option: Thermodynamic package allowing calculation of phase behavior of complex mixtures for upstream oil and gas facilities. Technology provided by Infochem.
- UniSim OLGAS Option: Industry-standard multiphase pipeline flow correlations for calculation of flow properties. Steady-state. Available in two phase and three phase versions. Technology provided by Scandpower.
- UniSim PIPESYS Option: Integrated pipeline model supporting single- and multi-phase flow in pipes. Technology provided by SPT Group Canada Ltd. (SPT Group).
- UniSim Black Oil Option: Thermodynamic package allowing calculation of phase behavior of petroleum fluids in a non-compositional manner. Technology provided by SPT Group.
- UniSim OLI Electrolyte Option: A thermodynamic engine allowing calculation of complex electrolyte systems in UniSim Design. Two options are available, the standard aqueous chemistry solver and a new Mixed Solvent engine allowing for more complex models. Technology supplied by OLI Systems Inc.
- UniSim OLI Corrosion Analyzer Option: A utility providing corrosion analysis for material streams in UniSim Design flowsheet. Technology provided by OLI Systems Inc. Requires the OLI Electrolyte Option.
- UniSim Intercorr Predict Option: A utility providing corrosion analysis for material streams in the UniSim Design flowsheet. Technology provided by Intercorr International Inc.

Interfaces

Interfaces enable the use of third-party packages from UniSim Design. Interfaces are available for the following third-party packages when purchased separately:

- UniSim OLGA Transient Interface: Enables OLGA to be used within UniSim Dynamic Option, allowing simulation of transient multiphase flow in pipelines. Requires OLGA from Scandpower.
- UniSim Pipesim Net Option: Integrated pipeline model supporting subsurface and surface systems. Technology provided by Schlumberger.

Supported Versions

Third-Party Software	Supported Version
DIPPR	May2014
OLI Systems Corrosion Analyzer	2.0.64
OLI Systems Engine	9.2.1
SPT Group PIPESYS	2.8.3.0
SPT Group BlackOil	3.0.7.200
Schlumberger PIPESIM	2012.1
Intercorr Predict-SW	3.0
HTRI	7
Schlumberger AMSIM	v2014.1.0.1
Infochem Multiflash	4.4.10
SPT Olgas	7.3.1
CAPE-OPEN	440.1
SPT OLGA Transient Flow	7.3.3*
PetEX IPM	9.0 Build 127

- OLGA Transient versions are released much more frequently than UniSim Design. The supported version listed has been tested at the time UniSim Design R440 was released. Newer OLGA versions are likely to function properly, but this is not guaranteed, and the use of a newer version is at the user's risk.

APPENDIX

D CERTIFICATE CONFIGURATION

D.1 Important Information on Certificate

1. It is necessary to obtain and configure digital certificates on the HAS servers and computers.

   You can use self-signed certificates or custom certificates (CA).

2. HTTPS certificate needs to be configured in the IIS for the https binding. The certificate with the subject name should match with fully qualified machine name used for the HTTPS certificate.
3. If the certificate is changed after installation, you need to run the **Preinstall Checker**. Preinstall Checker utility is used for configuring the certificates, and all the required permissions are provided automatically by the tool.

> **NOTE**
> We recommended you use **Preinstall Checker** utility for configuring any certificate.

4. Use only the Certificates that support **SHA256**algorithms. If you intend to use a CA certificate, then you need to request the CA certificate which supports the SHA256 algorithm.

   Only **Microsoft Enhanced RSA** and **AES Cryptographic Provider** supports **SHA256**.

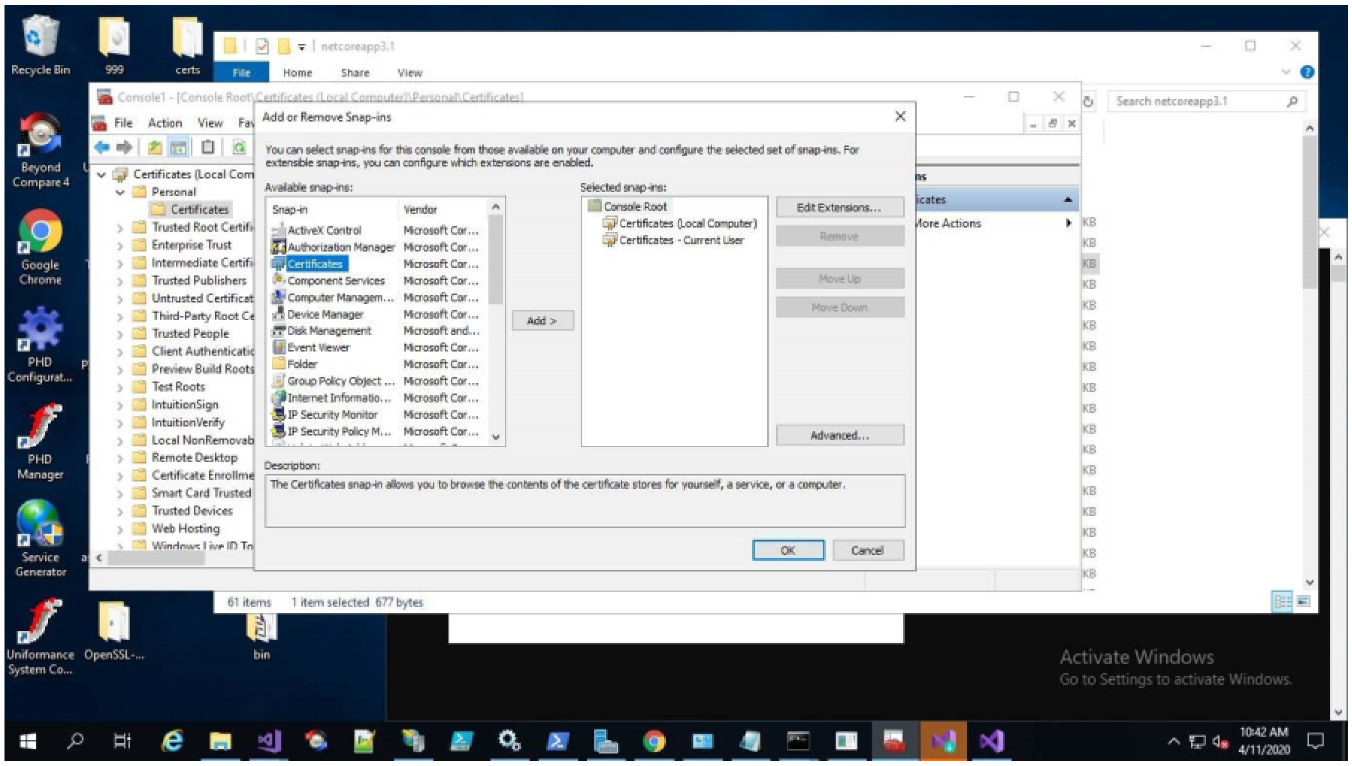
5. Cryptography Next Generation (CNG) certificates are not supported.

D.2 Configuring of Client Certificate and the Server Certificate for RabbitMQ (Trusted Certificate/Domain Controller)

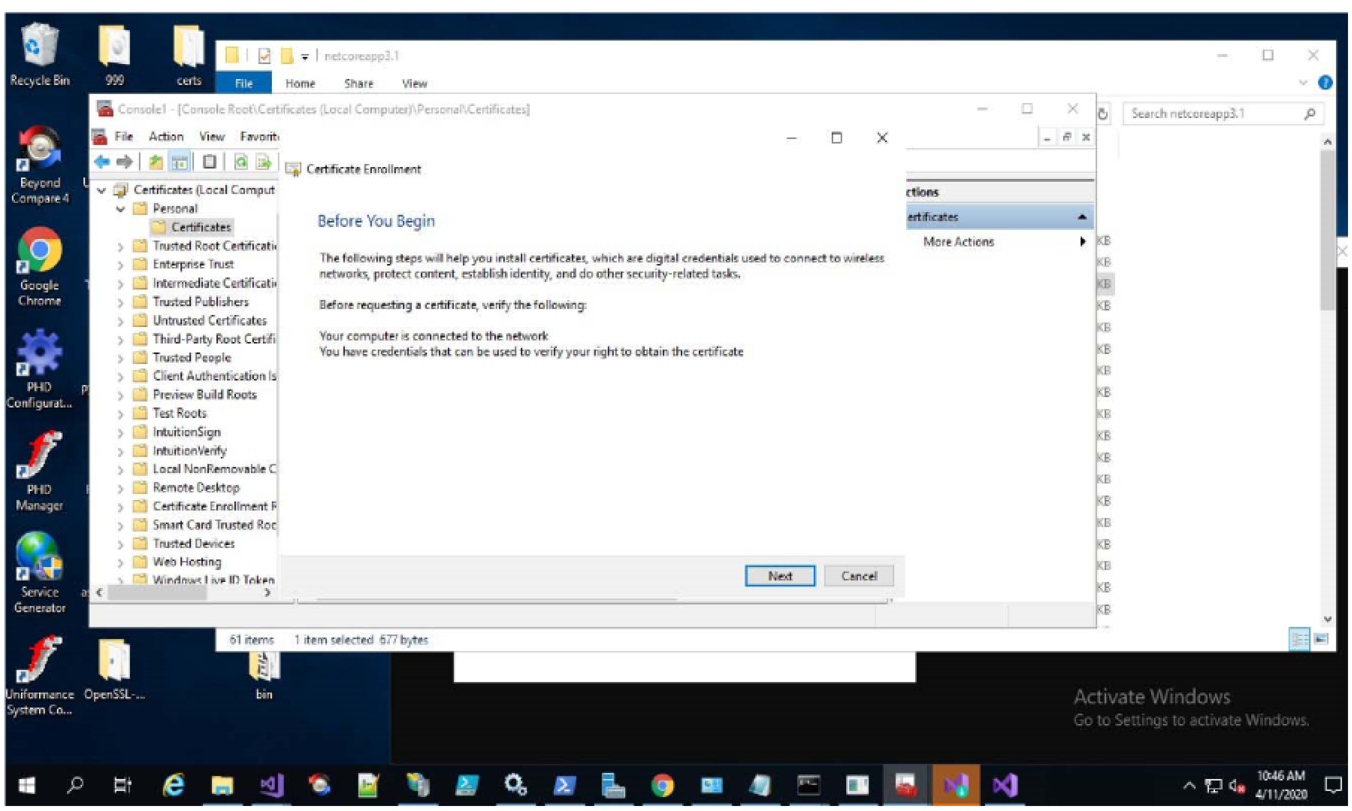
D.2.1 Configuring the Client and Server Certificate

Follow the below steps to configure the client and server certificate:

1. Open **Microsoft Management Console (MMC) > Add or Remove Snap-ins** and add a certificate for both the local and current computer.

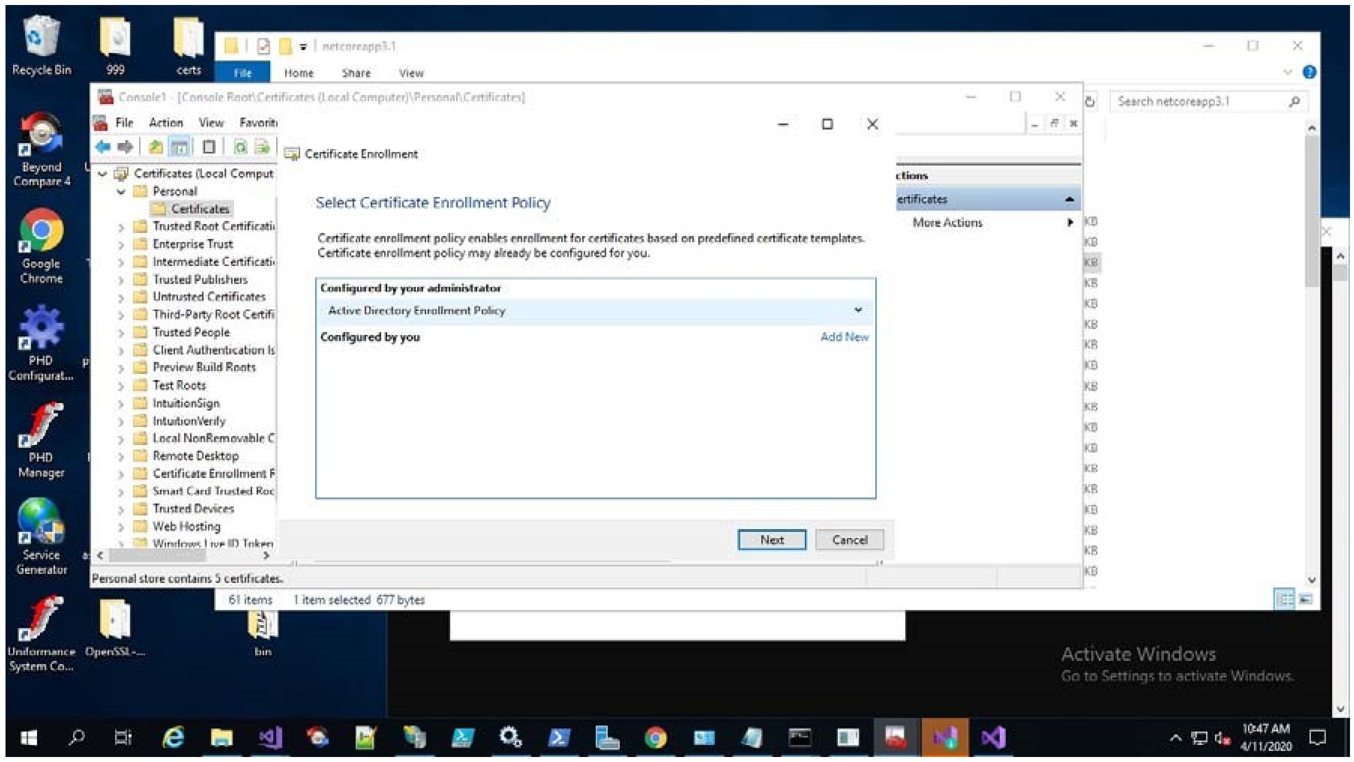


2. On the certificates (Local Computer) choose **Personal >Certificate** and click **All Tasks and Request New Certificate.**

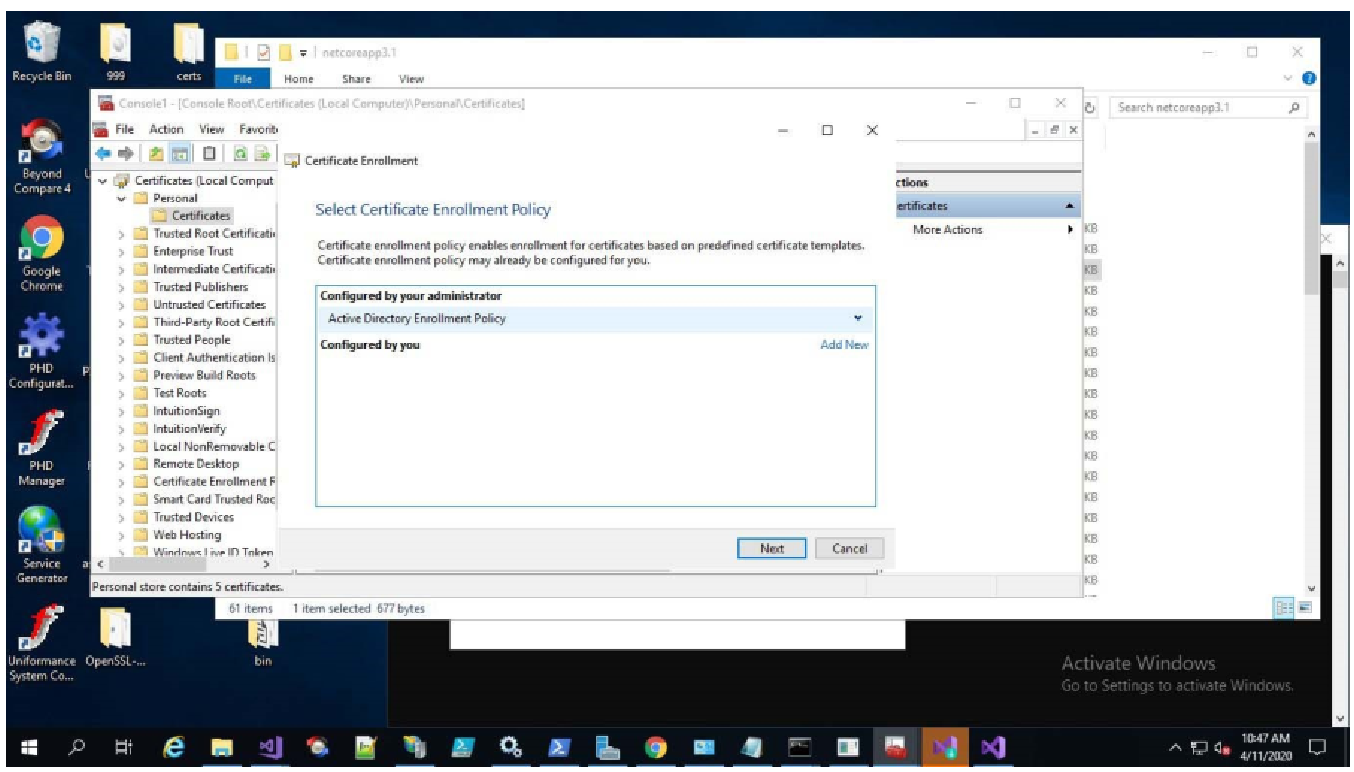


3. Follow the below steps in the certificate wizard for configuring a new certificate:

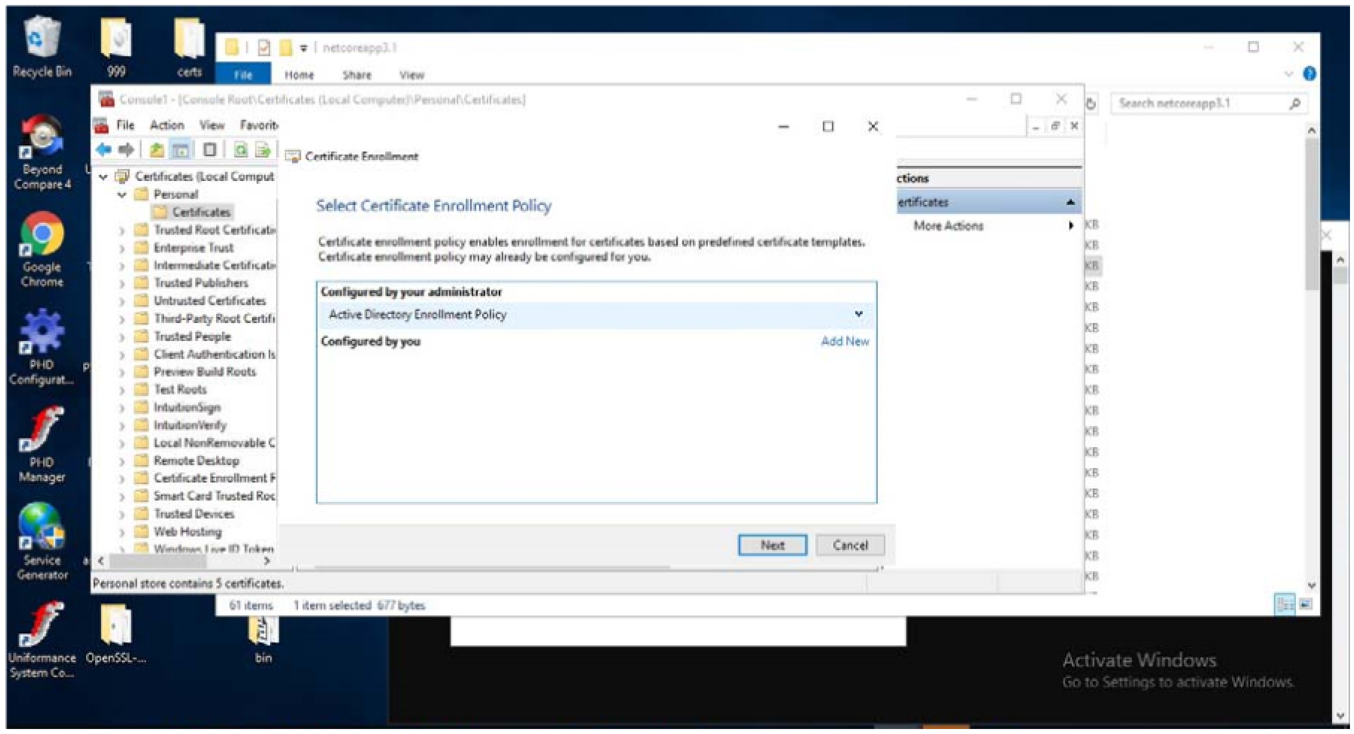
a. On the **CertificateEnrollment**, click **Next.**



b. Select **Active Directory Enrollment Policy** under **Configured by your administrator** and click **Next.**



c. Select **Active Directory Enrollment Policy** under **Request Certificate,** select **SSL Certificates**, and click **Enroll.**

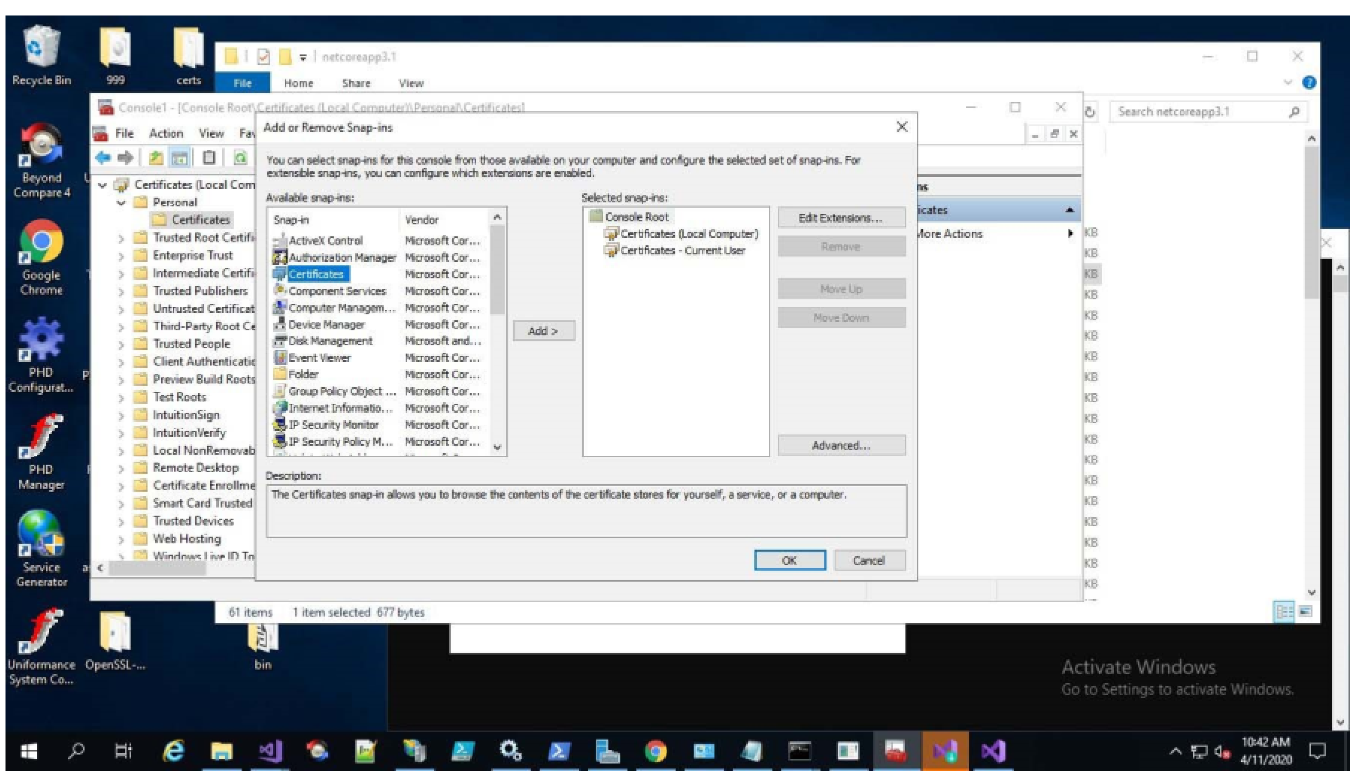


4. Export the SSL certificate as a server certificate or use any other server certificate for the same purpose (Password should be entered while exporting in pfx format).

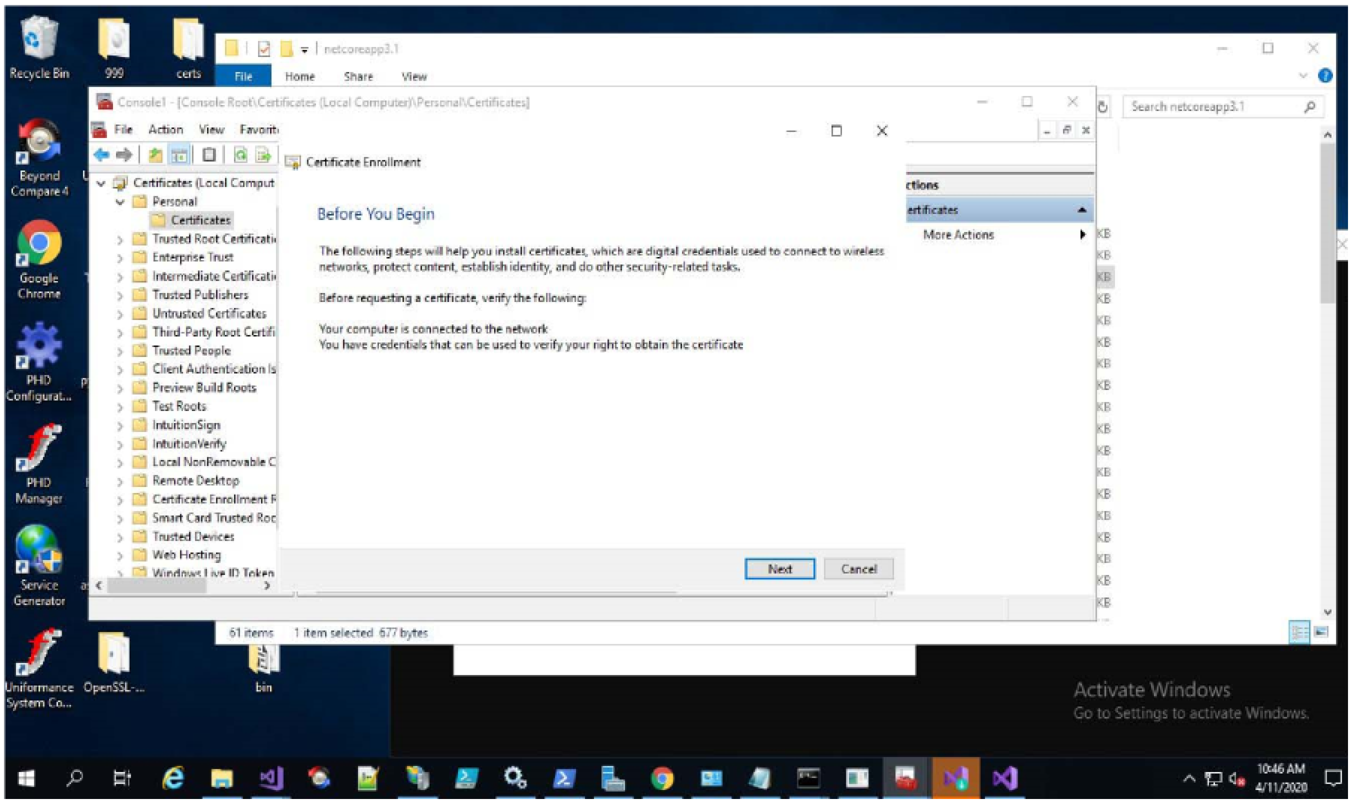
D.2.2 Configuring the Client Certificate

Follow the below steps to configure the client certificate:

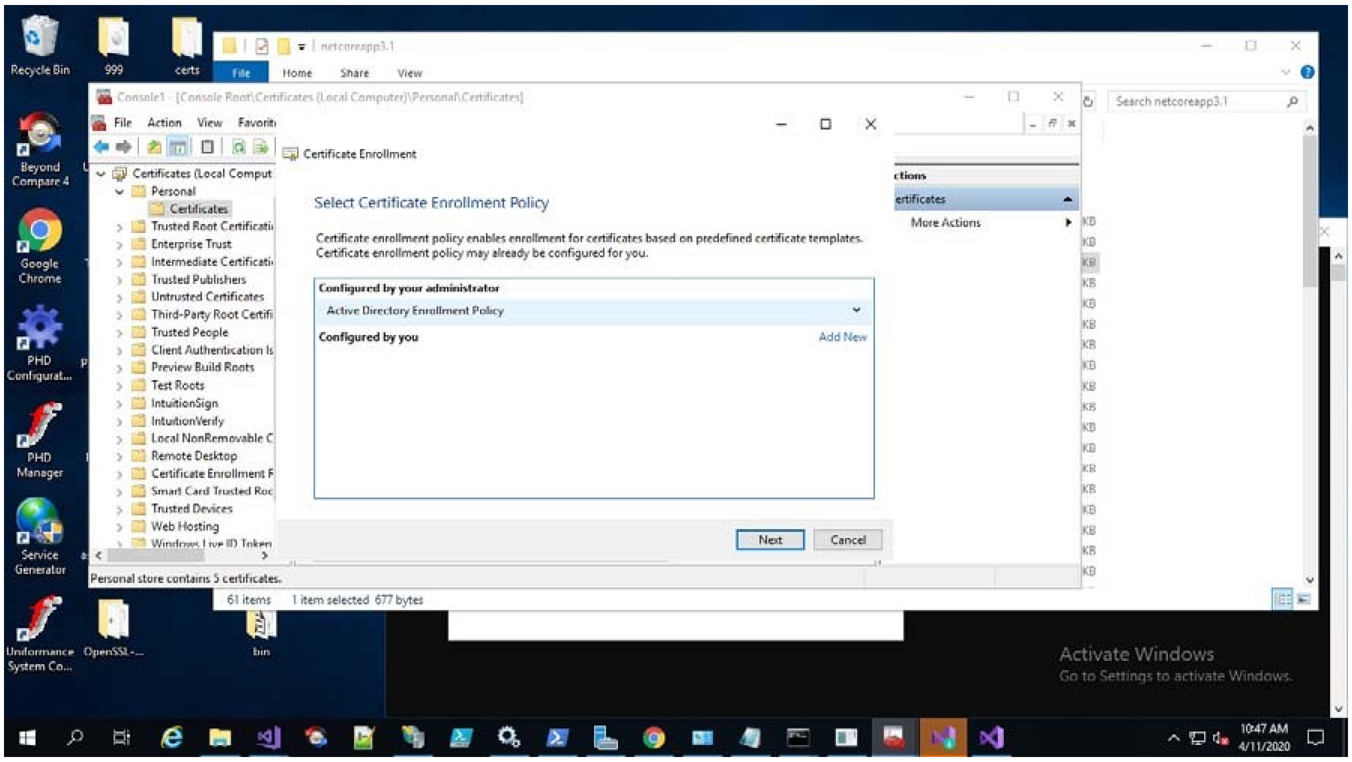
1. **Open Microsoft Management Console (MMC) > Add or Remove Snap-ins** and add a certificate for both the local and current computer.



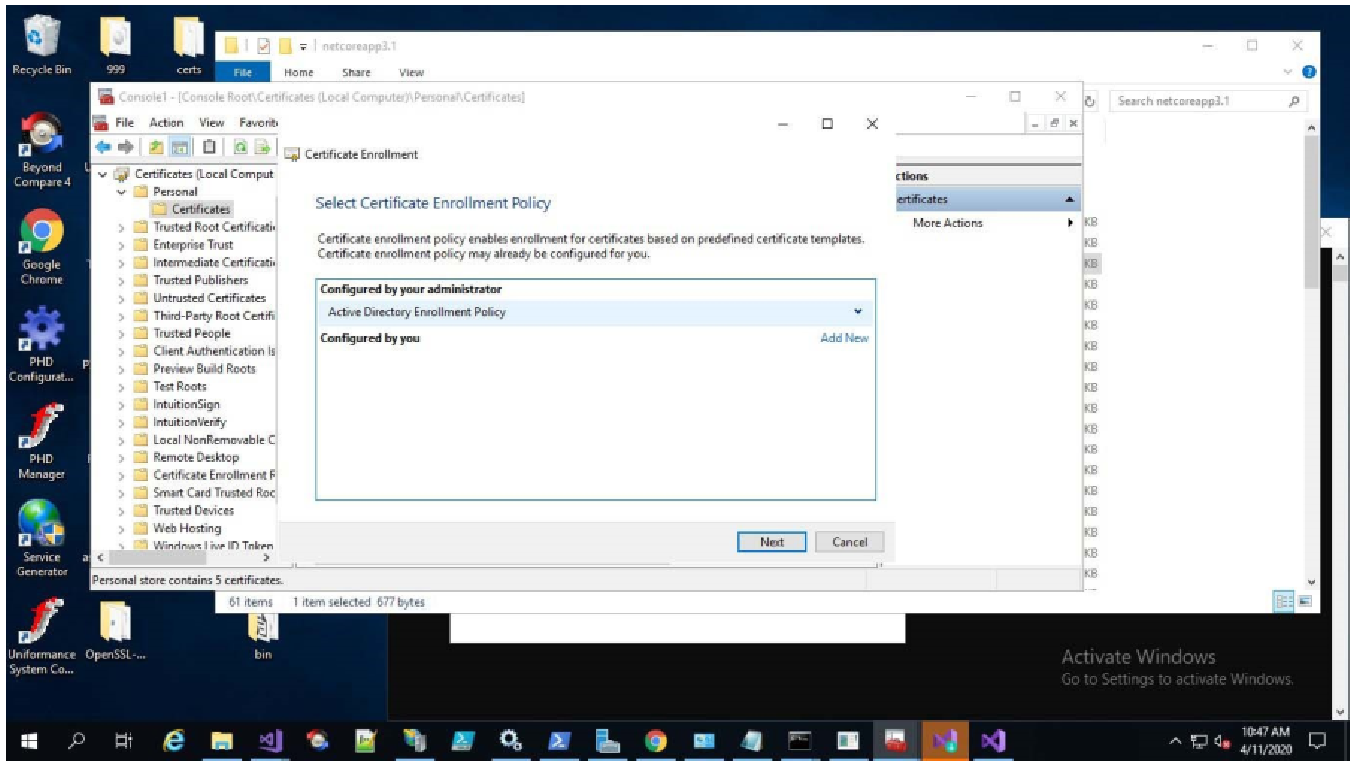
2. On the certificates (User Computer) choose **Personal>Certificate** and click **AllTasksandRequestNewCertificate.**



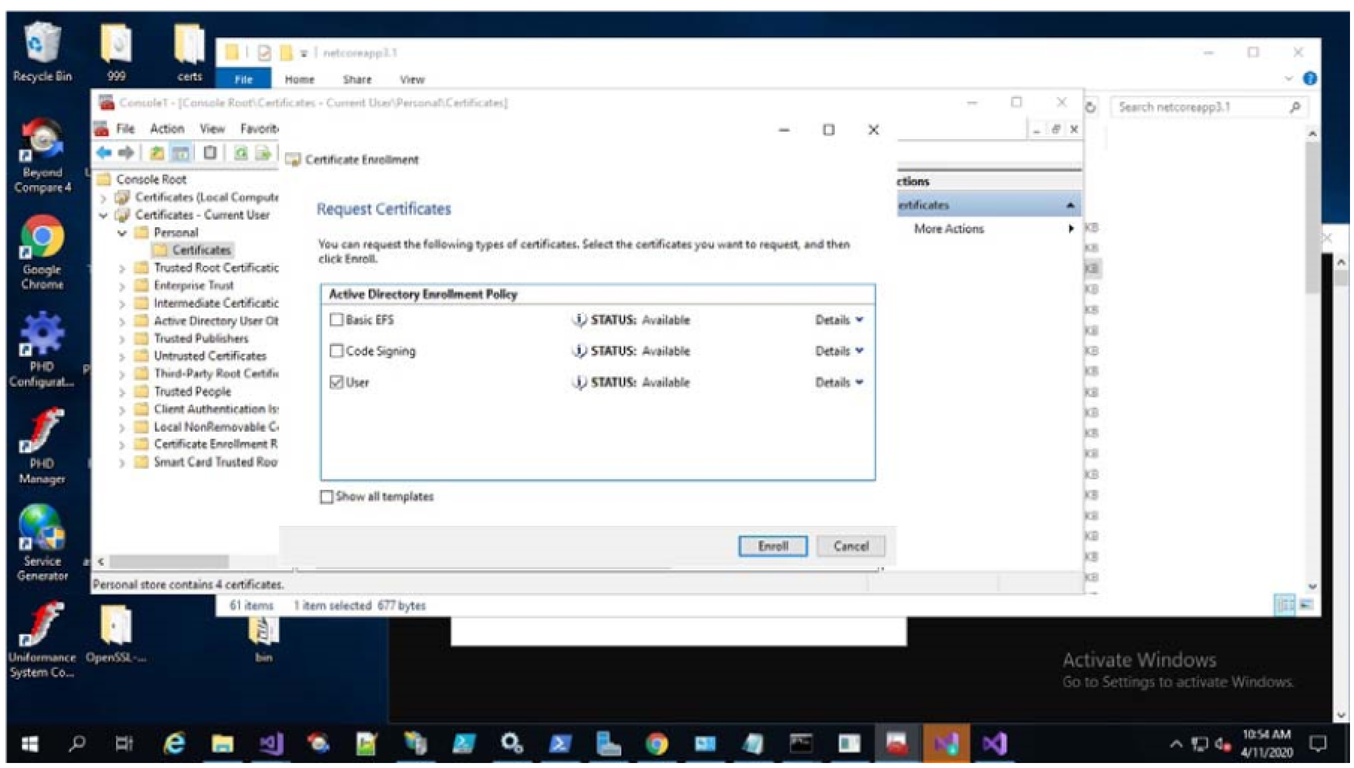
3. Follow the below steps in the certificate wizard for configuring a new certificate:
   a. On the **CertificateEnrollment**, click **Next.**



   b. Select **Active Directory Enrollment Policy** under **Configured by your administrator** and click **Next.**



c. Under **Active Directory Enrollment Policy**, select the **User** checkbox and click **Enroll.**



4. Export the SSL certificate as a client certificate or use any other server certificate for the same purpose (Password should be entered while exporting in .pfx format).
5. Follow the below steps to export the certificate:

a. Open the Console Root **(Start > Search > MMC)**.

b. Do all of the following to add the Certificates Snap-in:

   a. On the **File** menu, select **Add/Remove Snap-in.**

   b. Double-click **Certificates**

   c. Select **Computer Account** and click **Next.**

   d. Click **Finish.**

   e. Click **OK.**

c. Click **All Tasks** and select **Export.**

d. Complete the steps in the **Certificate Export Wizard**, and save the certificate in a location that is accessible or can be transferred to the other servers.

6. Open the command prompt and run the following commands:

```
<Install Location>\Honeywell\MES\AssetManager\Tools\InstallConfiguration
Honeywell.UAS.Configuration.exe -r -m -p "SerivceAccountPassword" -s
"pathtoServerCertificate" --rabbitserverpassword "ServerCertificatePassword"
-c "pathtoClientCertificate" --rabbitclientpassword
"ClientCertificatePassword"
```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APPENDIX

E PASSWORD ENCRYPTION

When there is an installation failure or reboot, you need to perform the following steps of password encryption before resuming the installation:

1. On the Asset Performance Management installation media, Select the **PasswordEncryption.exe** file in the Installer folder.
2. Right-click the PasswordEncryption.exe file and run as administrator.
3. Enter the password of the service account in the **Password** field and click **Encrypt**.

   The encrypted password is populated in the **Encrypted Password** field.
4. Click **Update Encrypted Password**.
5. Close the dialog box.

APPENDIX

F CONFIGURING SECURE SOCKET OPTION

1. **Port Number:** The default port number displayed is 25, you can change the port number as per the requirement.
2. **Secure Socket Option:** This setting will be read and used during SMTP connection and you can opt for the following options:
    a. **Auto (0)**: If the server does not support SSL or TLS, then the connection will continue without any encryption.
    b. **SslOnConnect (1)**: The connection should use SSL or TLS encryption immediately.
    c. **StartTls (2)**: Elevates the connection to use TLS encryption immediately after reading the greeting and capabilities of the server.

       If the server does not support the STARTTLS extension, then the connection will fail, and a **System Not Supported Exception** will be thrown.
    d. **StartTlsWhenAvailable (3)**: Elevates the connection to use TLS encryption immediately after reading the greeting and capabilities of the server, but only if the server supports the STARTTLS extension.

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